Drama and Redistribution:

The Impact of Telenovelas on Preferences for Redistribution in Latin America*

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Abstract

This paper examines how telenovelas depicting inequality affect support for redistribution in Latin America. Using a novel dataset of telenovelas aired from 1960 to 2024 in the region, I employed three Natural Language Processing methods to identify inequality telenovelas. My analysis focuses on instances where a new inequality telenovela is introduced during the LAPOP's fieldwork, ensuring that no other inequality telenovelas are simultaneously airing. Consistently, I observe a 4% reduction in support for redistribution following exposure to such telenovelas. These narratives seem to misrepresent the problems of inequality, portraying fewer issues than are present in reality, making individuals adjust their policy priorities. This study provides evidence of how entertainment media can shape the perceptions of inequality, thereby influencing policy preferences.

JEL Clasification: H11, H23, N16, P16, L82
Keywords: Redistribution, Media, Latin America, Inequality, Perceptions, Telenovelas

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1 Introduction

Inequality perceptions are central to understanding the demand for redistributive policies (Alesina and Giuliano, 2011; Stantcheva, 2021). These perceptions typically arise from individuals' direct exposure to inequality within their country or neighborhood (Alesina, Stantcheva and Teso, 2018; Alesina, Miano and Stantcheva, 2023; Domènech-Arumí, 2023). The portrayal of inequality in traditional media can skew individuals' perceptions of income inequality within their country, impacting their trust in others (Diermeier *et al.*, 2017), and shaping their beliefs about upward mobility (Kim, 2023). Consequently, these (mis)perceptions of inequality subsequently impact individuals' likelihood of approving redistributive policies. Despite this, there is limited understanding of how entertainment media, with its potential to present a fictional and potentially inaccurate representation of inequality, affects perceptions of or attitudes towards redistribution.

This paper examines how television shows depicting inequality affect support for redistribution in Latin America. I focus on telenovelas, commonly known as soap operas, a publicly and freely available TV show usually aired during prime time and central to Latin American culture. These shows appeal to a mass audience and usually display social norms, cultural identity, and the historical roots of each country (Lopez, 2002; Pastina, Rego and Straubhaar, 2003; Green, Brock and Kaufman, 2004; Antezana *et al.*, 2022; 2023). The cultural influence of telenovelas is evident in naming trends, as many newborns are named after popular characters (La Ferrara, Chong and Duryea, 2012). Beyond their entertainment value, telenovelas serve as powerful instruments for influencing individuals' behaviors and attitudes on various social issues throughout Latin America (Chong and La Ferrara, 2009; Bertrand, 2020; Gulesci, Lombardi and Ramos, 2024).

To estimate the effect of inequality telenovelas on support for redistribution, I collect a novel dataset of telenovelas aired in the Latin America from 1960 to 2024. This dataset contains information on each show's airtime, start and end dates, broadcasting channel, genre, and a plot synopsis. To identify telenovelas that specifically address issues of inequality, I employed Natural Language Processing (NLP) techniques on the synopses. I implement three distinct methods: (i) a dictionary-based approach that utilizes related terms extracted from ConceptNet, (ii) a vector embedding approach using the Universal Sentence Encoder (USE), (iii) a generative AI model capable of classifying content with no examples. Each of these methods has its strengths and limitations. The ConceptNet approach is the most intuitive way to capture inequality themes, but fails to capture more nuanced aspects. In contrast, the generative AI models are less transparent yet show the best performance in classification precision.

My empirical strategy centers on instances where a new inequality telenovela is introduced during the Latin American Public Opinion Project (LAPOP) fieldwork. I compare individuals currently exposed to an inequality telenovela (those interviewed after the telenovela's debut) with those who are not exposed (interviewed beforehand). The identification strategy encounters two key challenges. First, there is the concern that the groups interviewed before and after the telenovela's premiere differ significantly. To address this, I provide a balance table that reveals no significant differences in observable characteristics. Second, the validity of the results depends on the accurate identification of inequality themes by the NLP methods. To validate this, I show that the introduction of an inequality telenovela is not associated with other political and social issues like the support for minorities. Moreover, these methods prove robust to variations of parameters for each method. Lastly, I show that telenovelas identified as lacking inequality content show no effect of exposure on support for redistribution.

Exposure to an inequality telenovela reduces support for redistribution, with the estimated effects ranging from 3% to 5% reduction, depending on the categorization method. Taking an ensemble of all three methods, I find a statistically significant 3% reduction in support for redistribution after

being exposed to an inequality telenovela. This effect appears to reverse in cases where an inequality telenovela airs while another is already on-air. Under these conditions, exposure is associated with a significant increase in support for redistribution. This finding, however, is not robust to changes in the classification method, unlike the main finding.

The results suggest that inequality telenovelas misrepresent the inequality issues, leading individuals to alter their policy preferences based on this distorted depiction. For instance, when these telenovelas portray the disparity between the rich and the poor in a comedic light, they may diminish the perceived urgency of addressing inequality-related challenges. Additional evidence suggests exposure to an inequality telenovela shifts individuals' policy priorities. Exposed individuals report placing less importance on inequality and poverty, while focusing more on issues like the economy, violence, or government performance. Individuals with a keen interest in politics, and potentially more informed about economic conditions, experience no significant impact from inequality telenovelas. In contrast, those who identify as lower class exhibit an amplified reduction in support, indicating that these portrayals may lead them to perceive systemic inequality as less severe than it is.

Inequality telenovelas are also associated with changes in voting behavior. I merge my telenovela dataset with presidential election data across Latin America and test whether the premier of an inequality telenovela in the months leading to presidential elections affects voter behavior. I find that an additional inequality telenovela is associated with a 0.02 percentage point decrease in voter turnout. Furthermore, this exposure is also associated with a narrower victory margin for the winning candidate and a lower probability of incumbent turnover. These findings suggest that entertainment media can subtly influence political landscapes, potentially by dissuading voters to participate in their civic duties or generating a bias towards the status quo.

1.1 Literature Review

This work contributes to the literature that analyzes the role of media in shaping individuals' social attitudes and behavior. Mass media (e.g., news, entertainment, and social media) are the main drivers of economic perceptions (Soroka, 2014). Pioneering studies have focused on media bias in newspapers and radio and its impact on political outcomes (Besley and Burgess, 2002; Strömberg, 2004; Arceneaux and Johnson, 2013; Drago, Nannicini and Sobbrio, 2014; Gentzkow, Shapiro and Stone, 2015; Martin and Yurukoglu, 2017; Ash and Hansen, 2023). Recently, researchers have explored the effects of television news on various social and political behaviors, such as voter turnout and teenage education outcomes (Gentzkow, 2006; Gentzkow and Shapiro, 2008).

Entertainment media offers individuals a fictional or idealized economic reality and has been shown to shape various political attitudes by "transporting" the viewer into the fictional world and generating an emotional connection (Green, Brock and Kaufman, 2004; Holbrook and Hill, 2005; Morgan and Shanahan, 2010). Economists have shown the effects of entertainment media on the acceptance of domestic violence (Jensen and Oster, 2009), teenage pregnancy (Kearney and Levine, 2015), HIV attitudes (Banerjee, La Ferrara and Orozco-Olvera, 2019), approval of domestic violence (Banerjee, Ferrara and Orozco, 2019), and early educational outcomes (Kearney and Levine, 2019). Within entertainment media, telenovelas offer a unique perspective because of their role in Latin American culture (Lopez, 2002; Green, Brock and Kaufman, 2004; Antezana *et al.*, 2022; 2023). Outside of economics, the literature has focused on the contents of telenovelas and their relation to inequality (Tufte, 2000; Mayer, 2003; Coppini, Alvarez and Rojas, 2018). In economics, researchers have focused on the effects of telenovelas on demographic changes or support for minorities. For instance, La Ferrara, Chong and Duryea (2012) and Chong and La Ferrara (2009) show how the entrance of Rede Globo, which essentially introduced telenovelas to viewers in Brazil, decreased fertility and increased divorce filings, respectively. Moreover, Gulesci, Lombardi and Ramos (2024) provides evidence

suggesting that exposure to characters from the LGBTIQ+ community in telenovelas reduces support for said community in Latin American countries. The latter study highlights media representation's complex and potentially contradictory effects on attitudes toward specific communities or topics. This paper contributes to the literature on telenovelas and culture by focusing on the subject of inequality and considering both the content and the effects of telenovelas.

This paper also extends to the broader literature seeking to understand how perceptions of inequality influence the demand for redistributive policies (Alesina and Giuliano, 2011; Stantcheva, 2021). Alesina, Stantcheva and Teso (2018) explores the role of upward mobility in shaping perceptions of inequality, shedding light on how individuals' aspirations and opportunities for advancement influence their attitudes toward redistributive policies. Alesina, Miano and Stantcheva (2023) shows evidence that the (mis)perceptions of inequality about the intensity of immigration decrease support for redistributive policies. Domènech-Arumí (2023) argues that exposure to inequality within one's neighborhood can improve the approval of redistribution policies, adding to our understanding of the complex nature of individual experiences and redistributive preferences. Note that this part of the literature focuses on experimental designs. This study highlights the role of entertainment media in shaping individuals' perceptions of their own social class, uses a survey approach rather than experiments, and provides evidence outside of online experiments that (mis)perceptions of inequality are the main drivers of support for redistributive policies.

This paper contributes to the existing literature by shedding light on the role of telenovelas in shaping viewers' preferences for redistributive policies. Latin America has a long history of significant income inequality (Fergusson, Robinson and Torres, 2023) alongside lower levels of redistribution compared to developed economies (Ocampo and Gómez-Arteaga, 2018). Given the ongoing struggles with income distribution in the region, it's vital to grasp the impact of entertainment media on the survival of various redistributive policies through political approval.

A small part of this literature has studied the portrayal of inequality or upward mobility in the media and its impact on perceptions. Diermeier *et al.* (2017) shows that intense coverage of inequality in newspapers worsens individuals' perceptions of social fairness in Germany. Kim (2023) shows that exposure to programs that promote "rags-to-riches" narratives (e.g., American Idol or Shark Tank) reinforces Americans' belief in upward mobility through laboratory and online experiments. In contrast, our study focuses on preferences for redistribution instead of fairness or upward mobility, allowing for a more direct examination of individuals' support for policies addressing inequality.

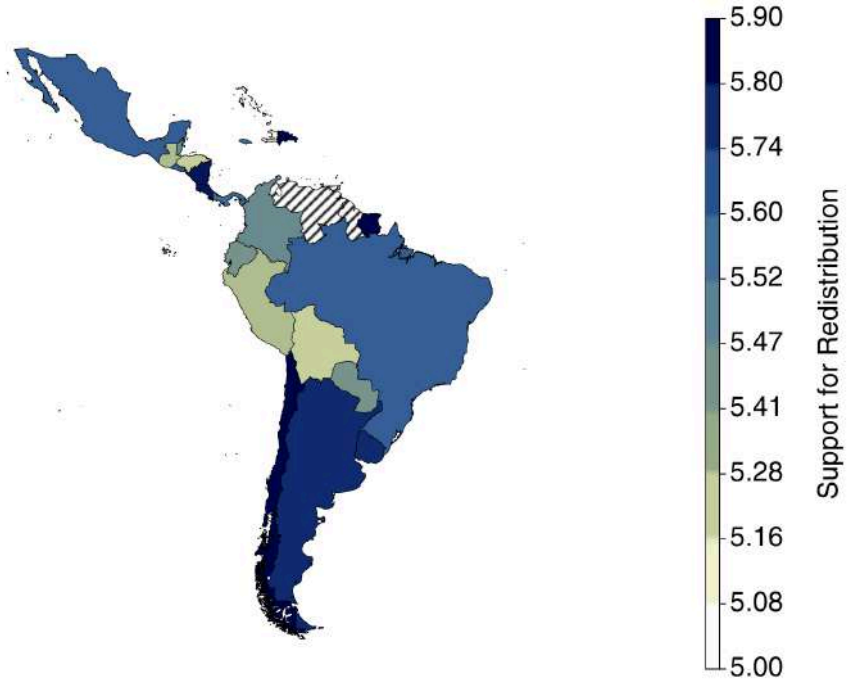
2 Data

2.1 Preference for Redistribution Surveys

Data on preferences for redistribution was sourced from the Latin American Public Opinion Project (LAPOP). The LAPOP has conducted numerous surveys across Latin America, collecting answers to a wide range of socio-political questions. To measure support for redistributive policies, individuals were asked to rank their agreement to the following statement from 1 to 7. *"The (Country) government should implement strong policies to reduce income inequality between the rich and the poor"*.[1] An answer of 1 indicates strong disagreement with the statement and 7 a strong agreement with the statement. This

[1]Some studies elicit preferences for redistribution using similar a question (Almas *et al.*, 2025). Others measuresupport for redistribution using alternative questions centered on the role of government to reduce inequality. Alesina and Giuliano (2011) uses as their main measure for support for redistribution the following question from the World Value Survey *"Some people think that the government in Washington should do everything to improve the standard of living of all poor Americans (they are at point 1 on this card). Other people think it is not the government's responsibility, and that each person should take care of himself (they are at point 5). Where are you placing yourself in this scale?"* Alesina, Miano and Stantcheva (2023) elicited support for redistribution by asking how much of the government's budget should be destined to social causes like education, and how much should the government tax the top 1% and the bottom 90%.

Figure 1: Average Support for Redistribution in Latin America



Notes: This figure displays the average responses to the support for redistribution question in LAPOP surveys. A score of 7 indicates strong agreement, while a score of 1 represents strong disagreement. For each country, I calculate the average response across the surveyed years, applying the provided population weights.

question is available across all survey waves in the sample, spanning from 2006 to 2023. Figure 1 shows the mean value of the outcome question across the survey waves in each country. There is a high level of support for redistribution in Latin America with an overall mean of 5.5 out of 7, with most of the individuals in each country concentrating on high values of support. The countries with the highest approval for redistribution are Chile, Nicaragua, and Costa Rica, while Haiti and Bolivia exhibit the lowest support.

2.2 Telenovelas

Data on telenovelas was collected using various online sources tailored to each country of interest. From each source, we gathered a written synopsis, the air time, the duration (air and end date), the number of episodes, the broadcasting channel, and the genre of the show (e.g. comedy, drama, or melodrama). Additionally, we extracted ratings for each telenovela from the IMDb website, indicating of how well-received each telenovela was by its audience. Telenovelas aired in Chile were gathered from the chilenovelas, a Wiki-like page, which provides comprehensive lists of telenovelas along the genre of telenovelas, ratings, duration, airing times, cast members, and synopses. Similar information for Brazilian telenovelas was sourced from teledramaturgia. For other Latin American countries, lists of telenovelas are available on Wikipedia and user's IMDb lists. Each synopsis was translated into English using OpenAI's gpt-4o-mini model, which provides a good quality translation from Spanish to English and performs reasonably well translating from Portuguese to English (Sanz-Valdivieso and López-Arroyo, 2023; Törnberg, 2023).[2]

A total of 1852 telenovelas were gathered from Chile, Brazil, Mexico, Colombia, Panama, and Argentina, with around 90% concentrated in Chile, Brazil, and Mexico. Table 1 shows some descriptive statistics for all the telenovelas in my sample. Around 34% of telenovelas are dramas or melodramas, around 17% are comedies and the rest are mix of comedy and drama. On average, a

[2]There are other other alternatives like DeepL and Google Translate that come at a higher cost. See (Hidalgo-Ternero, 2020) for a discussion on the Spanish to English translation of these two alternatives.

Table 1: Descriptive Statistics of Telenovelas

	mean	sd	min	max
Drama (%)	34.39			
Air time	16:38	07:09	10:00	23:15
N of episodes	129	76	1	1018
Synopsis word count	1719	1681	17	20784
IMDB Rating	6.95	1.12	1.80	9.40

Notes: This table provides descriptive statistics for all collected telenovelas between 1960-present in Chile, Mexico, Argentina, Colombian, Panama, and Brazil. TV ratings refer to Nielsen ratings, which measure the percentage of households that, on average, tuned in to watch the show throughout its entire run. IMDb ratings represent user ratings for each telenovela, as listed on the IMDB website.

telenovela airs around 16:38, consists of 129 episodes, and an IMDB user rating of 6.95 out of 10. The word count for each synopsis varies significantly, with a mean of around 2,000 words, but with some telenovelas having as few as 66 words and others exceeding 10,000 words.

The discrepancy in synopsis in word length might pose a challenge, as I aim to determine whether these narratives explore themes related to socio-economic differences among their characters. On average, a typical telenovela synopsis is extensive, averaging around 2,000 words, comparable to a couple of paragraphs. This length offers an overview of the plot with sufficient descriptions of the characters and the plot. However, there are notable exceptions within the dataset. Two telenovelas feature summaries that are less than 100 words, providing a brief overview of the plot, which may overlook subtler expressions of inequality. At the other end of the spectrum, the maximum word count for a synopsis reaches approximately 20,000 words, comparable to a small book. Four telenovelas have summaries exceeding 10,000 words, often including detailed descriptions of various seasons or encompass the entirety of the series. While rich in content, they introduce several themes making it challenging to extract a single central theme. Nevertheless, short and extensive synopses only account for less than 1% of all collected telenovelas, with the vast majority centered around the mean length.

2.3 How is inequality displayed in Telenovelas?

Telenovelas can visually show inequality through the choice of sets and wardrobe; audibly through the character's expressions and how they speak; and lastly, through the ideas with the script and dialogue. Pobre Rico, a popular comedy television series aired in Chile in 2012, narrates the story of two babies swapped at birth between a wealthy and a middle-class family, illustrating social class differences.[3] It juxtaposes the challenges a teen gas station worker faces with the lifestyle of the son of one of the most influential families in the country. The first paragraph of the translated synopsis can be found below:

First Paragraph of *Pobre Rico*'s Synopsis
Freddy Pérez and Nicolás Cotapos have led a normal life until now. They are two happy boys and are very comfortable in each of their worlds. Meanwhile, El Rucio, as Freddy is known, lives with his mother Eloísa and his sister Megan in poorer economic conditions. Nicolás lives with his parents Máximo and Virginia, and with his sister Julieta, in a wealthy family. The Cotapos are owners of the Cotapos Holding and Cotapos Airlines, while the Pérez are workers at a gas station.

[3]This telenovela draws inspiration from the classic tale of *The Prince and the Pauper*.

Figure 2: Visual Social Class Differences in *Pobre Rico*

Panel A: *Cotapos* (rich) house

Panel B: *Pérez* ("poor") house

Panel C: *Cotapos* (rich) living room

Panel D: *Pérez* ("poor") living room

Notes: these screenshots were taken from the first episode "Cambiados al Nacer" of Pore Rico available on YouTube

This telenovela illustrates the economic gap between the Cotapos (rich) and Pérez (poor) families through their sets and script. Figure 2 displays screenshots taken from the first episode of the show. Panels A and C portray the Chilean wealth, characterized by a spacious residence adorned with several windows and a fence, alongside modernly decorated living rooms, ownership of at least two cars, and a maid. The Cotapos family represents the elite, belonging to the top 0.1% of the income distribution; they own one of the largest corporations and have a direct line to the president. In stark contrast, panel B and D present the reality of low-middle-income families, with modest house featuring only two small windows, the absence of cars, and worn-down living spaces. The Pérez family works at a gas station and, while facing their own struggles, live in a fully equipped home, owns a car, and maintains a complete nuclear family.

A common pattern across telenovelas is to show inequality by focusing on the struggle between the top 1% earners and the middle class, as producers often aim to appeal to a mass audience, particularly the working class. *Pobre Rico* does not adequately address the struggles faced by the most impoverished members of society and, instead, focus on the disparities between the wealthy elite and the gas station workers. By omitting the lowest social class, these telenovelas may present a distorted reality of the problem of inequality, which in turn can lead to the (mis)perception of inequality in a society.

3 Measuring Inequality in Telenovelas

To identify telenovelas that portray inequality, I utilize three natural language processing (NLP) techniques to analyze the information about the content of the telenovelas using their synopsis. The synopsis of telenovelas usually contains the main plot points of the story, the primary characters, and

the challenges they face, providing a good representation of the content of the first few episodes.[4] Each method will categorize telenovelas into two groups. Inequality telenovelas are those telenovelas that each method predicts its plot contains social class conflict. No-inequality telenovelas are the ones in which social class conflict is not present or is not the central conflict.

Using NLP is essential for several reasons. Human categorization, while valuable, is often unreliable, as it can introduce subjective bias (Grimmer and Stewart, 2013; Evans and Aceves, 2016; Dhar *et al.*, 2021; Rathje *et al.*, 2024). Additionally, manually categorizing a large number of telenovelas is both time-consuming and costly. In contrast, machine-based categorization, though not without its imperfections, offers a more efficient and scalable solution. These techniques have been employed in previous research, and evidence suggests that they correlate with human categorization (Törnberg, 2023; Michalopoulos and Rauh, 2024; Rathje *et al.*, 2024). I employ three specific methods using the telenovela synopses as input: (i) a dictionary approach using related words extracted from ConceptNet, (ii) a vector embedding approach utilizing the Universal Sentence Encoder (USE), and (iii) categorization through ChatGPT, a generative AI capable of classifying with few or even zero examples.

3.1 ConceptNet

The first method is based on a simple idea: the frequency of words such as "poor," "humble," "rich," and "worker" indicates that the telenovela focuses on inequality. I employ a bag of words or dictionary approach to find out which telenovelas use these words the most. I clean the text of each synopsis by removing stop words, numbers, and proper nouns and extract the lemmas for each word[5]

The ConceptNet score is the frequency of the inequality words that appear in each telenovelas synopsis normalized by the text lemma length. Let D represent an inequality dictionary that contains words that represent socio-economic inequality, chosen by the researcher. The $\text{ConceptNet score}_t$ for any telenovela t with a cleaned synopsis of lemma length n_t, and each lemma within that synopsis is w_i where $i = 1, 2, ..., n_t$ is measured by

$$\text{ConceptNet score}_t = \sum_{i}^{n_t} \frac{I(w_i \in D)}{n_t}$$

where $I(w_i \in D)$ is an indicator function taking the value of 1 if the lemma w_i is part of the inequality

Table 2: Related Words to "poor" Using ConceptNet

root	related terms		
	en	es	pt
poor	poor	pobre	pobre
	poorer	pobrete	pobres
	slummy	pobretones	pobretão
	resourceless	pobretón	
		pobres	

[4]Other researches have relied on show synopsis to evaluate the content of the TV shows. Kim (2023) used the summaries to identify rags-to-riches contests shows (e.g. Shark Tank) which ordinary American participate to gain economic benefits through hard work and effort. However, telenovela synopses often provide incomplete plot overviews, as they may omit evolving conflicts and themes, leading to a fragmented understanding of the full story arc, but rather a representation of the starting conflict.

[5]Lemmatization is used to capture different forms of the same word. For example, "break," "breaks," "broke," "broken," and "breaking" are all variations of the lemma "break."

dictionary and 0 otherwise.[6]

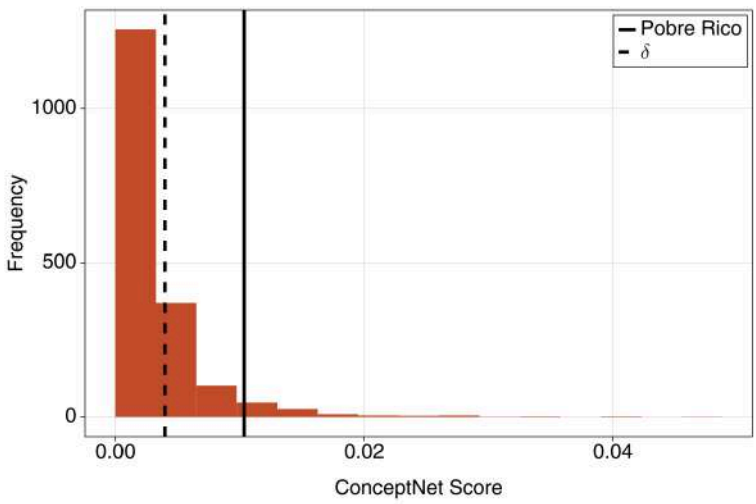
To build the dictionary D, I begin by compiling a list of root words that typically refer to social class conflict, such as "economic," "wealthy," "inequality," and "social class."[7] Using these root words, I then search for related terms in ConceptNet, a knowledge graph of natural languages that show the relations between words in any language. For example, in Table 2 contains the root word "poor" and some of the related terms found in English (EN), Spanish (ES) , and Portuguese (PT). The word "poor" is related to terms are "poor", "poorer," "resourceless," and "slummy," in English, as well as "pobre" and "pobretón" in Spanish which translates to "poor".

Any telenovela with a $\text{ConceptNet score}_t$ higher than a threshold δ will be considered an inequality telenovela. The chosen threshold is equal to 1.5 standard deviations bellow the $\text{ConceptNet score}_{pr}$ of *Pobre Rico*. This will capture more subtler ways in which inequality represented by less frequency of inequality words than *Pobre Rico*.[8]

Figure 3 shows the histogram of the ConceptNet scores across the telenovelas in the sample. The mean score of 0.0027 with a standard deviation of 0.0042. Pobre Rico has a score of 0.011, indicating that the average telenovela in the dataset is about two and a half standard deviations away from Pobre Rico in terms of social class-related content. The chosen threshold δ is set to 0.004, corresponding to 438 inequality telenovelas in all my sample.

To provide context for what these scores represent, consider the telenovela *Chipe Libre*, which has a score of 0. This story revolves around a couple taking a break and exploring new romances, leading to complicated relationships that challenge their views on love and fidelity, with no significant social class conflict depicted. A telenovela with social class conflict in their plot but doesn't have frequent inequality words is *Corazón Rebelde*. This telenovela has a score of 0.006 and tells the story of a group of scholarship students in an elite high school tormented by their wealthy classmates. *Asi en el Barrio Como en el Cielo* is the telenovela with the highest score of 0.039, with a plot paralleling Pobre Rico with a stronger emphasis on the class differences between the two families.

Figure 3: ConceptNet Score Distribution



Notes: This figure shows the ConceptNet scores of the telenovelas in my sample. The ConceptNet score is the frequency of all the words in the built inequality dictionary, normalized by the lemma count of each synopsis. Inequality telenovelas are those with a ConceptNet score $\geq \delta$. The threshold $\delta = 0.004$ was constructed taking 1.5 standard deviations away from the ConceptNet score of Pobre Rico.

[6]Example B.1 provides an example of how the ConceptNet score is calculated.

[7]For the full list of root words and their related terms please see Table Table B.3 and Table B.4 .

[8]I consider different δ's, with 2 and 1 standard deviations aways from *Pobre Rico*, in the robustness tests.

3.2 Vector Embeddings

The second method employs the Universal Sentence Encoder (USE) to assign a vector to each synopsis, enabling the identification of telenovelas that closely resemble Pobre Rico. The USE is a machine learning model developed by Google that effectively transforms sentences and phrases into vector representations (Cer *et al.*, 2018). It is part of a broader family of models designed to convert words, text, and documents into vectors, commonly referred to as vector embeddings. These vector representations play a crucial role in various of natural language processing tasks, including classification and semantic similarity. The algorithm processes text to produce a 512-dimensional vector, which captures the input's context and semantic information. This capability extends beyond simple word matching, a limitation exhibited by the ConceptNet method. Another distinguishing feature of the USE is its comprehensive approach; unlike traditional methods that concentrate on individual words, the USE considers the entire sentence. This allows the model to grasp subtleties and contextual elements more effectively.[9]

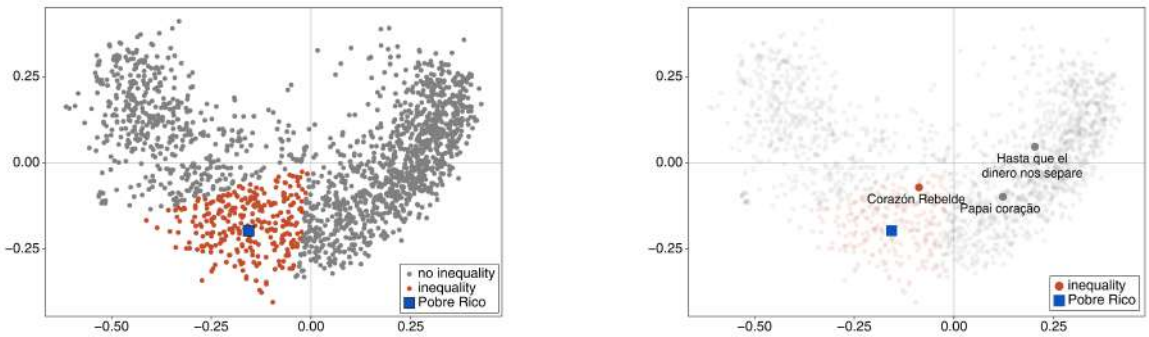
In this study, I apply the USE to the translated synopses of various telenovelas and keep the first 2 principal component using the Kernel PCA algorithm (Schölkopf, Smola and Müller, 1998).[10] Subsequently, I calculate the cosine similarity between the telenovela Pobre Rico and other telenovelas. The cosine similarity, cs, is formally defined as follows

$$cs\left(A, \overrightarrow{\text{Pobre Rico}}\right) = \frac{A \times \overrightarrow{\text{Pobre Rico}}}{\|A\| \times \| \overrightarrow{\text{Pobre Rico}} \|}$$

where A is the vector embedding of the synopsis of a telenovela, $\overrightarrow{\text{Pobre Rico}}$ is the 2-dimentional vector embedding for the Pobre Rico synopsis, and $\|A\|$ is the Euclidean norm of the 2-dimentional vector embedding of A. The cosine similarity ranges from −1 to 1, where the 1 shows that A has the same direction as the $\overrightarrow{\text{Pobre Rico}}$ vector, while a −1 shows that A points to the opposite direction, and a 0 indicates that A is perpendicular to $\overrightarrow{\text{Pobre Rico}}$. Although this metric has a mathematical interpretation, what does it mean for a telenovela to be opposite or perpendicular to *Pobre Rico*?

Figure 4 plots the principal components derived for each telenovela's vector embedding. The telenovela *Hasta que el dinero nos separe* is an opposite vector to *Pobre Rico* with a similarity score

Figure 4: Telenovela Data Latent Space *Pobre Rico*



Panel A: full sample

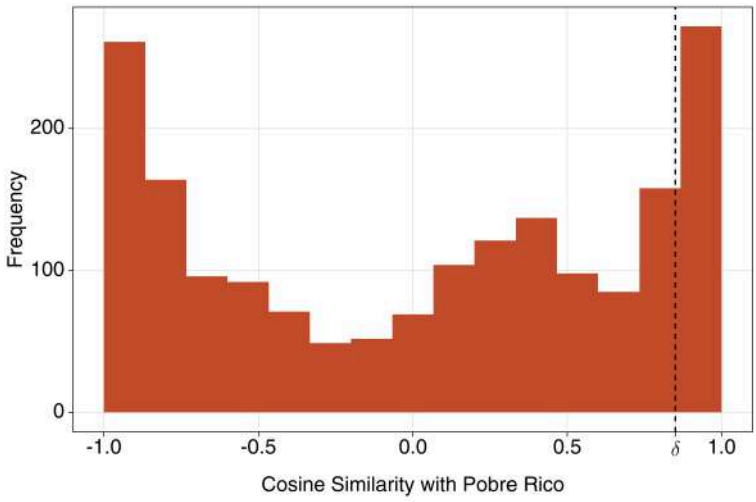
Panel B: select examples

Notes: This figure displays the first two principal components extracted form the vector embeddings of each telenovela and shows those telenovelas that are most similar to *Pobre Rico*. Inequality telenovelas are those with a similarity score $\geq .85$.

[9]For instance, the USE can differentiate between the various meanings of the word "capital" in sentences such as "The capital of Denmark is Copenhagen" and "Denmark will need to invest in capital to get out of the crisis." Other vector embedding models like GloVe or the TF-IDF assign the same vector to "capital" independent of the context (Pennington, Socher and Manning, 2014).

[10]Applying any dimensionality reduction techniques is a common practice in the machine learning literature that improving performance, visualization, and preventing overfitting (Roweis and Saul, 2000; Singh *et al.*, 2022).

Figure 5: Cosine Similarity with *Pobre Rico* Distribution



Notes: This figure displays the histogram of the similarity between a telenovela and the benchmark telenovela *Pobre Rico*. Inequality telenovelas are those with a similarity score $\geq .85$.

of -0.99. This telenovela is focused on the love of a poor merchant with a rich sales manager, and although there is an inequality element, it is not the main theme of the telenovela. Thus, opposite vectors to *Pobre Rico* might represent those stories with a component of inequality, but not at the center stage. An example of a telenovela with a perpendicular vector is *Papai coração*. This telenovela is centered around the struggles of a single father and his relationship with his children. *Corazon Rebelde* stands as a telenovela with vectors in the same direction, detailing the story of a group of poor students in an elite highschool that suffer from the harassment of their rich classmates.

Inequality telenovelas are those with a similarity score exceeding a threshold, delta, set to 0.85. Figure 5 illustrates the histogram of similarity scores, revealing a mean similarity of 0.03 with a standard deviation of 0.7. The distribution in the figure indicates that most telenovelas cluster at both extremes, either extremely similar to *Pobre Rico* or have opposite vectors.

Although *Pobre Rico* is centered around inequality issues, it's synopsis also incorporates law and family components which the vector embeddings also capture. Therefore, this method can mislabel a telenovela focused on inequality but in another setting, like Corazón Rebelde which achieved a high score with the ConceptNet score, but was labeled as no-inequality with vector embeddings with a similarity of 0.52 as its plot revolves around students and not families.

3.3 ChatGPT

Generative AI tools, such as OpenAI's gpt models, have shown remarkable capabilities in categorizing text into groups not present in their training data (Wang, Pang and Lin, 2023).[11] This versatility makes these models appropriate for identifying inequality in telenovelas. Other researchers within the fields of social sciences, psychology, and economics have similarly employed them in different contexts with demonstrated reliability and even outperforming human experts in some cases (Törnberg, 2023; Rathje *et al.*, 2024; Michalopoulos and Rauh, 2024).[12]

[11]Text generative AI models, such as OpenAI's gpts, are based on Large Language Models (LLMs) which are probabilistic models trying to predict $P(\text{word}|\text{previous word})$ and create the most likely sentences given some previous words, sentences, or prompts. For a brief overview of these models I recommend 3Blue1Borwn's YouTube video "Large Language Models explained briefly", Welch Labs YouTube video The moment we stopped understanding AI and for a more detailed explanation to read Chapter 1.3 and 1.10 of (Jurafsky and Martin, 2024).

[12]For instance, Michalopoulos and Rauh (2024) utilized OpenAI's `gpt-3.5-turbo` to classify film synopsis into those with risk-taking attitudes and those with more traditional gender roles.

Example 1 *Chocolate com Pimenta*	
ChatGPT CoT: The narrative frequently highlights socio-economic class differences. Ana Francisca is initially portrayed as poor, which leads to her being mistreated and humiliated by others, particularly by the wealthier characters like Olga and Danilo's family	**Extract from synopsis:** Danilo falls in love with Aninha [Ana Francisca], to the despair of Olga and his family, who reject her because she is poor. With the help of Bárbara, first lady of Ventura and Danilo's aunt, Olga plots to humiliate her rival.

In this project, I use OpenAI's gpt-4o model to classify telenovelas based on whether they portray social class conflict five times.[13] The model reads the translated synopses and outputs the classification result, a confidence score (50-100), and the model's chain of thought. Those telenovelas in which the model labels them as inequality with more than 90% confidence in 3 out of 5 times are considered as inequality telenovelas, and no-inequality telenovelas otherwise.[14]

I look at the provided chain of thought to understand the model's reasoning and performance. Figure 6 presents a word cloud that visually represents the vocabulary utilized in the reasoning during the categorization process. The most prominent words in the cloud, such as "socio," "economic," and "class," indicate that the model appropriately grasps the fundamental aspects of the topic at hand. Other frequently occurring words, including "difference," "topic," "conflict," "theme," "highlight," and "contrast," suggest that the model recognizes the importance of distinguishing between a display of different social classes conflict and conflict

A concern of generative AI models is their potential to hallucinate, make up a story that does not reflect the content in the synopsis. However, requesting the model's chain of thought mitigates this issue (Ashwin, Chhabra and Rao, 2023). In the task of identifying social class conflict, the model's reasoning matches extracts of the synopsis. For example, Example 1 shows the model's reasoning also matching with an excerpt of Chocolate com Pimenta. The fight betwee Ana Francisca and her mother-in-law was because of Ana's lower social class. Another example is Meu Pedacinho de Chão, that tells the story of a devoted teacher who arrives in a small village and challenges the oppressive

Figure 6: ChatGPT Reasoning Word Cloud

Notes: This figure displays the histogram of the similarity between a telenovela and the benchmark telenovela *Pobre Rico*. Inequality telenovelas are those with a similarity score $\geq$ .85.

[13]Generative A.I. models are stochastic by nature (Chann, 2023), meaning that the models can give you a different output from the same input prompt. Thus I opted to produce the results 5 times and then get an agreement between the different responses.

[14]The threshold δ for this methods is the confidence level acceptance set to 90%.

Example 2 *Meu Pedacinho de Chão*	
ChatGPT CoT: The summary describes characters from different socio-economic backgrounds: the humble townspeople, the wealthy Colonel and his family, and an orphan boy. The disdain of the Colonel for the orphan boy and the conflicts arising are directly linked to social class differences.	**Extract from synopsis:** Teacher Juliana arrives in the small town of Santa Fé to teach the children and is faced with a humble people who are cowed by the excesses of Colonel Epaminondas Napoleão, an arrogant man who dictates the rules in the region and solves everything by shouting and using weapons.

rule of a powerful colonel while navigating romantic entanglements with both the colonel's son and a loyal farmhand amid rising tensions. The model's reasoning, detailed in Example 2, correctly identifies the conflict between the wealthy, oppressive colonel and the humble townspeople, thus categorizing it as an inequality telenovela.

3.4 Performance

Descriptive statistics for all inequality telenovelas identified by different methods is presented in Table 3. Overall, inequality telenovelas tend to have a slightly higher percentage of dramas, similar episode count, and air time compared to an average telenovela. Additionally, on average, telenovelas centered on inequality appear to achieve higher television ratings than the typical telenovela while maintaining comparable IMDb ratings. Synopsis word count does vary significantly across methods, with ConceptNet showing a similar length to an average telenovela, vector embeddings with less, and ChatGPT with more.

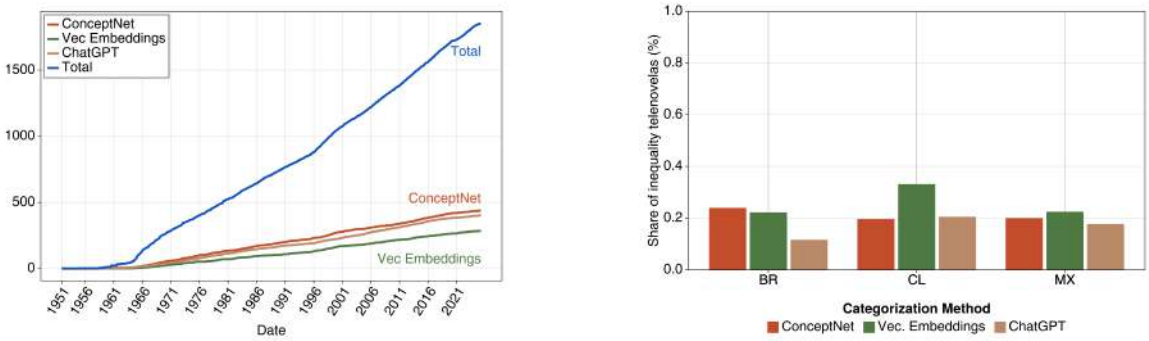
Inequality telenovelas are common in Latin America, but they are not the most frequent type of narrative. Panel A of Figure 7 shows the cumulative time series of all the telenovelas collected in the sample. The cumulative sum of social telenovelas by all categorization methods is well below the cumulative line of all telenovelas. This suggests that social class conflict is a common theme

Table 3: Inequality Telenovelas Descriptive Statistics by Method

Panel A: Full sample (N = 1852)					Panel B: ConceptNet (N = 438)				
	mean	sd	min	max		mean	sd	min	max
Drama (%)	34.39				Drama (%)	32.88			
Air time	16:38	07:09	10:00	23:15	Air time	16:09	07:36	14:00	23:15
N of episodes	129	76	1	1018	N of episodes	135	87	10	807
Synopsis word count	1719	1681	17	20784	Synopsis word count	1421	1081	75	9162
IMDB Rating	6.95	1.12	1.80	9.40	IMDB Rating	6.97	1.18	2.10	9.40
Panel C: Vector Embeddings (N = 284)					Panel D: ChatGPT (N = 402)				
	mean	sd	min	max		mean	sd	min	max
Drama (%)	29.58				Drama (%)	38.56			
Air time	16:58	06:51	16:00	23:10	Air time	17:11	06:34	14:00	23:00
N of episodes	130	75	10	604	N of episodes	140	65	2	459
Synopsis word count	956	271	407	1779	Synopsis word count	2423	2454	78	20784
IMDB Rating	6.95	1.06	3.20	9.40	IMDB Rating	6.98	1.08	2.10	9.10

Notes: This table provides descriptive statistics for all collected telenovelas between 1960-present in Chile, Mexico, Argentina, Colombian, Panama, and Brazil. TV ratings refer to Nielsen ratings, which measure the percentage of households that, on average, tuned in to watch the show throughout its entire run. IMDb ratings represent user ratings for each telenovela, as listed on the IMDB website.

Figure 7: Inequality Telenovelas Across Time and Countries



Panel A: Cumulative over time

Panel B: Across countries

Notes: Panel A shows the cumulative number of collected telenovelas and inequality telenovelas with all three categorization methods across time. Panel B shows the share of inequality telenovelas out of all telenovelas aired in a particular country.

when producing telenovelas in LATAM but is far from being the most common theme. Panel B of Figure 7 shows that inequality telenovelas as a percentage of all telenovelas is around 30% for Chile, Mexico, and Brazil. Thus, individuals in Latin America are not constantly exposed to these type of telenovelas.[15]

3.4.1 Model performance

To evaluate how each method performed in identifying inequality, I compare the model's label prediction to a personal categorization of 220 telenovelas from Chile and Brazil. I chose four commonly used measures in the Machine Learning literature: precision, recall, accuracy and f-score (Jurafsky and Martin, 2024). Let tp be the number of true positives, where a true positive happens when both the model and my categorization predict inequality label. In addition, let tn, fp, fn, bet he true negative, false positive, and false negative respectively. As an illustration, a false positive (fp) is when the model predicts a no-inequality telenovela while I label that telenovela as inequality. The formal definition of each metric is presented bellow.

$$\text{accuracy} = \frac{tp + tn}{tp + tn + fp + fn} \qquad \text{precision} = \frac{tp}{tp + fp}$$

$$\text{recall} = \frac{tp}{tp + fp} \qquad \text{f-score} = 2 \times \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$

Accuracy measures the overall proportion of correct predictions, but it may be misleading if one class (inequality or not) dominates. Precision evaluates the quality of positive predictions, indicating how many telenovelas classified as inequality are correct, which is important to minimize false positives. Recall assesses the model's ability to identify all inequality-focused telenovelas, reflecting its sensitivity to true positives. The F-Score balances precision and recall, providing a single metric to evaluate the model's effectiveness, particularly when both false positives and false negatives matter or the classes are imbalanced. For this study, precision is the most relevant measure of model performance, as any false positive (misclassifying a telenovela as inequality when it is not) could bias the estimation of exposure to inequality-themed content.

Table 4 shows the performance metrics for each categorization method. ChatGPT has the best performance in precision and accuracy, which are most important for the empirical estimation. This method only produces false positives in 5% of the cases and incorrectly labels telenovelas in 25%. The

[15]These three countries contain 98% telenovelas in our sample. The remainder 2% are telenovelas from Colombia, Argentina, Peru and Venezuela which were omitted.

Table 4: Inequality Prediction Performance by Method

Categorization Method	Accuracy	Precision	Recall	F-Score
ConceptNet	0.70	0.70	0.41	0.52
Vec Embeddings	0.62	0.53	0.19	0.28
ChatGPT	0.75	0.80	0.48	0.60

Notes: This table shows the model performance based on my own categorization of telenovelas that display inequality aired in Chile. A higher number indicates better performance in all four measures.

other methods, ConceptNet and vector embeddings, have much lower precision and lower accuracy. Overall, the worst-performing model is the vector embeddings across all metrics.

Another way to evaluate the models' performance is by examining the most important words for each category and method. To measure word importance, I follow Kessler (2017), which considers the frequency of each word within specific categories. This method assigns greater importance to words that are frequent in one category but infrequent in another. Table 5 displays the top 10 important words for each category, inequality and no-inequality, according to the categorization method used. Both ConceptNet and ChatGPT highlight expected terms such as "poor," "rich," "neighborhood," "humble," and "social." However, with the vector embeddings model there does not appear to be a clear representation of the most significant words related to inequality. The most important words in the no-inequality category do not reflect inequality, with the exception of a few mentions in the vector embeddings category. Therefore, one could anticipate the impact on preferences for redistribution would be smaller using the vector embedding classifier.

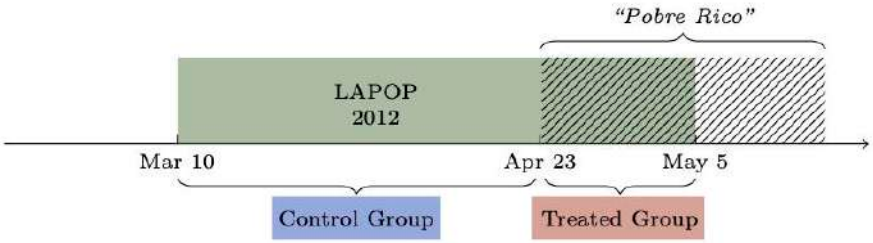
Each of these three methods each has clear advantages and limitations. The dictionary-based approach using ConceptNet offers simplicity and transparency, but it may potentially missed nuanced meanings and it's dependent on the researchers choice of the inequality dictionary. The vector embeddings provide a more sophisticated alternative, capturing contextual meanings and handling synonyms effectively, which allows for better generalization. But its complexity obscures the interpretation of what telenovelas are similar to *Pobre Rico*. In addition, it has the worst performance. Lastly, the ChatGPT method offers high flexibility and the ability to understand context and subtle language features with minimal examples. However, this method is somewhat more opaque than the vector embeddings, making it difficult to interpret its reasoning.

Table 5: Most Important Words by Label and Categorization Method

Inequality			No inequality		
ConceptNet	Vec. Embeddings	ChatGPT	ConceptNet	Vec. Embeddings	ChatGPT
economic	student	poor	people	return	mysterious
social	teacher	class	murder	mansion	passion
poor	case	refuse	travel	plan	past
rich	form	rich	search	receive	search
class	drug	neighborhood	grow	home	heart
employee	doctor	humble	little	go	happy
student	group	social	get	house	story
wealthy	hotel	house	truth	suspect	crime
own	personal	accept	go	promise	secret
neighborhood	obsess	money	find	accept	people

Notes: This figure shows the term (or word) importance for each category following Kessler (2017). Each method has a different definitions for the social category.

Figure 8: *Pobre Rico* Timeline



4 Empirical Strategy

To assess the impact of inequality telenovelas on preferences for redistribution, this study focuses on inequality telenovelas that were aired during the fieldwork of the Latin American Public Opinion Project (LAPOP) survey, ensuring that no other inequality telenovelas were on the air concurrently. This enables a comparison of individuals' responses to the redistribution question before and after the telenovela's debut.

For example, the 2012 LAPOP's Chilean fieldwork spanned from March 10th to May 5th, coinciding with the debut of Pobre Rico on April 23rd. Figure 8 illustrates the timeline of these events. The control group, or not exposed to an inequality telenovela, are those who were interviewed between March 10th and April 22nd. The treatment group, or those exposed to an inequality telenovela, are those who were interviewed between April 23rd and May 5th. Cases in which an inequality telenovela that ends within the same survey fieldwork as another inequality telenovela aired were discarded to avoid any contamination effect from the end of inequality telenovelas. This method provides a clear comparison, ensuring that individuals were not exposed to any other inequality telenovelas, at least since the start of the fieldwork.

Table A.1 presents descriptive statistics for each telenovela alongside the categorization methods employed for their identification in Panel A, and Panel B offers a concise one-line summary for each telenovela. *Os Dias Eram Assim*, *Pobre Rico*, and *Tengo todo excepto a ti* highlight the conflict between social classes, while *Tranquilo Papa* and *La fiscal de hierro* does not seem to have a strong inequality component. This shows the limitation of the vector embeddings method, which categorizes these telenovelas as similar to Pobre Rico in other aspects of the plot. Given this context, one would expect that inequality telenovelas identified by the vector embeddings method would attenuate the effect on support for redistribution than other methods.

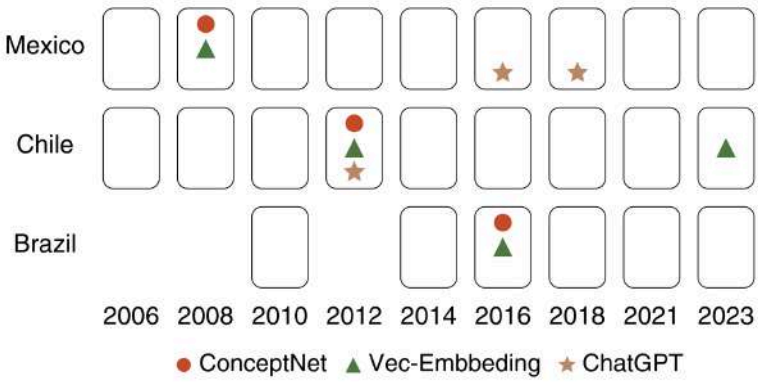
Figure 9 shows the categorization methods for each respective country and wave from the LAPOP survey that fulfills the previous identification criteria. The methods that exhibit the greatest variation across time and countries are ConceptNet and vector embeddings, which utilize 3 to 4 telenovelas in their respective analyses. Notably, the vector embeddings method identifies inequality telenovelas within the same survey wave across different coun- tries. ChatGPT only identified Pobre Rico as an inequality telenovela that meets the estab- lished identification criteria.

Empirical model: To estimate the causal effect of telenovelas that display social class differences and the preferences for redistribution, I run the following OLS regression:

$$Y_{icwt} = \alpha + \gamma \text{InequalityTV}_{cwt} + \beta X_i + \omega d_{t^*} + \delta_{cw} + \varepsilon_{icwt}$$

where Y_{icwt} is the individual answer to the support for redistribution question for individual i at interview date t residing in country c in the survey year wave w. The support for redistribution question is a 1-7 Likert scale question asking individuals how much they agree with the statement "The government should take measures to reduce differences in income levels." The answers are recoded to a 0-1 variable, where 1 indicates that the individual agrees with the statement and 0 oth-

Figure 9: Sources of Variation Across LAPOP Waves



Notes: This figure depicts the variation in the treatment across the LAPOP surveys. Each rectangle represents the fieldwork for a country and year, while the points indicate that at least one telenovela aired that year meets the identification criteria for that categorization method. For a full description of the telenovelas, please refer to Table A.1

erwise. X_i is a set of individual controls which include gender, age, if they live in an urban location, employment status, education, religion, relative income bracket, and marital status. Hausman and Rapson (2017) suggest to include a distance to the treatment date, d_{t^*}, in order to reduce time bias in this type of empirical framework. This control gives more weight to those individuals interviewed closer to the premiere date of the telenovela making for a more precise estimate of the short-term causal effect. δ_{ct} are wave × country fixed effects. My treatment variables are $\text{InequalityTV}_{cwt}$ that takes a value of 1 if a inequality telenovela is currently on the air at interview date t in country c and takes a value of 0 if there are no inequality telenovelas currently on the air at interview date t in country c.

With the established identification criteria, γ can be interpreted as the short-run effect of the exposure to an inequality telenovela compared to no exposure to inequality in telenovelas on preferences for redistribution.[16] This limitation is due to the LAPOP fieldwork's duration, which averages around two months. Given that inequality telenovelas typically start in the middle-end of the fieldwork, individuals in the treatment group are exposed to these telenovelas for at most one month. As a result, the findings reflect short-run effects on support for redistribution, primarily influenced by the impressions formed during the early episodes. The synopses of the identified telenovelas are well-aligned with the themes presented in the first episodes, ensuring that individuals are exposed the central theme of inequality identified by each categorization method.

The timing of the survey's fieldwork may lead to observable differences in between of individuals interviewed before and after the premiere of the inequality telenovela.[17] Table 6 presents a balance table for each categorization method, with each estimate corresponding to regressions using the set of individual characteristics as the dependent variable and the independent variable is the treatment variable, $\text{InequalityTV}_{cwt}$, with country and wave fixed effects. Individuals interviewed after the premiere of the social telenovela categorized by ConceptNet have a lower income levels, although no statistically significant differences between the two groups. With the vector embeddings method we find similar trends. Lastly, with the ChatGPT method we find that the treatment group

[16]Gulesci, Lombardi and Ramos (2024) uses a similar specification for their analysis. In particular, their treatment variable is the cumulative exposure to telenovelas with characters from the LGBTQ+ community. In contrast, I explore the idea of current exposure or salience effect, since inequality in telenovelas is more common than characters representing minorities.

[17]For instance, in the fieldwork of wave 2012 in Chile, they targeted Santiago, the main capital which houses almost half the population of the country, earlier than other regions in the country.

Table 6: Balance Table by Method

Method	Female	Age	Education	Married	Catholic	Employment	Urban	Income
ConceptNet (N=1636)	−0.008	−0.104	0.035	0.003	−0.005	0.003	−0.008	−0.182
	(0.024)	(0.889)	(0.270)	(0.010)	(0.031)	(0.034)	(0.036)	(0.246)
Vec. Embeddings (N=1071)	−0.006	−0.083	0.035	0.009	−0.010	0.004	0.014	−0.115
	(0.024)	(0.912)	(0.274)	(0.010)	(0.032)	(0.036)	(0.033)	(0.253)
ChatGPT (N=1476)	−0.048	−0.198	−0.025	0.003	−0.078***	−0.017	−0.011	0.065
	(0.033)	(0.850)	(0.336)	(0.009)	(0.028)	(0.019)	(0.052)	(0.251)
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wave Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country × Wave Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: catholics is a dummy for whether the individual is catholic or not. employment status refers to whether the individual is in the working or unemployed. Standard errors are clustered at interview date level.

has a slightly lower share of females, are located in rural areas, have higher income, and are less likely to be catholic. I directly control for these observable characteristics to avoid any bias due to the slight sample imbalance.

One might intuitively expect lower-income individuals would be more likely to support redistribution, suggesting a potential upward bias in my estimates. Similarly, a more rural population tends to be lower income, and thus, one should also expect an upward bias.[18] Considering all these potential biases, our estimates are most likely upwardly biased and the results still suggest that exposure to a social telenovela decreases support for redistribution.

5 Results

Rather than presenting results from each categorization method separately, I report an ensemble estimate that synthesizes the strengths of all three approaches. Each method captures distinct dimensions of inequality in telenovelas and has its own strengths and limitations—ConceptNet focuses on explicit terminology, vector embeddings capture contextual similarity, and ChatGPT leverages nuanced language understanding. By combining these methods, the ensemble approach provides a more robust and comprehensive estimate of the effect, reducing reliance on any single method and mitigating individual weaknesses. The ConceptNet approach focuses on explicit inequality-related terms, vector embeddings capture broader contextual similarities, and ChatGPT leverages nuanced language understanding. To combine their strengths and reduce reliance on any single method, I constructed an ensemble using principal component analysis (PCA) on the validation data, assigning weights to each method's coefficient. The PCA analysis indicated that most of the weight should be placed on ChatGPT (around 0.6), with about 0.3 assigned to ConceptNet and the remaining 0.1 to vector embeddings. This approach provides a more robust and balanced estimate of the effect.

Table 7 shows the ensemble estimates from the main specification. Across all methods and the ensemble, exposure to an inequality telenovela is associated with reduction in support for redistribution. This negative effect remain negative after including country-wave fixed effects, individual controls, and time distance to treatment shock, although the significance of the dissapears in some cases. The ConceptNet method shows a 4.9% decrease in support for redistribution, with higher sample size with other telenovelas and other countries. The vector embeddings we find the smallest and not statistically significant of 4.7% reduction. This is an expected result since this method

[18]There is no intuitive relation between support for redistribution and gender, however when you look at the data there is not a clear difference between mean and women. Empirically, there is is a difference of −0.02 points (se: 0.007, p_value = 0.02) in a scale from 1 to 7 between men and women.

Table 7: Exposure Effect of Inequality Telenovelas by Method

	Support for Redistribution (1-7)							
	ConceptNet		Vector Embeddings		ChatGPT		Ensamble	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Exposure to Inequality TV	−0.433***	−0.275**	−0.290***	−0.243	−0.263***	−0.219	−0.332***	−0.245***
	(0.101)	(0.128)	(0.092)	(0.154)	(0.070)	(0.154)	(0.054)	(0.064)
Controls		Yes		Yes		Yes		Yes
Country FE		Yes		Yes		Yes		Yes
Wave FE		Yes		Yes		Yes		Yes
Country × Wave FE		Yes		Yes		Yes		Yes
N	3329	1636	4891	1071	4517	1476	12319	2443
Adjusted R^2	0.035	0.046	0.028	0.038	0.012	0.028	0.010	0.027
Control Mean	5.847	5.871	5.678	5.678	5.710	5.811	5.811	5.821

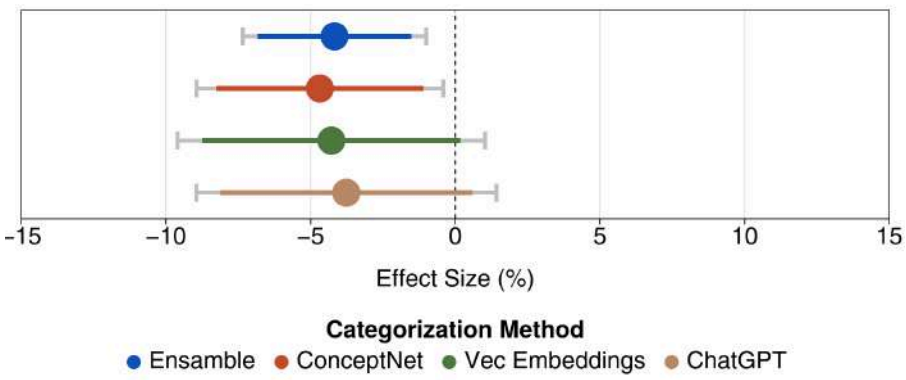
Notes: The outcome variable is the individuals answer to the following statement: "The (Country) government should implement strong policies to reduce income inequality between the rich and the poor." An answer of 7 indicates strong agreement and 1 strong disagreement with the previous statement. The ensemble refers to the weighted mean and standard deviation of the coefficients from the three methods using their PCA weights. The regression controls include gender, age, urban, employment status, education, religion, income, marital status, and a time trend. Clustered standard errors at the interview date level are showed in parenthesis.

misidentified some inequality telenovelas. The ChatGPT method show a 4.3% reduction in preferences for redistribution after being exposed to only Pobre Rico. Taking an ensamble of the three methods, I find a statistically significant 4.2% reduction in support for redistribution after being exposed to an inequality telenovela. This study's findings suggest that exposure to an inequality telenovela reduces individuals' preferences for redistribution in the short run. Although small, these results provide evidence that what is shown in the entertainment media influences policy relevant topics, like support for redistribution.

5.1 Intensive Treatments

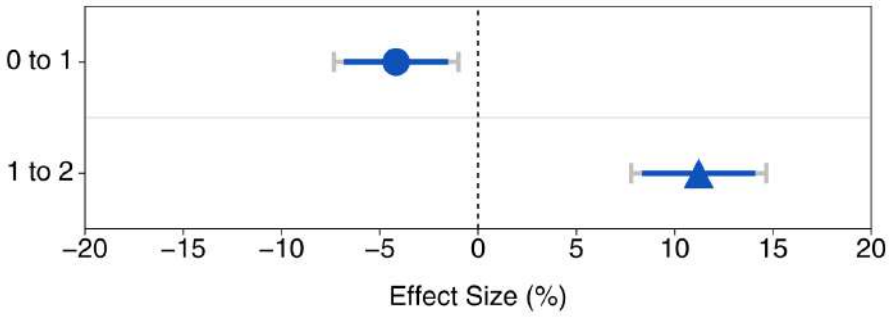
The empirical strategy employed to measure the intensive effects of adding extra inequality telenovelas follows a similar approach described previously. However, I now narrow the focus to cases where N inequality telenovelas are continuously airing throughout the entire fieldwork of the surveys. This restriction eliminates any cases in which one or more telenovelas finish mid-fieldwork.

Figure 10: Exposure Effect Coefficient Plot



Notes: This figure shows the effect size compared to the mean of the group with no exposure to an inequality telenovela. The outcome variable is the individual's answer to the following statement: "The (Country) government should implement strong policies to reduce income inequality between the rich and the poor." An answer of 7 indicates strong agreement and 1 strong disagreement with the previous statement. The ensemble refers to the weighted mean and standard deviation of the coefficients from the three methods using their PCA weights. Controls: gender, age, urban, employment status, education, religion, income, marital status, and a time trend. Standard errors are clustered at interview date level.

Figure 11: Intensive Exposure Effects of an Inequality Telenovela



Notes: This figure shows the effect size relative to the mean of the control group. For the "0 to 1" case, the control group consists of those with no exposure, while for the "1 to 2" case, the control group is composed of those with one exposure, and so on. The outcome variable is the individuals answer to the following statement: "The (Country) government should implement strong policies to reduce income inequality between the rich and the poor." An answer of 7 indicates strong agreement and 1 strong disagreement with the previous statement. Controls: gender, age, urban, employment status, education, religion, income, marital status, and a time trend. Standard errors are clustered at interview date level.

Then, I will analyze the effect of an additional inequality telenovela that debuts during the course of the fieldwork conditional of being exposed to N inequality telenovelas. For example, Figure A.1 shows that during the Mexican LAPOP 2021 survey, *Como Dice el Dicho* was in the air when *Si nos Dejan* premiered on June 1st. This situation stands in contrast to the case of Pobre Rico, which had no other social class-themed telenovelas airing concurrently during its premiere.

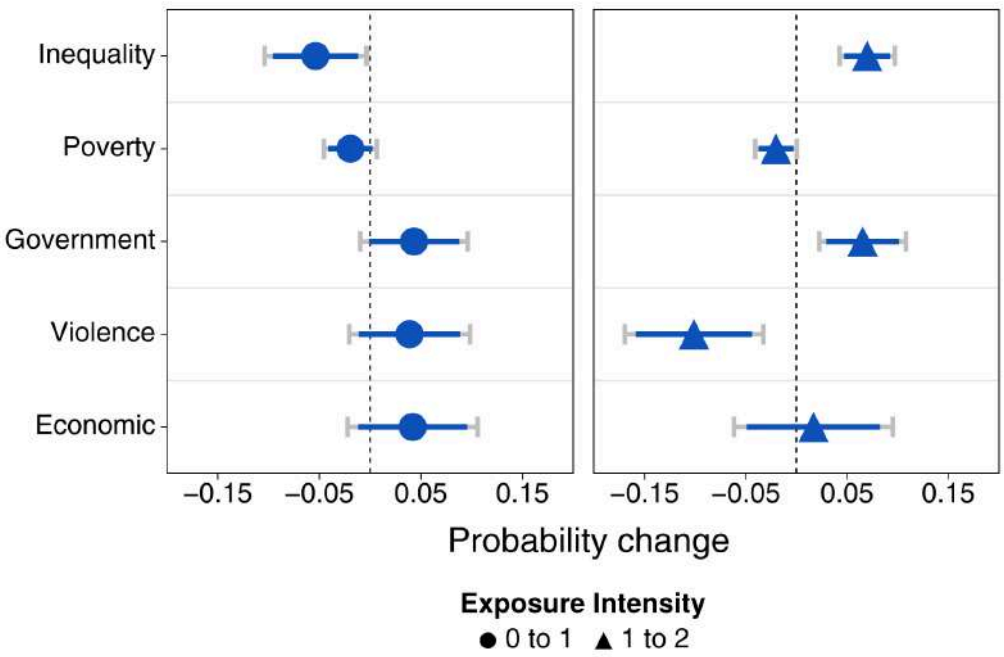
I will estimate a regression using the main specification replacing the treatment with $\text{InequalityTV}^{N}_{cwt}$ that takes a value of 1 if $N+1$ inequality telenovelas are currently on the air at interview date t in country c in the survey years w and takes a value of 0 if there are N inequality telenovelas currently on the air at interview date t in country c that meet the identification cases. Where γ in these cases will be the short-run effect of the exposure to an inequality telenovela compared to being exposed to N inequality telenovelas on preferences for redistribution.

Figure 11 displays the effect of introducing an additional inequality telenovela when individuals are already exposed to N inequality telenovelas. Only three telenovelas met the identification criteria when N = 1, one per method. No inequality telenovelas met the criteria when 2 or more inequality telenovelas were airing concurrently. In contrast to the negative effect observed with no prior exposure, adding an additional inequality telenovela is associated with a positive effect on support for redistribution. Specifically, the ensemble estimate suggests a 10% increase in support for redistribution. The cumulative effect from no exposure to two inequality telenovelas is a 6% increase in support for redistribution. This pattern may indicate that initial exposure to inequality telenovelas reduces support for redistribution, but further exposure in a context where inequality is already salient can actually increase support. However, this effect is not robust to changes in the thresholds used in each method, as documented in Figure 13.

5.2 Mechanisms

The reduction of support for redistribution after being exposed to an inequality telenovela is somewhat counterintuitive. Intuitively, when inequality is featured in media, individuals become more aware of its issues, prompting a shift in their preferences toward supporting redistributive policies and assuming that increased exposure to inequality would naturally lead to a heightened urgency to address it. However, the impact of media representation on public perception relies heavily on how inequality is portrayed. When telenovelas depict inequality as a significant and pressing issue, they have the potential to inform viewers and garner increased support for redistribution as the audience becomes more attuned to the severity of the problem. Conversely, if these shows downplay the challenges of inequality or offer overly simplistic solutions, viewers may absorb this distorted

Figure 12: Exposure Effects of Inequality Telenovelas on National Concerns



Notes: The outcome variables in the y axis is the individual's answer to the following statement: "What is the most important issue facing the country today?" The ensemble refers to the weighted mean and standard deviation of the coefficients from the three methods using their PCA weights. Controls: gender, age, urban, employment status, education, religion, income, marital status, and a time trend. Standard errors are clustered at interview date level.

portrayal. Such (mis)representation can lead individuals to underestimate the severity of inequality, ultimately lowering their priority to supporting redistributive policies. Thus, how inequality is represented in media is crucial in shaping public attitudes toward addressing it.

I will provide two pieces of evidence to support that inequality telenovelas give a distorted (mis)representation of inequality, which leads individuals to deprioritize redistribution policies, when there is no exposure. First, I will show that exposure to inequality telenovelas changes how important individuals perceive inequality as a main national issue. Second, I will explore the heterogeneous effects of exposure to an inequality telenovela, analyzing individual characteristics and political attitudes.

5.2.1 Interest in social problems

Individuals exposed to inequality telenovelas perceive social concerns as less significant at first impact and more significant in subsequent exposure. Figure 12 In fact, the magnitudes of the lower likelihood correspond to the magnitude of the effects on the support for redistribution. At first impact, individuals seems to substitute their social concerns with those about the government, violence, and the economy. This finding is consistent with the idea that inequality telenovelas do not frame inequality as a major problem, but rather as a conflict between the richest and the middle class. For instance, in the telenovela Pobre Rico, inequality is presented in a comedic light, highlighting the absurdities of a wealthy lifestyle from the perspective of a middle-class teenager. In addition, poverty concerns are unchanged after the exposure to inequality telenovelas since inequality in telenovelas focuses on the conflict between the richest and the middle class.

5.2.2 Perceived social class and interest in politics

Table 8 presents regression estimates of the interactions between socioeconomic status and political variables with the exposure to inequality telenovelas. The analysis reveals two key findings present in both first and second exposure. First, individuals' perceptions of their social class matter more than their actual income, as shown in Panel A. Income has little impact on attitudes toward redistribution,

while individuals' social class perceptions do. Individuals who perceive themselves as belonging to a low social class are more likely to support redistribution than those who see themselves as middle or upper class. However, exposure to inequality telenovelas further reduces this support among the perceived low social class. The extent of this effect varies significantly: ConceptNet and vector embeddings estimate a 2% decrease in support, while ChatGPT estimates a much larger reduction of 24%. This suggests that telenovelas may provide (mis)information about social disparities, possibly making viewers feel their problems are less severe or leading them to identify with higher social classes, thereby reducing their concern about inequality in society. This finding aligns with existing literature indicating that (mis)perceptions of inequality are significant drivers of support for redistribution, as highlighted in (Stantcheva, 2021).

Second, interest in politics serves as a counterbalance to the (mis)information about inequality telenovelas. Panel B reveals the heterogeneous effects of political leaning and interest in politics on support for redistribution. The estimates suggest that left-leaning individuals are more inclined to advocate for redistribution policies, but there is no statistically significant change after the exposure. However, the estimates indicate that individuals with a high interest in politics show a -0.02% to -5% reduction in their support for redistributive policies after the exposure to inequality telenovelas. This finding implies that those who are more engaged and likely better informed about the actual state of inequality in their country are less affected by external narratives. Consequently, their policy preferences remain unchanged, or at least less affected by the (mis)representation of inequality.

5.3 Robustness

The results presented earlier depend heavily on the ability of the methods to identify inequality telenovelas. To enhance the validity of these findings, I conduct three robustness checks. I start by adjusting the thresholds applied in all three methods to determine whether the results remain consistent. Next, I examine whether telenovelas categorized as definitively not addressing inequality have any impact on support for redistribution. Lastly, I explore changes in other outcomes, such as attitudes toward a minority running for president with the exposure to inequality telenovelas.

5.3.1 Alternative Thresholds (δs)

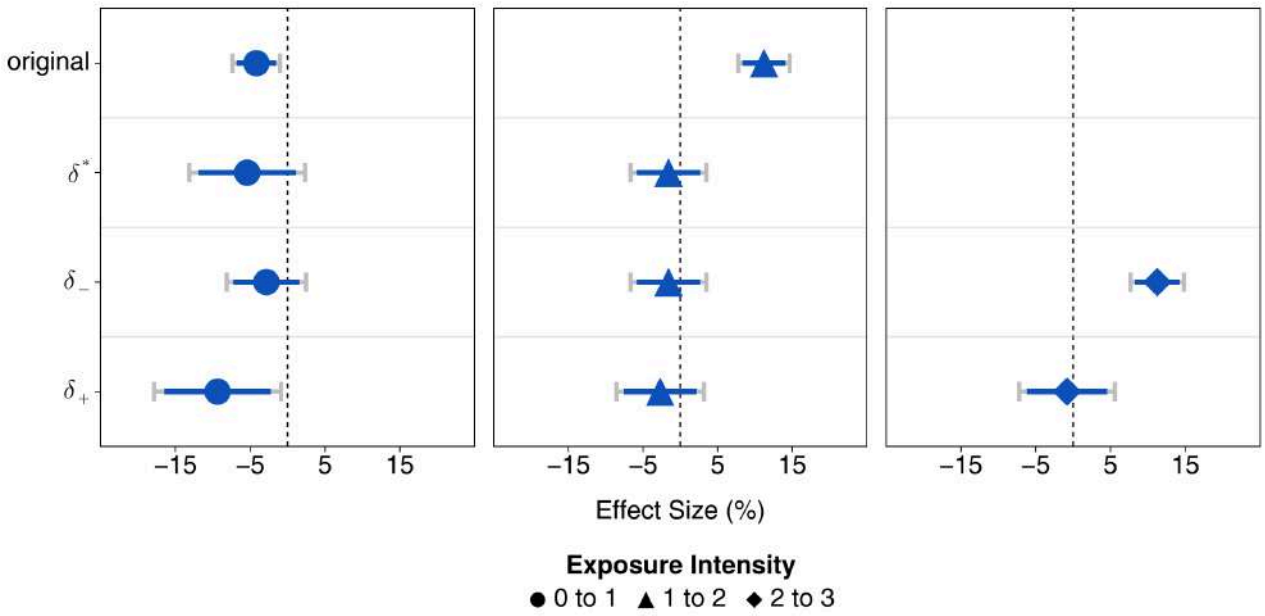
The categorization methods employed all incorporate specific thresholds. For the ConceptNet method, the threshold was set at a score of two standard deviations from the score of *Pobre Rico*. In the vector embeddings approach, inequality telenovelas are those with a similarity score to *Pobre Rico* greater than 0.85. Meanwhile, in the ChatGPT method, only those telenovelas that achieved a confidence level of 90, with the highest agreement among a set of five, were included.

I subsequently modified these thresholds in three ways: first, I optimized the threshold to maximize the precision of each model δ^*; second, I increased the threshold δ^+; and third, I reduced the threshold δ^-. The purpose of optimizing the threshold δ^* is to examine the scenario where false positives are minimized, while the latter two methods aim to assess the sensitivity of the findings to the original thresholds.[19]

Figure 13 compares results using both the original and modified thresholds for all intensive treatments. When $N = 0$, changes in the threshold maintain the negative effect, but with higher standard errors. In particular, increasing the threshold, making it more strict for a telenovela to be classified as inequality, yields a larger 8% decrease in support for redistribution, while reducing the threshold across all methods results in a smaller, non-significant negative effect. For the case of $N = 1$, the result loses its positive effect and significance, suggesting that the intensive effect is not robust and should be interpreted with caution. Lastly, although no telenovelas meet the identification criteria at the original threshold for higher N, some do with increased or decreased thresholds, but

[19]Table A.2 shows the different thresholds for each categorization method.

Figure 13: Exposure Effects to Inequality Telenovela with Alternative Thresholds



Notes: The outcome variable is the individual's answer to the following statement: "The (Country) government should implement strong policies to reduce income inequality between the rich and the poor." An answer of 7 indicates strong agreement and 1 strong disagreement with the previous statement. The original category displays the effect size with my chosen δ. The category δ^* optimizes the thresholds for each category to maximize the precision of each model. The category δ_- and δ_+ decreases and increases the original threshold δ, respectively. Controls: gender, age, urban, employment status, education, religion, income, marital status, and a time trend. Standard errors are clustered at interview date level.

the results are mixed. Overall, this suggests that the effects on redistribution occur primarily when there is no prior exposure to inequality telenovelas, and findings for intensive treatments should be taken with caution.

5.3.2 Placebo Treatments

Another test to enhance the validity of the findings is to examine whether those telenovelas predicted by my methods to exhibit no inequality have any effect on support for redistribution. These telenovelas include, for instance, those with a ConceptNet score of 0, those telenovelas with vector opposite or perpendicular to Pobre Rico, and telenovelas for which ChatGPT indicates no socio-economic class conflict with 90% confidence in at least 3 out of 5 evaluations. Figure 14 splays the effect of these placebo no-inequality telenovelas. As anticipated, the effects are statistically insignificant for both $N = 0$ and $N = 1$ cases. This suggests that the telenovelas identified by these methods provide a reasonably accurate representation of those narratives that do not depict inequality.

5.3.3 Placebo Other Outcomes

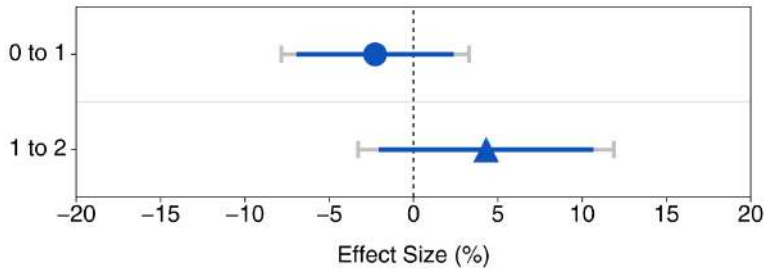
I test whether exposure to inequality telenovelas affects support for other social issues unrelated to economic redistribution. Table 9 presents estimates from the main specification using outcome variables measuring individuals' willingness to support minority candidates for president. As expected,

Table 8: Heterogeneous Effects of Exposure to an Inequality Telenovela

	Support for Redistribution									
	$N = 0$					$N = 1$				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Exposure to Inequality TV^N	−0.243*	−0.232	−0.374**	−0.299*	−0.238	0.655***	0.622***	1.010***	0.503***	0.525**
	(0.145)	(0.163)	(0.166)	(0.172)	(0.379)	(0.094)	(0.123)	(0.122)	(0.109)	(0.251)
Left Wing		0.118					0.159			
		(0.133)					(0.175)			
Exposure to Inequality $\text{TV}^N \times$ Left Wing		0.024					0.197			
		(0.199)					(0.158)			
High Pol. Interest			0.030					0.497***		
			(0.099)					(0.191)		
Exposure to Inequality $\text{TV}^N \times$ High Pol. Interest			0.395**					−0.870***		
			(0.184)					(0.216)		
Income $\leq$ \$100				−0.181					−0.247	
				(0.170)					(0.191)	
Exposure to Inequality $\text{TV}^N \times$ Income $\leq$ \$100				0.145					0.791***	
				(0.231)					(0.225)	
Perc. Low Class					0.414***					0.274*
					(0.135)					(0.163)
Exposure to Inequality $\text{TV}^N \times$ Perc. Low Class					−0.813**					−0.570
					(0.347)					(0.386)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country × Wave Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance to air date $t^{\{(*)\}}$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Combined N	2443	2180	2432	2443	1271	2176	1855	2174	2176	908
Average R^2	0.039	0.036	0.045	0.042	0.065	0.029	0.036	0.049	0.031	0.029
Control Mean	5.821	5.756	5.827	5.821	5.887	5.563	5.550	5.563	5.563	5.688

Notes: The outcome variable is the individuals answer to the following statement: "The (Country) government should implement strong policies to reduce income inequality between the rich and the poor." An answer of 7 indicates strong agreement and 1 strong disagreement with the previous statement. I consider respondent with Income < 100 if they earn less than 100 U.S. dollars every month, and those that Perc. Low Class if they classify themselves as low-middle class or low class. Left Wing are those individuals that have a political leaning from 1 (left) to 10 (right) less than 4. High Political Interest refers to those individuals that have at least some interest in politics. Controls: gender, age, urban, employment status, education, religion, income, marital status, and a time trend. Clustered standard errors at the interview date level are showed in parenthesis.

Figure 14: Exposure Effect of Non-inequality Telenovelas on Support for Redistribution



Notes: This figure shows the effect size relative to the mean of the control group for those telenovelas considered a placebo for each method. A placebo in ConceptNet are those with a score of 0, for Vec. Embeddings is either having a similarity score near 0 or less than −0.85 (opposite), for ChatGPT will be those where the model agree's with confidence that they don't display any inequality. The outcome variable is the individuals answer to the following statement: "The (Country) government should implement strong policies to reduce income inequality between the rich and the poor." An answer of 7 indicates strong agreement and 1 strong disagreement with the previous statement. Controls: gender, age, urban, employment status, education, religion, income, marital status, and a time trend. Standard errors are clustered at interview date level.

exposure to inequality telenovelas shows no impact on these unrelated social issues—all coefficients remain small, centered around zero, and statistically insignificant across all levels of prior telenovela exposure.

These placebo tests provide additional validation that the categorization methods successfully isolate telenovelas with inequality as their central theme, rather than capturing broader social content. This specificity is important because previous research has shown that telenovela content can influence attitudes toward the social groups they portray. For example, Gulesci, Lombardi and Ramos (2024) found that exposure to telenovelas featuring LGBTQ+ characters decreases support for candidates from that community in Latin America. In contrast, my results show no effect of inequality telenovelas on support for minority presidential candidates, confirming that these shows specifically address economic themes rather than broader social representation issues.

6 Effect on Voting Behavior

This section examines whether the influence of inequality telenovelas extends beyond shaping attitudes toward specific policy issues, and instead has broader effects on political behavior. The portrayal of inequality in telenovelas introduces a politically charged narrative into popular media,

Table 9: Effects of Inequality Telenovela on Placebo Outcomes

	$N = 0$			$N = 1$		
	Approval	Running President		Approval	Running President	
	LGBTQ+	Women	LGBTQ+	LGBTQ+	Women	LGBTQ+
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure to Inequality TV^N	0.004	0.026	0.011	−0.065	−0.125	−0.052
	(0.044)	(0.036)	(0.036)	(0.041)	(0.150)	(0.048)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Country × Wave Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Distance to air date t^*	Yes	Yes	Yes	Yes	Yes	Yes
Combined N	3348	4100	3269	2211	2211	925
Average R^2	0.092	0.081	0.036	0.129	0.088	0.046
Control Mean	0.508	0.643	0.225	0.511	0.591	0.262

Notes: The controls include gender, age, urban, employment status, education, religion, income, marital status, and a time trend. Clustered standard errors at the interview date level are showed in parenthesis.

which may impact how viewers engage with electoral processes. Potential mechanisms include increased salience of economic disparities, shifts in perceptions of fairness or legitimacy, and changes in how voters assess candidates or political parties. While these pathways are plausible, establishing a direct causal link between media exposure and voter behavior requires further research. Here, I provide empirical evidence on the association between exposure to inequality telenovelas and electoral outcomes, highlighting the need for future studies to unpack the underlying mechanisms.

To investigate whether exposure to inequality telenovelas influences electoral outcomes, I merged presidential election data from the National Elections Database (Marx, Pons and Rollet, 2025) with my telenovela dataset for Latin American countries. For each election, I calculated the number of inequality telenovelas airing in the 1, 3, and 6 months preceding election day.

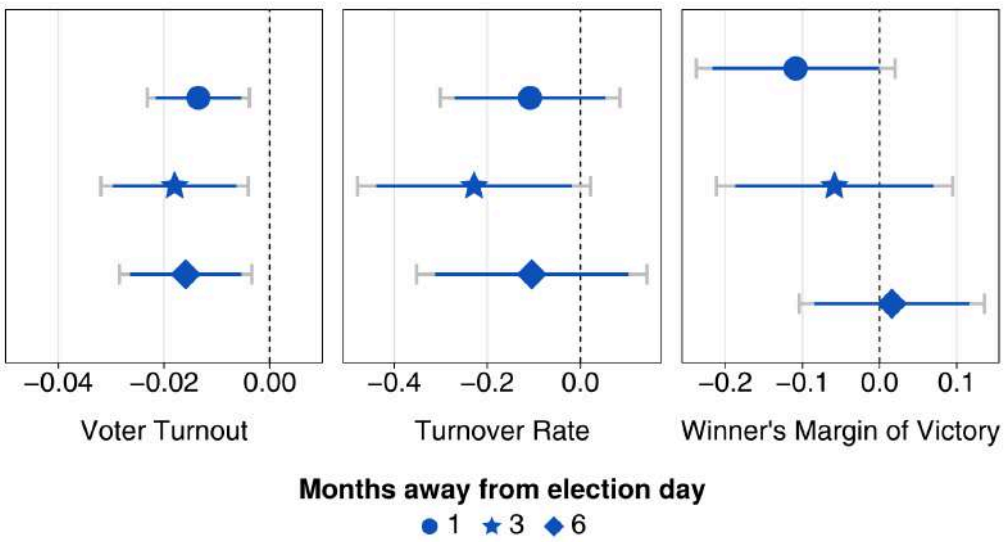
The estimated regression model is as follows:

$$Y_{ct} = \alpha + \gamma \text{IneTV}_{c,t} + \beta X_c + \delta_c + \varepsilon_{c,t}$$

where Y_{ct} is the electoral outcome for country c at time t, which can be total votes cast, vote share of the winning candidate, or the victory margin. The treatment variable, IneqTV_{ct}, indicates if there was an inequality telenovela airing in the months leading up to the election. X_c includes country-level controls such as GDP per capita and population size, and δ_c are country fixed effects to account for country differences. The coeficient γ will then measure the correlation between an electoral outcome and if there was an inequality telenovela airing in the months leading up to the election.

Figure 15 show the estimated coeficients for voter turnout, turnover rate, and the winner's margin of victory. The regressions reveal that the presence of a inequality telenovelas in the months leading up to elections has a correlation with electoral outcomes. An additional inequality telenovela airing in the months before an election is associated with a 0.02 percentage point decrease in voter turnout. This is consistent with the idea that individuals put less weight social concerns and those on the margin of voting are less likely to turn out. Exposure to inequality telenovelas leading up to election day is also associated with a narrower victory margin for the elected president and a reduction in incumbent turnover, suggesting that these shows may subtly influence electoral competitiveness and the likelihood of political change.

Figure 15: Exposure Effect and Voting Behavior



Notes: Voter Turnout is the number of voters divided by the voting-age population. Turnover rate is the likelihood of an incumbent being replaced. Winner's margin of victory refers to the difference in vote share between the winning candidate and the runner-up. Controls: GDP per capita, population size, and other macroeconomic variables for that country and year of the election. Clustered standard errors at the interview date level are showed in parenthesis.

7 Conclusion

This paper demonstrates that the fictional narratives of entertainment media can have tangible consequences for economic policy preferences and political outcomes. Using a novel dataset of Latin American telenovelas and a quasi-experimental design, I show that exposure to inequality-themed storylines paradoxically reduces public support for redistribution and is correlated with decreased voter turnout and tighter electoral competition. The proposed mechanism is that these portrayals distort viewers' understanding of economic realities. This effect is particularly pronounced among lower-class individuals and varies by narrative style. For instance, comedic representations appear to reduce the perceived severity of the issue. Methodological checks confirm the robustness of these findings, which are specific to inequality content. These findings contribute to our understanding of how culture shapes economic and political behavior, highlighting that popular media acts not just as a mirror of societal norms, but as a powerful force that can actively mold them. The stories we consume on screen can change minds, and in doing so, they can change the world outside the screen as well.

Overall, this research underscores the influential role of entertainment media in shaping public perceptions and policy preferences. It highlights the necessity for a deeper understanding of how narratives in popular culture can sway societal attitudes toward inequality and redistribution. Future research should concentrate on developing improved methods for analyzing the content of telenovelas. The behavioral economics framing literature (Tversky and Kahneman, 1981) suggests that how information is presented, whether in a positive or negative light, affects individual behavior differently. Consequently, discerning whether inequality is portrayed negatively or positively is crucial for understanding its impact on public attitudes.

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Table A.1: Description of Telenovelas that match the identification from 0 to 1

Panel A: Telenovela Statistics							
Title	Country	Air Date	Episodes	Genre	ConceptNet	VecEmbeddings	ChatGPT
Os Dias EramAssim	brazil	2017-04-17	88	drama	x	x	
Pobre Rico	chile	2012-04-23	227	comedy	x	x	x
Tengo todo excepto a ti	mexico	2008-02-11	50	comedy-drama	x	x	
Un poquito tuyo	mexico	2019-02-25	79	comedy			x

Panel B: One-sentence Summary	
Os Dias Eram Assim	*In 1970s Brazil, a young idealistic doctor, Renato, falls in love with a wealthy girl, Alice, only to be separated by political turmoil and deceit, leading to years apart until their love rekindles upon an unexpected reunion.*
Pobre Rico	*Freddy and Nicolás, teenagers from different socioeconomic backgrounds, discover they were swapped at birth, leading to a legal mandate to switch lives with their biological families, as they navigate complex family dynamics and the pursuit of acceptance.*
Tengo todo excepto a ti	*Years after parting ways due to economic hardships, former lovers Carlos and Rebeca, now both married with competing families and magazines, reignite their past connection, leading to conflicts and suspicions of infidelity that unravel both their households.*
Un poquito tuyo	*A successful family man, Antonio, realizes the detrimental effects of spoiling his family and decides they must fend for themselves while becoming entangled with Julieta, a woman with her own dramatic past.*

Notes: These telenovelas meet the identification criteria, provided that no other inequality telenovelas currently airing when these telenovelas started during the same fieldwork period. These telenovelas are used to measure the impact of increasing exposure from zero to one telenovelas that display inequality

8 Appendix

A. Additional Tables and Figures

B. NLP Method Details

B.1 ConceptNet

Tables Table B.3 and Table B.4 present the root words selected for the dictionary and the corresponding related terms found in English (EN), Spanish (ES), and Portuguese (PT), utilizing the ConceptNet database. To access the database I sent requests to ConceptNet API, where you can search for the word "poor" and their related words in Spanish by visiting http://api.conceptnet.io/c/en/poor?filter=/c/es using the following syntax

```
http://api.conceptnet.io/c/ROOT_LANG/WORD?filter=/c/TARGET_LANG
```

Figure A.1: *Si nos Dejan* timeline

Table A.2: Thresholdss by Method

Categorization Method	original	δ^*	δ_-	δ_+
ConceptNet	0.900	0.927	0.850	0.950
Vec Embeddings	0.0040	0.0104	0.0018	0.0061
ChatGPT	0.850	0.931	0.750	0.950

Where ROOT_LANG is the language of the root word, WORD is the root word itself, and TARGET_LANG is the language you want to get related words from. As an example, for the word "poor" in Spanish, I used the URL `http://api.conceptnet.io/c/en/poor?filter=/c/es`. On the website you can find a json file structure and under the key related you get the list of all related words for poor in Spanish with a certain relationship weight, which I chose to be ≥ 0.5.

The chosen root words embody themes of social class conflict depicted in popular telenovelas such as *Pobre Rico* and *Rebelde*. These shows vividly portray the stark inequalities present in various social classes. For example, from *Pobre Rico*, I extracted terms including "poor," "wealthy," "rich," "owner," and "worker," all of which highlight the dynamics of social disparity. Similarly, words like "social class," "elite," and "distinguished" were sourced from *Rebelde* further illustrating the narratives of class distinction and conflict. Most of the Spanish and Portuguese terms identified through ConceptNet are synonyms of their English counterparts, mirroring the translated root words. For instance, "pobre" in both Spanish and Portuguese serves as the literal translation of the word "poor." Additionally, other terms in Spanish, such as "pobrete" or "pobreton," represent alternate expressions of "pobre."

Example B.1 illustrates the ConceptNet score in action. Suppose that the chosen inequality dictionarty $D = \{\text{poor, poorer, slummy, resourceless}\}$, and take a sentence from the synopsis of *Pobre Rico* and *Mama Mechona*, a telenovela about a mother and son going to college at the same time which does not display social class differences. The telenovela *Pobre Rico* then gets a score of $\frac{1}{6}$, since only one word from the dictionary (the word poor) matches among 6 total unique words. The telenovela *Mama Mechona* will get a score of zero since neither "poor", "poorer," "resourceless," nor "slummy" appear in it's synopsis among 10 unique words.

Example B: ConceptNet Score	
Pobre Rico score $= \frac{1}{6}$	*Mama Mechona* score $= \frac{0}{10}$
Freddy lives in **poorer** economic conditions.	Macarena, the mother of a freshman, is also a student.

B.2 Vector Embeddings

Vector embeddings are a type of representation for words in a continuous vector space, where each word is mapped to a unique vector of real numbers. This technique helps capture the semantic meanings of words based on their contexts in which they appear. Essentially, words that have similar meanings or are used in similar ways are positioned closer together in this multidimensional space.

Common techniques for generating word embeddings include Word2Vec, GloVe, Transformers, and even large language models like OpenAI's ChatGPT (Jurafsky and Martin, 2024). These methods employ large datasets containing text to learn the relationships between words based on co-occurrences and context. Once generated, these embeddings can be used in various natural language processing tasks, such as sentiment analysis, translation, and information retrieval, enhancing machines' understanding of human language.

Take for example Figure B.2 where the words "apple," "tree," "grape," and "vine" are encoded in a 2 dimensional space. Vector embeddings are designed to maintain some semantic relationship between words. In this case, it aims to maintain the following relationship: "a apple grows from a tree, and a grape grows from a vine." Taking the relationship GROWS by substracting tree from apple, and adding grape we can get vine. Another common example is the relationship between the words "king," "male", "queen," and "female", with vectors queen $\approx$ king $-$ male $+$ female.

Table B.3: ConceptNet Inequality Terms List

root		related terms		
		EN	**ES**	**PT**
poor	→	poor	pobre	pobre
		poorer	pobrete	pobres
		slummy	pobretones	pobretão
		resourceless	pobretón	
			pobres	
inequality	→	inequality	desigualdad	desigualdade
		inequity	inequidad	inequação
		disparity	inecuaciones	eqüidade
		unequal	disparidad	igualdade
		gini coefficient	igualdad	disparidade
economic	→	economic	económicas	econômica
		noneconomic	economía	economia
		economy	fisiócratas	ecónomo
		macroeconomic	macroeconomico	macroeconómica
		socio economic	socioeconómico	financeiros
rich	→	rich	ricas	rica
		superrich	rico	rico
		richen	opulento	opulento
		richer	pudiente	
wealthy	→	wealthy	adinerado	ricas
		inferior	acaudalada	rico
			acaudalado	ricos
			pudiente	
			adinerada	

Vector embeddings have also been used to keep relationship with other objects in economics. Garcia-Jimeno and Parsa (2024) uses the academic network of economists and assigned them a vector to each theorist to find who are the other economists are most likely to be their co-author. Another example is Hansen uses a big data set of firm characteristics and input output structures to assign a vector to each firm. This exercise surprisingly captures the geographical distances of each firm without having that as an input.

B.3 ChatGPT

The prompt used for this categorization exercise was:

Prompt 1: Inequality Extraction Prompt

```
You are a helpful research assistant. Classify each TV show summary if the plot is
centered around socio-economic class differences. Please show your reasoning step by
step of the categorization. Also provide a confidence level where 50 (uncertain) and 100
(perfectly certain).
```

Table B.4: ConceptNet Inequality Terms List Continued

root		related terms		
		EN	**ES**	**PT**
owner	→	owner	propietarias	proprietária
		proprietor	dueño	dono
		possessor	empresaria	proprietario
		landlord	poseedor	possuidor
		owns	amo	amo
worker	→	worker	cuadrillero	operário
		laborer	laburante	obrador
		employee	trabajadora	trabalhador
		farmworker	obrero	assalariado
		manual_laborer	empleado	empregado
elite	→	elite	elite	elite
		élite	élite	escol
		elitist	elitista	nata
social class	→	social	social	social
		sociability	macrosocial	sociabilidade
		societal	ecosocial	socialização
		social_network	socialista	socialismo
		socialist	socialismo	sociabilizar
distinguished	→	distinguished	distinguido	distinta
		illustrious	egregio	assinalada
		esteemed	prestigioso	eminente
		renowned	destacado	notável
		prestigious	ilustre	marcante

Figure B.2: ConceptNet Inequality Terms List

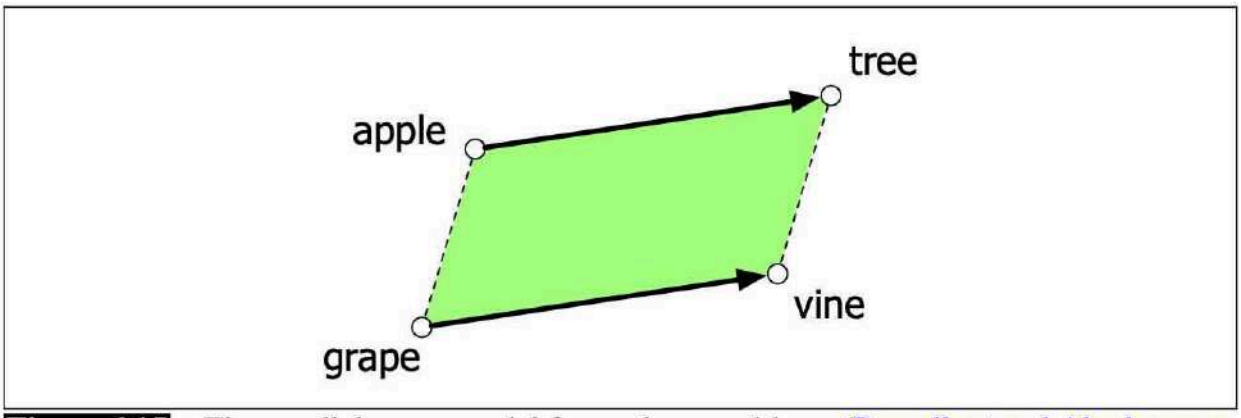

Figure 6.15 The parallelogram model for analogy problems (Rumelhart and Abrahamson, 1973): the location of $\overrightarrow{\text{vine}}$ can be found by subtracting $\overrightarrow{\text{apple}}$ from $\overrightarrow{\text{tree}}$ and adding $\overrightarrow{\text{grape}}$.

The prompt can be divided into three distinct parts that each serve a specific purpose for maximizing the model's efficacy.

First, the instruction, "You are a helpful research assistant," while seemingly lacking contextual depth, is significant in enhancing the performance of generative AI. Research conducted by Salewski *et al.* (2023) indicates that assigning a particular role to the model enables it to concentrate on specific segments of its knowledge base while adopting an expert perspective on the topic at hand. This role-based approach helps align the AI's responses with the desired format and depth of information. Second, my request for the model to explore socio-economic class differences, rather than just issues of inequality, is intentional. My aim is to highlight how characters from distinct economic backgrounds are portrayed in conflict with one another, thereby reflecting the nuances of inequality that arise from these disparities. This focus ensures that the theme of conflict stemming from economic differences remains central to the narrative at least at the beginning of each episode of the telenovelas under analysis. Third, I request that the model provide a confidence level alongside its reasoning, employing a chain of thought (COT) approach. Although this method is informal, the inclusion of confidence levels allows me to gauge the model's certainty regarding whether it identifies inequality or not. This mechanism is helpful in filtering out cases where the model expresses uncertainty. While a formal definition of this confidence could be derived from the log-probabilities of the categorization, implementing such a framework has proven challenging. Nevertheless, encouraging the model to articulate its chain of thought can enhance its performance and mitigate the incidence of hallucinations, leading to more accurate and reliable outputs (Ashwin, Chhabra and Rao, 2023).

Generative AI models are inherently stochastic, which means they can produce different responses to the same prompt, even when the temperature setting is fixed at 0. This unpredictability raises interesting questions about the underlying mechanisms of these models. Although experts have not fully pinpointed the reasons for this behavior, two primary factors are often considered the most likely culprits Chann (2023).

First, computational floating-point inaccuracies in GPUs could contribute to the variability in responses. These inaccuracies arise from the way numerical calculations are performed on GPUs, which can lead to slight discrepancies in the results. Even minimal differences in calculations can propagate through the model's computations, resulting in varied outputs. Second, randomness in the tie-breaking rules for words with equal probabilities is another potential source of this stochastic behavior. When the model encounters multiple words that share the same probability for selection, it may rely on randomization techniques to make a decision. This randomness can result in different words being chosen each time, contributing to the overall variability of the responses.

To mitigate the randomness inherent in generative AI models, or even better, to harness it effectively, I maintain OpenAI's default temperature setting. This choice allows the model to retain a level of creativity in its analyses, which is essential for nuanced interpretations. To further structure the results and enhance reliability, I ask the same prompt five times. By doing so, I can identify the most agreed-upon categorization from the responses provided. This approach not only enables me to aggregate insights from multiple runs but also ensures that the findings are robust and reflective of a consensus among the varied outputs. Additionally, I keep detailed records of each iteration's results, allowing others to verify the outcomes. This transparency fosters trust in the analytical process and enables further scrutiny or replication of the findings.

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ORIGINAL ARTICLE

Economic Inquiry

Trade and ethnolinguistic differences: A replication and extension

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Abstract

We revisit the role of trade on long-run inter-ethnic linguistic differences. Dickens (2022) hypothesized that neighboring languages are more similar when agriculture provides potential gains from *inter-ethnic* trade. Since his empirical approach confounds inter- and intra-ethnic trade, we replicate his main analysis using improved measures of potential inter-ethnic gains from trade. Our results confirm the role of trade in inter-ethnic linguistic differences while showing the difficulty of disentangling the role of inter- versus intra-ethnic trade. We also provide an open-source computational framework to replicate both sets of results, which others can use to produce original replicable economic research.

KEYWORDS
geographic information systems, inter-ethnic trade, intra-ethnic trade, linguistic distances, mapping, replication, trade

JEL CLASSIFICATION
F10, O10, Z10

1 | INTRODUCTION

A growing literature has identified ethnolinguistic groups and their characteristics as central to our understanding of contemporary economic development, conflict, and other social, political, and economic phenomena (Acemoglu et al., 2020; Alesina et al., 2003; Alesina & La Ferrara, 2005; Depetris-Chauvin & Özak, 2020a; Easterly & Levine, 1997; Michalopoulos & Papaioannou, 2013). In turn, this has prompted the search of deep historical roots that explain the differences between these groups and their languages (Depetris-Chauvin & Özak, 2020b; Galor et al., 2016, 2018; Michalopoulos, 2012).

In an important recent contribution to this literature, Dickens (2022) studies the effect of potential gains from *inter-ethnic* trade on linguistic distances. He hypothesizes that neighboring ethnolinguistic groups, which interacted more often due to historical *inter-ethnic* trade opportunities, became more culturally similar. To test this hypothesis, he

Abbreviations: CSI, Caloric Suitability Index; FAO, Food and Agriculture Organization of the United Nations; GAEZ, Global Agro-Ecological Zones.

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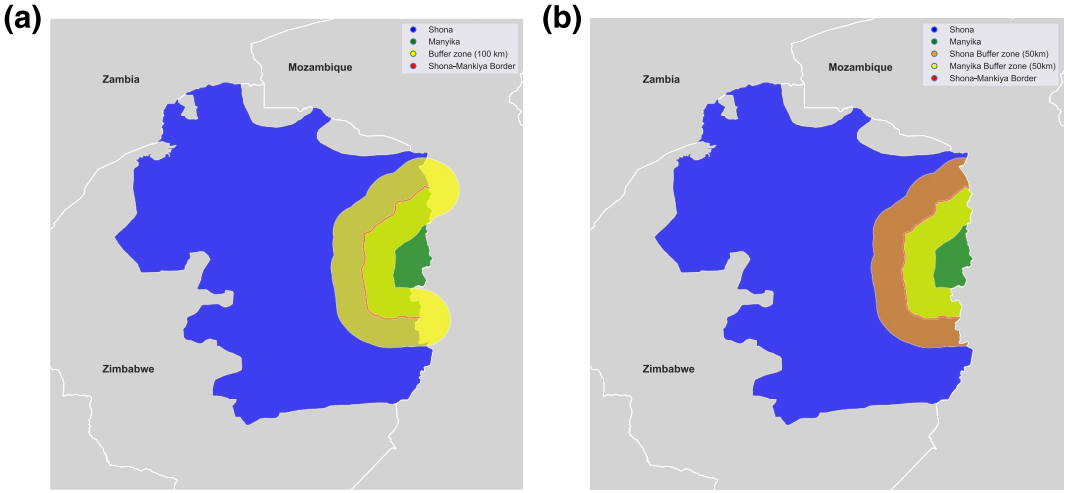

FIGURE 1 Buffer zones. (a) Original buffer zones (Replication, Ethnologue v16) (b) alternative buffer zones (Ethnologue v16). The figure shows two versions of buffer zones around the Shona-Manyika border in Zimbabwe. Panel (a) depicts the original buffer zones used in the replication, constructed using a 50 km radius around the border based on Ethnologue v16 data. Panel (b) presents alternative buffer zones, splitting the 100 km area into two distinct 50 km buffers for each group to capture potential differences in local conditions across ethnic groups. The Shona (blue) and Manyika (green) territories are highlighted, with each buffer zone shown in varying shades to distinguish the proximity to the ethnic boundary and the side of each border in which the buffer is located.

analyzes differences in a fundamental cultural trait—language. In particular, he explores whether linguistic distances - based on lexicographic distances (Wichmann et al., 2016) between spatially adjacent languages still spoken in the world in 2009 (Lewis, 2009)- were affected by potential gains from trade *between* ethnolinguistic groups. He proxies these potential gains using the standard deviation of agricultural productivity using the Caloric Suitability Index (CSI) in a 50 km buffer around the borders where their homelands meet (see Figure 1a).[1,2] Taking advantage of the spatial and temporal variation in agricultural productivity, he identifies the effect of the standard deviation of agricultural productivity on linguistic distances. Specifically, following Galor and Özak (2015, 2016), he exploits the quasi-random changes in productivity generated in the course of the Columbian Exchange (Crosby, 2003) as a natural experiment to identify their effect on linguistic distances, finding that larger standard deviations decreased linguistic distances between neighboring ethnolinguistic groups. While this main result is well identified and very robust, one major drawback of his analysis is his measure of potential gains from trade, which (as we show below) confounds *inter-ethnic* and *intra-ethnic* or more generalized trade.

To understand his hypothesis and the empirical issues better, consider two neighboring ethnolinguistic groups (like the Shona and Manyika depicted in Figure 1). Dickens (2022) argues through the lens of a Ricardian trade model, that if agricultural productivity differs *between* these two groups, then there will exist potential gains from *inter-ethnic* trade due to specialization. The larger these differences, the larger those potential gains, and, thus, more trade between these groups. So, the presence and frequency of inter-ethnic trade should have increased inter-ethnic contact and linguistic exchange, which, in turn, should have decreased linguistic differences between these two groups. In his empirical test, Dickens (2022) uses the standard deviation of agricultural productivity in the buffer around the border where these ethnic groups meet (yellow area in Figure 1a) as his proxy for potential for inter-ethnic trade. But, clearly, the potential for *inter-ethnic* trade should depend on the difference in productivity *between* ethnic groups around the border and not on the standard deviation of productivity in the buffer as a whole, that is, it should depend on the differences in agricultural productivity between the orange and yellow areas in Figure 1b.

In the present paper, we investigate whether his empirical results are reproducible and replicable, and further test his hypothesis using this alternative approach to measuring the potential gains from *inter-ethnic* trade. Our focus in this replication is less on the statistical robustness of the main results (which tends to be the focus of many replication exercises) or the reproducibility using the data and code produced by the author,[3] and more on the reproducibility and replicability of the empirical findings from original sources (which given the geographical nature of the analysis tends to be a more difficult task), and the empirical implementation of his theory. Specifically, (1) we reconstructed his main independent variables from original sources (Galor & Özak, 2015, 2016; Lewis, 2009), (2) we reconstructed his main independent variables using an updated version of the ethnolinguistic map he used to create the buffers in his analysis,

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TABLE 1 Reproduction of Table 1 in Dickens (2022) standardized betas.

	Dependent variable: Lexicostatistical linguistic distance ∈ (0, 1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Δ in std CSI in buffer (post-1500)	−0.114	−0.113	−0.094	−0.117	−0.111	−0.078
	(0.034)	(0.038)	(0.038)	(0.038)	(0.037)	(0.041)
	[0.001]	[0.003]	[0.013]	[0.002]	[0.003]	[0.056]
std CSI in buffer (pre-1500)	−0.094	−0.093	−0.071	−0.066	−0.062	−0.046
	(0.033)	(0.034)	(0.040)	(0.033)	(0.039)	(0.042)
	[0.005]	[0.005]	[0.071]	[0.042]	[0.116]	[0.281]
Δ in average CSI in buffer (post-1500)		−0.003	0.002	−0.018	−0.004	0.027
		(0.036)	(0.035)	(0.036)	(0.035)	(0.037)
		[0.932]	[0.958]	[0.616]	[0.918]	[0.452]
Average CSI in buffer (pre-1500)		−0.005	−0.002	0.005	0.017	0.037
		(0.024)	(0.024)	(0.025)	(0.025)	(0.025)
		[0.847]	[0.927]	[0.846]	[0.481]	[0.134]
Geography controls	No	No	Yes	No	Yes	Yes
Spatial controls	No	No	No	Yes	Yes	Yes
Language family FEs	Yes	Yes	Yes	Yes	Yes	Yes
Country FEs	No	No	No	No	No	Yes
Adjusted R^2	0.248	0.248	0.259	0.264	0.278	0.369
Observations	8402	8402	8402	8402	8402	7291

Note: Unit of observation: border buffer zone (100 km). This table establishes the negative and statistically significant effect of variation in land productivity on a language pair's lexicostatistical linguistic distance. Geography controls include mean elevation, ruggedness, mean temperature and its standard deviation, mean precipitation and its standard deviation, and the prevalence of malaria. Spatial controls include logged distance to the nearest coast, country border, lake, major river and minor river, logged distance between group centroids, the absolute difference in latitude and longitude, logged land area, and logged population. SEs are double clustered at the level of each language group and are reported in parentheses.

(3) using the same original and updated sources, we constructed alternative (and perhaps more appropriate) measures of potential gains from inter-ethnic trade. In particular, instead of using the standard variation of agricultural productivity around the border, we proxy the potential for *inter-ethnic* trade by exploiting differences in agricultural productivity *across the border* (Figure 1b), which (as shown below) should better capture the gains from inter-ethnic trade. We further explore the role of inter-ethnic trade by analyzing the effect of cross-border differences in agricultural productivity in all crops underlying the CSI measures.

Using these various new measures, we successfully replicated his main result (column 5 of Table 1 in his paper), which we use for our analysis here. We provide a more complete replication in the repository with the new data and code. Our reconstructed independent variables, based on the same and updated underlying raw data, are very highly correlated with the ones provided in the replication package and generate almost identical results. Thus, we provide a successful direct replication of his work. Moreover, we find that our improved measures of potential gains from inter-ethnic trade generate weaker results in terms of sign and size of the coefficient of interest, while having almost identical explanatory power. Since our new measures capture mainly inter-ethnic trade potential, while the original measures reflect generalized or intra-ethnic trade, our results suggest that it is trade in general which matters for cultural convergence.

Besides contributing to our understanding of the role of trade in cultural convergence, our paper also contributes to the literature on replication in economics and other social sciences. In particular, we provide an open-source computational environment that can be used to replicate our and the original results of Dickens (2022), allowing the replicators to be replicated. Additionally, our tools and methods can be used to replicate other research, especially studies that employ geographical data or generate maps. Moreover, our tools can be used for teaching and learning purposes, especially for students, policy-makers, and other individuals in developing countries, who may not have

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access to proprietary software (ArcGIS, Stata, etc.) or sophisticated computational resources. In particular, to help others replicate and learn from our replication analysis, we have also created scripts to recreate our computational environment using conda/mamba, Deepnote, Docker, and MyBinder. All our data and code are available in the OPENICPSR repository https://doi.org/10.3886/E209325V2 and in OSF https://osf.io/k3p7g/.

2 | REPRODUCIBILITY

Dickens (2022) estimates the effect of his measures of potential gains from trade (standard deviation of agricultural productivity in the buffer around the common ethnolinguistic border) using a cross-section of all spatially adjacent language pairs and ordinary least squares regressions (his most comprehensive specification includes a large set of geographical and spatial controls, language family and country fixed effects). He double-clustered standard errors at the level of each language group to account for dependence between observations. His main results are presented in his Table 1 (p. 967). Specifically, in Column (1) he "reports within-family estimates for pre-Columbian land productivity variation and the post-Columbian change in land productivity variation. Both estimates enter with the expected negative sign and are statistically significant at the 1% level. This says that ethnic groups separated across high-variation regions are more similar in language than groups separated across low-variation regions. In particular, a one standard deviation increase in pre-1500 land productivity variation decreases linguistic distance by 1.7 percentage points, while a standard deviation increase in productivity variation at the onset of the Columbian exchanges implies a 2.0 percentage points decrease in linguistic distance." (p. 967) Additionally, he shows that after accounting for all controls except country fixed effects (column 5) "[t]he coefficient on pre-1500 land productivity variation retains the expected sign, but loses significance at standard levels. Whereas the coefficient of interest—the effect of land productivity variation resulting from the Columbian Exchange in the post-1500 period—retains statistical significance with the expected negative sign, and is statistically equivalent to the unconditional estimate in column (1)." (p. 968).

We downloaded the replication package provided by the author to the journal.[4] The code and data worked without any issues. We were able to reproduce the results of the paper using Stata-MP version 17. In order to improve reproducibility and provide educational tools based on these analyses for individuals who may not have access to this paid software, especially individuals in developing countries, we replicated these results using free tools. Specifically, we used the open-source programming language Python and various statistical and graphical packages based on it that are freely available.[5,6]

Our Table 1 reproduces the results of Table 1 in the original paper and shows the OLS estimates using standardized beta coefficients to simplify interpretability of the results (Table A1 in the Appendix reproduces the original Table 1 without this transformation) of the following equation:

$$\begin{aligned} LD_k = \beta_0 + \beta_{std}^{1500} CSIstd_k + \beta_{std}^{change} \Delta CSIstd_k \\ + \beta_{mean}^{1500} CSI_k + \beta_{mean}^{change} \Delta CSI_k + X_k' \Phi + \lambda_{l_i(k)} + \theta_{l_j(k)} + \delta_{c(k)} + \varepsilon_k, \end{aligned} \tag{1}$$

where LD_k is the linguistic distance in border k, $CSIstd_k$ is the standard deviation of CSI (pre-1500) and $\Delta CSIstd_k$ is its change (Δ) (post-1500) in buffer k, and CSI_k and ΔCSI_k are the average pre-1500 and its change post-1500 in CSI in buffer k. The results in column (5) of Table 1 suggest that after accounting for geographical and spatial controls, and language family fixed effects, a one standard deviation increase in the change in land productivity variation (post-1500) decreases linguistic distances by 0.11 standard deviations (p-value =0.003) and a one standard deviation increase in land productivity variation (pre-1500) decreases linguistic distances by 0.06 standard deviations (p-value =0.116). To simplify the presentation, we will focus on this same specification in the rest of the replication analysis.

3 | REPLICATION

3.1 | Direct replication using original and updated data

We also replicated the construction of the main independent variables used in his analysis. We assumed the original study employed the *langa.shp* file from the World Language Mapping System, which provides polygons for the

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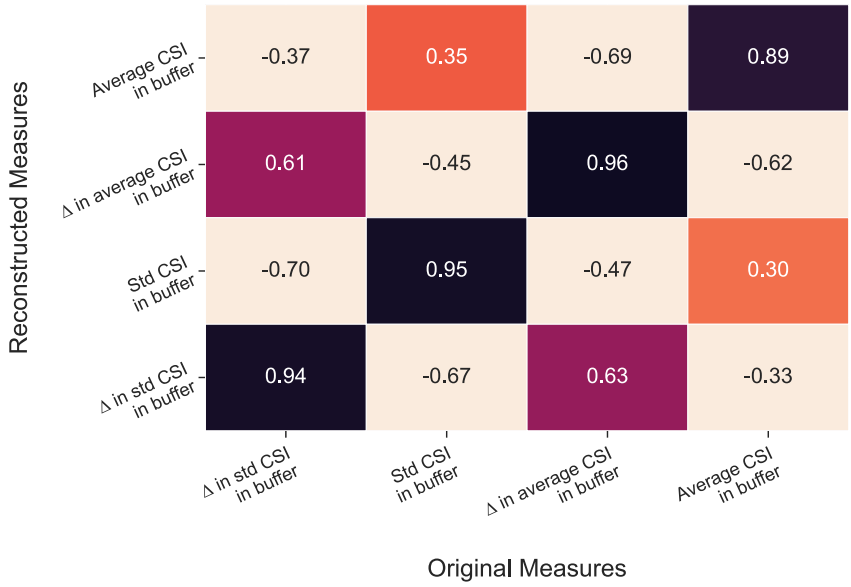

FIGURE 2 Correlation between original and reconstructed independent variables. The figure presents the correlation matrix between original and reconstructed independent variables for different measures of agricultural productivity based on CSI data within buffer zones. Each cell reports the correlation coefficient between the corresponding reconstructed and original variable based on Ethnologue (v16). Variables include the average CSI, change in average CSI, standard deviation of CSI, and change in standard deviation of CSI within the buffer. High correlations (close to 1 or −1) indicate a strong similarity between the original and reconstructed measures, while low correlations suggest distinct patterns in their behavior.

homelands of languages spoken across the world in 2009 (version 16) and 2014 (version 17) based on the Ethnologue (Lewis, 2009; Lewis et al., 2014). While the paper explains clearly most steps involved in the construction of the data, it did not clarify the geographical projection used in the analysis, which provided us with one degree of freedom. We followed the most straightforward (though perhaps not the most correct) strategy and used the original projections of the raw data in the construction.[7] Additionally, Galor and Özak (2015, 2016) provide two distinct Caloric Suitability Indices for the pre-1500 period, but Dickens (2022) does not clarify which one he used. Thus, we have a second degree of freedom, and we use their pre-1500 measure that includes Asian crops in Africa for our main replication analyses.[8]

Once we had these data sources and had made choices regarding the projections to be used, we followed a similar strategy to him to construct buffers around the borders. In the construction of borders and buffers, (i) following Dickens (2022), we excluded pairs in which the border reflected the same language in two countries (e.g., the language Abron is spoken both in Ghana and Côte d'Ivoire, Spanish in many neighboring Latin American countries, and similar for Arabic in the Middle East and North Africa). Clearly, including these would have been problematic and bias the analysis.[9] We also identified one language that had the same name (NAME2 in the original data), was spoken in two countries, but had different ISO codes. We decided to treat it as a unique language and exclude it from the data construction.[10] None of these language pairs were included in the (original) empirical analyses since they were excluded due to missing linguistic distance data. (ii) We excluded pairs in which the homeland of one language was (fully) contained inside the other, since this may reflect recent changes in the location of speakers and may bias the results.[11]

We reconstructed the data using both versions 16 and 17 of the Ethnologue. We found that between version 16 and 17, Ethnologue dropped many languages, which may have become extinct, or by mistake (e.g., version 17 does not have a polygon for Spanish in Paraguay). Additionally, some new languages appeared in the data set (not clear whether these are languages that did not have a polygon in version 16 or which became more vibrant). Specifically, version 17 has 414 languages not contained in version 16, while version 16 has 537 languages not contained in version 17. These differences explain the main differences in the samples for replication. In the main text, we focus on version 16, but all the results for version 17 are available in the appendix and replication code.

Our reconstructed independent variables, based on the same underlying raw data, are very highly correlated with the ones provided in the replication package, as shown in Figure 2. Similar high correlations exist between the original data and our reconstructed measures based on version 17 of the Ethnologue, as shown in Figure B1 in the Appendix. In

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TABLE 2 Replication and extension of main result.

	Dependent variable: Lexicostatistical linguistic distance ∈ (0, 1)						
	Replication					Robustness	
	Original data			Reconstructed data		Alternative measure	
	Original sample (1)	v16 (2)	v17 (3)	v16 (4)	v17 (5)	v16 (6)	v17 (7)
Δ in std CSI in buffer (post-1500)	−0.111	−0.111	−0.136	−0.140	−0.190		
	(0.037)	(0.037)	(0.040)	(0.042)	(0.042)		
	[0.003]	[0.003]	[0.001]	[0.001]	[0.000]		
Std CSI in buffer (pre-1500)	−0.062	−0.062	−0.076	−0.064	−0.105		
	(0.039)	(0.039)	(0.041)	(0.041)	(0.042)		
	[0.116]	[0.114]	[0.066]	[0.120]	[0.013]		
Δ in average CSI in buffer (post-1500)	−0.004	−0.003	0.017	0.007	0.039	−0.034	−0.022
	(0.035)	(0.035)	(0.035)	(0.043)	(0.043)	(0.031)	(0.031)
	[0.918]	[0.922]	[0.622]	[0.872]	[0.354]	[0.275]	[0.482]
Average CSI in buffer (pre-1500)	0.017	0.018	0.020	−0.003	0.012	0.008	0.006
	(0.025)	(0.025)	(0.026)	(0.032)	(0.033)	(0.024)	(0.025)
	[0.481]	[0.476]	[0.436]	[0.926]	[0.706]	[0.747]	[0.807]
Δ in cross border CSI difference (post-1500)						−0.049	−0.059
						(0.025)	(0.025)
						[0.049]	[0.016]
Cross border CSI difference (pre-1500)						0.001	−0.002
						(0.022)	(0.021)
						[0.965]	[0.935]
Geography controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spatial controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Language family FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.278	0.278	0.280	0.279	0.282	0.277	0.279
Observations	8402	8400	7560	8400	7560	8400	7560

Note: Unit of observation: border buffer zone (100 km). This table establishes the negative and statistically significant effect of variation in land productivity on a language pair's lexicostatistical linguistic distance. Geography controls include mean elevation, ruggedness, mean temperature and its standard deviation, mean precipitation and its standard deviation, and the prevalence of malaria. Spatial controls include logged distance to the nearest coast, country border, lake, major river and minor river, logged distance between group centroids, the absolute difference in latitude and longitude, logged land area and logged population. SEs are double clustered at the level of each language group and are reported in parentheses.

Table 2, we explore how the reconstructed data and samples affect his main result from Table 1 column 5 in the original paper. Specifically, column (1) replicates the original result, while columns (2) and (3) show the result of constraining the original data to the samples of language pairs we have in the reconstructed data with qualitatively similar results, albeit a bit stronger effects in the reduced sample of version 17. Columns (4) and (5) replicate the analysis using our reconstructed measures and find almost identical results, although the estimated effects are always larger. Specifically, using the original data and sample, a one standard deviation increase in the change in land productivity variation (post-1500) decreased linguistic distances by 0.111 standard deviations (p-value =0.003), while it decreases it by 0.14 standard deviations (p-value =0.001) in the reconstructed Ethnologue version 16 data, and by 0.190 standard deviations (p-value <0.001) in the reconstructed Ethnologue version 17 data. Thus, there does not seem to exist significant differences

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between the replication and original analyses. This result is further confirmed in Figure B2a, which explores the effects of changing buffer sizes or accounting for country-fixed effects.

4 | ROBUSTNESS—INTER-ETHNIC VS GENERALIZED TRADE POTENTIAL

Dickens's (2022) "aim [...] is to shed light on an unexplored channel of cultural change: inter-ethnic trade" (p. 953). His main hypothesis, general theory, and model are based on the idea that "the gains from trade are a function of land productivity variation *between* groups" (p. 959, emphasis added). Thus, the idea is that linguistic distances should decrease due to *different* groups coming together to exchange *different* types of goods they specialize in.[12] Yet, neither his measure of inter-ethnic trade potential, the standard deviation of agricultural productivity, nor the evidence he presents based on it, *directly* provide support for this *interethnic* aspect of trade, since they fail to capture cross-ethnic economic specialization. In particular, the ethno-linguistic group level analyses, which show that a larger standard deviation of agricultural productivity is associated with more trade and less conflict (Section 4 of the paper, especially Table 4 and Figure 9), is similar and in line with previous research that has established the positive effect of diversity (ecological, geographical, and population) within an ethno-linguistic group (or its homeland) on the emergence of economic specialization, trade, and states (Depetris-Chauvin & Özak, 2020b; Fenske, 2014), which can directly reduce linguistic distances across languages in a region. Thus, his border level analyses may reflect the negative effect of *generalized* trade on linguistic distances even in cases where there is no cross-ethnic specialization. Specifically, it may very well be the case that linguistic distances decrease because people from *different* ethnic groups interact when trading *similar* goods, that is, these interactions are not driven by ethnic level specialization. While this would have no impact on the empirical analyses themselves, as agricultural variation would decrease linguistic distances, it would affect their interpretation.

To understand this concern better, Figure 3 shows that the potential for *inter-ethnic* trade should depend (as in his model and theory) on the difference in productivity *between* ethnic groups around the border and not on the variation of productivity in the buffer as a whole, which should reflect incentives for generalized trade.[13] In particular, these figures depict two artificial buffers around an ethnic border, which is assumed to have the same mean and standard deviation in productivity, that is, they would have the same trade potential according to the original measure. But, clearly, inter-ethnic trade potential should be higher in Figure 3b than in Figure 3a since it provides stronger incentives for cross-ethnic economic specialization.

To explore this issue further, as a final step, we construct measures of the difference in productivity pre-1500 and its change post-1500 between the two sides of each buffer (as depicted in Figure 1b) and replicate the main specification using these. Specifically, for each buffer, we compute the absolute difference in productivity pre-1500 between both sides of the border, that is, for each part of the buffer that belongs to a specific homeland, and the change in this

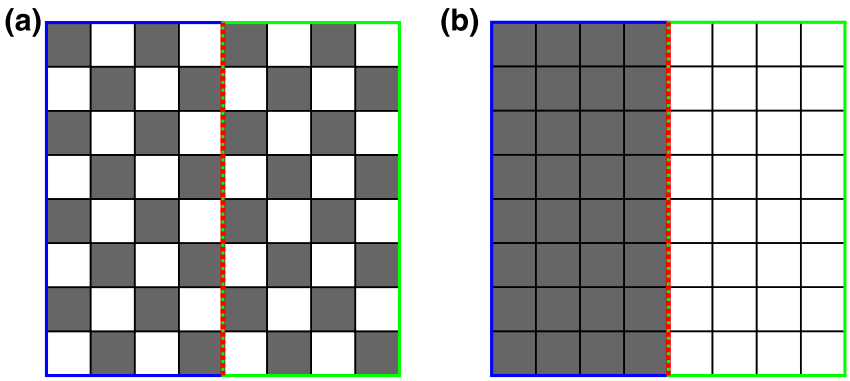

FIGURE 3 Productivity and inter-ethnic trade potential in buffer around border. (a) Similar productivity and low potential (b) different productivity and high potential. The figure illustrates how inter-ethnic trade potential is influenced by productivity differences *between* ethnic groups at the border, rather than by overall productivity variation within the buffer region. Both panels have the same mean and standard deviation of productivity, yet trade potential differs based on inter-ethnic productivity disparity. Panel (a) shows similar productivity levels on both sides of the border, implying low trade potential. In contrast, Panel (b) depicts distinct productivity levels across the border, suggesting higher trade potential.

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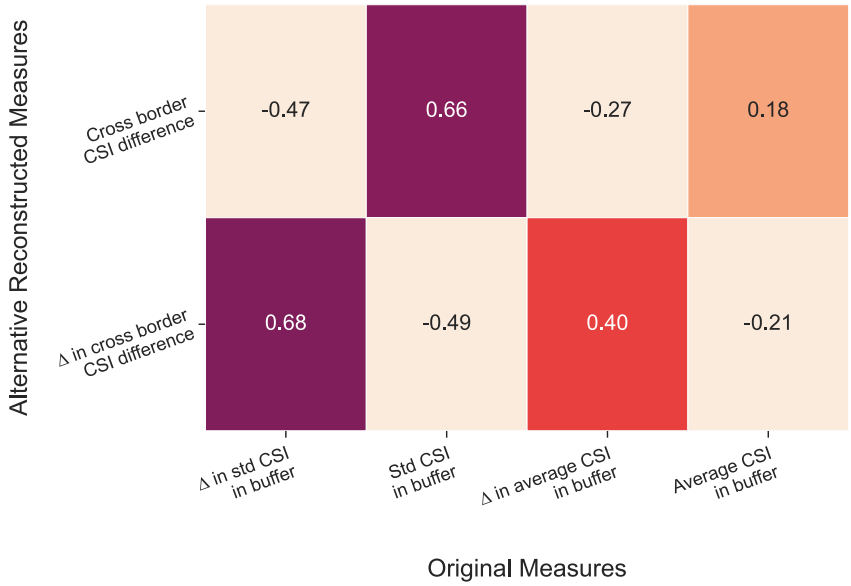

FIGURE 4 Correlation between original and alternative measures of potential gains from inter-ethnic trade based on Ethnologue v16. The figure presents the correlation matrix between original and alternative measures of potential gains from inter-ethnic trade based on Ethnologue (v16) data. Each cell shows the correlation coefficient between the corresponding original and alternative measure. Variables include cross-border CSI differences, changes in cross-border CSI differences, as well as within-buffer average CSI, changes in average CSI, standard deviation of CSI, and changes in standard deviation of CSI. High correlations indicate similarity between the original and alternative measures, while low correlations reflect significant differences.

difference post-1500. For example, given the border between ethnicities A and B, our new measure is $\left|CSI_k^A - CSI_k^B\right|$ where CSI_k^i is the average CSI on side $i = A, B$ of the buffer. We similarly compute its change due to the Columbian Exchange. Using these measures, we estimate the following regression

$$\begin{aligned} LD_k = \beta_0 + \beta_{alt}^{1500} CrossBorder_k + \beta_{alt}^{change} \Delta CrossBorder_k \\ + \beta_{or}^{1500} CSI_k + \beta_{or}^{change} \Delta CSI_k + X_k' \Phi + \lambda_{l_i(k)} + \theta_{l_j(k)} + \varepsilon_k, n \end{aligned} \tag{2}$$

where $CrossBorder_k$ and $\Delta CrossBorder_k$ are the new measures for border k. Clearly, a significant effect of these differences, that is, $\beta_{alt}^{1500} < 0$ and $\beta_{alt}^{change} < 0$, would be prima facie evidence for the role of inter-ethnic trade. Columns (6) and (7) in Table 2 show the results of this analysis, in which we replace the original proxies of inter-ethnic trade (standard deviation of agricultural productivity and its change) with our new measures. Specifically, the results show the effect of differences in productivity across the border pre-1500 and its change post-1500 on linguistic distances after accounting for all the same controls as in the main specification we have been focusing on in this replication (i.e., Column (5) in Table 1 of the original study). The estimates suggest that a one standard deviation increase in the changes post-1500 in the differences in agricultural productivity across the two sides of the borders decreases linguistic distances by 0.049 (p-value $= 0.049$) and by 0.059 (p-value $= 0.016$) using data from Ethnologue version 16 and 17 respectively. This is strong direct evidence that inter-ethnic trade does affect linguistic distances.

Still, the results may reflect the forces of generalized trade. In particular, as Figures 4 and B3 show, the correlation between our alternative measures of inter-ethnic trade potential based on cross-border differences and the original measures of agricultural productivity variation both in the pre- and post-1500 periods is quite high with correlations above 0.65.[14] Clearly, given these high correlations, disentangling the two forces is difficult.[15] To further explore this issue, we construct crop-specific measures of inter-ethnic trade potential. Specifically, for all crops in Galor and Özak (2016, 2015), we constructed measures of the cross-border difference in crop productivity (similar to our alternative proxy based on CSI). Unlike the original measures, we do not have changes, since the changes were based on the introduction of new crops. Using these measures, we reestimated Equation (2) using crop-specific instead of CSI cross-border differences. The results are shown in Figure 5. In particular, in Figure 5a, we plot the coefficient and standard error when we include each crop individually in Equation (2), that is, it plots the coefficients of 40 individual

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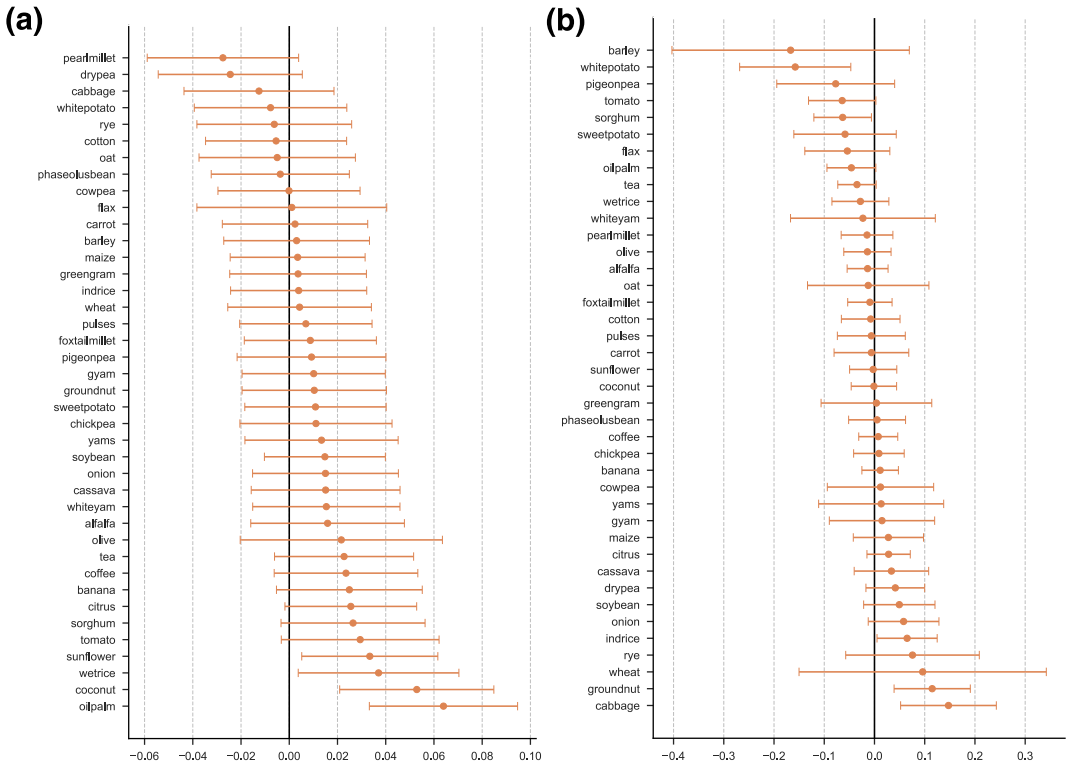

FIGURE 5 Coefficient plots—crop specific estimations. (a) Cross border difference (individual regressions) (b) cross border difference (joint regression). The figure displays coefficient plots for crop-specific estimations using cross-border differences. Panel (a) shows the results from individual regressions for each crop, while Panel (b) presents the results from a joint regression including all crops. Each coefficient estimate is represented by a dot, and the horizontal lines indicate the 95% confidence intervals. Positive coefficients suggest that higher productivity differences are associated with greater linguistic distances, while negative coefficients imply that higher productivity differences are linked to smaller linguistic distances. The variation in coefficient estimates across crops highlights the heterogeneous effects of cross-border differences on productivity.

regressions. It shows that most coefficients are not statistically different from zero (except sunflower, wet rice, coconut, and oil palm, which are significant, but have the wrong sign). In Figure 5b, we plot the coefficients when we include *all* crops jointly with similar results. In this case, white potato and sorghum have negative and statistically significant coefficients, while groundnut and cabbage have positive and statistically significant coefficients.

Thus, the results fail to provide strong evidence in favor of the inter-ethnic aspect of trade and may reflect the role of generalized trade instead. Nonetheless, these results need to be taken with a grain of salt, since there are many issues that may still bias the analysis. In particular, the location of these contemporary linguistic borders may not reflect historical ethnic borders (Depetris-Chauvin & Özak, 2020a), so that our measures may reflect poorly the historical conditions that may have generated inter-ethnic specialization. Moreover, our measures may not fully capture the incentives to specialize, for example, across crops (see, e.g., Blouin & Dyer, 2021). Additionally, it is important to remember, that even if endowments are similar, inter-ethnic specialization and trade may still have evolved and driven linguistic convergence. Clearly, more research is needed.

5 | CONCLUSION

Dickens (2022) presents novel and important evidence to understand the cultural convergence between ethno-linguistic groups and the role trade and geography play in it. Following his methodology, we have successfully replicated his central result establishing the negative effect of the standard deviation of agricultural productivity on linguistic distances. We have shown the results are robust to changes in the construction method and the underlying linguistic maps finding that the empirical analyses remained qualitatively unchanged, although the estimated effects tended to be larger.

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As an extension of his work, we have delved deeper into the role of inter-ethnic as opposed to generalized trade in this result. Since both types of trade should increase contact between individuals, they should decrease linguistic distances due to increased contact among individuals. We argue that the original measures used in the study failed to properly distinguish between these two forces and construct novel measures of potential gains from inter-ethnic trade at ethno-linguistic borders. Using our novel measures, we obtained qualitatively similar although quantitatively smaller effects. Also, we find that these results are not robust to accounting for the original measures. This may reflect the fact that it is generalized trade and not inter-ethnic trade that drives linguistic convergence, or various shortcomings of the new measures. Fully understanding the role of inter-ethnic trade may require different strategies like building more sophisticated measures that account for potential mismeasurement issues (Depetris-Chauvin & Özak, 2020a) or proper crop specialization across ethnic groups (Blouin & Dyer, 2021).[16] Clearly, more research is needed to understand this important subject.

ACKNOWLEDGMENTS

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in OPENICPSR at https://doi.org/10.3886/E209325V2, reference number openicpsr-209325 (Gonzalez & Özak, 2024).

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ENDNOTES

[1] That is, buffers with a diameter of 100 km.

[2] The Caloric Suitability Indices (CSI) capture the variation in potential crop yield across the globe, as measured in calories per hectare per year. Moreover, in light of the expansion in the set of crops that are available for cultivation in the course of the Columbian Exchange, the CSI indices provide a distinct measure for caloric suitability for the pre-1500 and the post-1500 era. They are based on a set of 48 crops and data from the GAEZ project (see Galor & Özak, 2015, 2016).

[3] Both us and, independently, the Institute for Replication were successfully able to reproduce the original study using the replication package provided by the author on the journal's website. We did not find any differences between the published results and the ones we obtained using his data and code. The replication code and data is available from the website of the Economic Journal (https://doi.org/10.1093/ej/ueab065). The code reproduces all tables and most figures (except maps), although it does not save the output to a usable file.

[4] Downloaded from https://doi.org/10.1093/ej/ueab065 on October 4, 2022.

[5] Our reproduction uses Jupyter notebooks and performs the same analyses in Python and Stata. To execute cells in the notebooks that use Stata, this software needs to be installed on the computer.

[6] We also used Julia, another open-access programming language, to replicate Dicken's (2022) tables and figures.

[7] Both the linguistic maps for Ethnologue (Lewis, 2009; Lewis et al., 2014) and the Caloric Suitability Index (Galor & Özak, 2015, 2016) are provided in the WGS84, that is, EPSG:4326, geographic coordinate system. These are not the best projections to use in a global analysis given the underlying data and question. Using a projected coordinate system like a cylindrical equal area projection (ESRI:54034) would have been preferable. Yet, previous experience suggests that using either of these generates highly correlated measures so we do not consider this a major issue. Moreover, it provides us with an alternative robustness analysis, which clearly could be improved by selecting different projections in the construction.

[8] Our replication code and data reconstructs the data using all versions of the CSI data.

[9] In the original replication file, we found 111 pairs of languages that were the same language across two different countries (Stata command: tab identifier if langIso1 = = langIso2). Yet, these pairs are never used in the analyses in the original paper.

[10] The language is Marwari which is spoken in India and Pakistan (see notebook for references).

[11] Our sample using version 16 of the Ethnologue map differed from the original study by only 1 observation, which is the pair BZX-MLI-FFM-MLI (Bozo, Hainyaxo and Fulfulde, Maasina in Mali), where one is strictly contained in the other. The replication data contains a variable called overlap, which is used in Table A2 in the original paper to explore the robustness to excluding languages that overlap. We assumed these would include the ones we excluded due to being one contained in the other, but this variable identifies many more language pairs as overlapping. We find that 2848 of the language pairs with linguistic distance data that are identified as overlapping according to the original study are not identified as overlapping in our procedure. These include cases like KHK-MNG-CMN-CHN, AAC-PNG-DBY-PNG, and AAL-CMR-KOT-CMR, which the Ethnologue map does not show as overlapping. Without the code used to generate

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the original data we could not determine how these overlaps were identified and why our results differed in this aspect. We did not explore this issue further since it had no impact on the main regression we analyze here.

[12] One aspect that is not studied here nor in the original paper is how those differences come about. Specifically, whether the convergence is due to horizontal or vertical transmission of culture, and whether the type of trade can affect transmission via one channel or the other. We thank referee 2 for this insight.

[13] Equivalently, if one were to do the analysis using the complete homelands, it should be driven by the difference in productivity between the homelands and not by the variations across both homelands.

[14] This could be expected due to the way the data is constructed and the relation between the standard deviation across the whole buffer and difference in the mean across both sides of the border.

[15] For example, in a horse race, the effect of our measures of inter-ethnic trade potential fall between 77% and 80% and become statistically insignificant with p-values over 0.6. On the other hand, the original measures, which reflect generalized trade, remain basically unchanged and remain statistically significant at the 1% level.

[16] It should not be difficult to use the code we provide in our repository to extend these analyses further.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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APPENDIX A

TABLE A1 Reproduction of Table 1 in Dickens (2022).

	Dependent variable: Lexicostatistical linguistic distance ∈ (0, 1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Δ in std CSI in buffer (post-1500)	−0.100	−0.100	−0.083	−0.103	−0.098	−0.065
	(0.030)	(0.034)	(0.034)	(0.034)	(0.033)	(0.034)
	[0.001]	[0.003]	[0.013]	[0.002]	[0.003]	[0.056]
std CSI in buffer (pre-1500)	−0.061	−0.061	−0.046	−0.043	−0.040	−0.029
	(0.022)	(0.022)	(0.026)	(0.021)	(0.026)	(0.027)
	[0.005]	[0.005]	[0.071]	[0.042]	[0.116]	[0.281]
Δ in average CSI in buffer (post-1500)		−0.001	0.001	−0.005	−0.001	0.008
		(0.010)	(0.010)	(0.010)	(0.010)	(0.010)
		[0.932]	[0.958]	[0.616]	[0.918]	[0.452]
Average CSI in buffer (pre-1500)		−0.001	−0.001	0.001	0.004	0.009
		(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
		[0.847]	[0.927]	[0.846]	[0.481]	[0.134]
Geography controls	No	No	Yes	No	Yes	Yes
Spatial controls	No	No	No	Yes	Yes	Yes
Language family FEs	Yes	Yes	Yes	Yes	Yes	Yes
Country FEs	No	No	No	No	No	Yes
Adjusted R^2	0.248	0.248	0.259	0.264	0.278	0.369
Observations	8402	8402	8402	8402	8402	7291

Note: Unit of observation: border buffer zone (100 km). This table establishes the negative and statistically significant effect of variation in land productivity on a language pair's lexicostatistical linguistic distance. Geography controls include mean elevation, ruggedness, mean temperature and its standard deviation, mean precipitation and its standard deviation, and the prevalence of malaria. Spatial controls include logged distance to the nearest coast, country border, lake, major river and minor river, logged distance between group centroids, the absolute difference in latitude and longitude, logged land area and logged population. SEs are double clustered at the level of each language group and are reported in parentheses.

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TABLE A2 Replication and extension of main result non-standardized coefficients.

	Dependent variable: Lexicostatistical linguistic distance ∈ (0, 1)						
	Replication					Robustness	
	Original data			Reconstructed data		Alternative measure	
	Original sample (1)	v16 (2)	v17 (3)	v16 (4)	v17 (5)	v16 (6)	v17 (7)
Δ in std CSI in buffer (post-1500)	−0.111	−0.111	−0.136	−0.140	−0.190		
	(0.037)	(0.037)	(0.040)	(0.042)	(0.042)		
	[0.003]	[0.003]	[0.001]	[0.001]	[0.000]		
Std CSI in buffer (pre-1500)	−0.062	−0.062	−0.076	−0.064	−0.105		
	(0.039)	(0.039)	(0.041)	(0.041)	(0.042)		
	[0.116]	[0.114]	[0.066]	[0.120]	[0.013]		
Δ in average CSI in buffer (post-1500)	−0.004	−0.003	0.017	0.007	0.039	−0.034	−0.022
	(0.035)	(0.035)	(0.035)	(0.043)	(0.043)	(0.031)	(0.031)
	[0.918]	[0.922]	[0.622]	[0.872]	[0.354]	[0.275]	[0.482]
Average CSI in buffer (pre-1500)	0.017	0.018	0.020	−0.003	0.012	0.008	0.006
	(0.025)	(0.025)	(0.026)	(0.032)	(0.033)	(0.024)	(0.025)
	[0.481]	[0.476]	[0.436]	[0.926]	[0.706]	[0.747]	[0.807]
Δ in cross border CSI difference (post-1500)						−0.049	−0.059
						(0.025)	(0.025)
						[0.049]	[0.016]
Cross border CSI difference (pre-1500)						0.001	−0.002
						(0.022)	(0.021)
						[0.965]	[0.935]
Geography controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spatial controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Language family FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.278	0.278	0.280	0.279	0.282	0.277	0.279
Observations	8402	8400	7560	8400	7560	8400	7560

Note: Unit of observation: border buffer zone (100 km). This table establishes the negative and statistically significant effect of variation in land productivity on a language pair's lexicostatistical linguistic distance. Geography controls include mean elevation, ruggedness, mean temperature and its standard deviation, mean precipitation and its standard deviation, and the prevalence of malaria. Spatial controls include logged distance to the nearest coast, country border, lake, major river and minor river, logged distance between group centroids, the absolute difference in latitude and longitude, logged land area, and logged population. SEs are double clustered at the level of each language group and are reported in parentheses.

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APPENDIX B

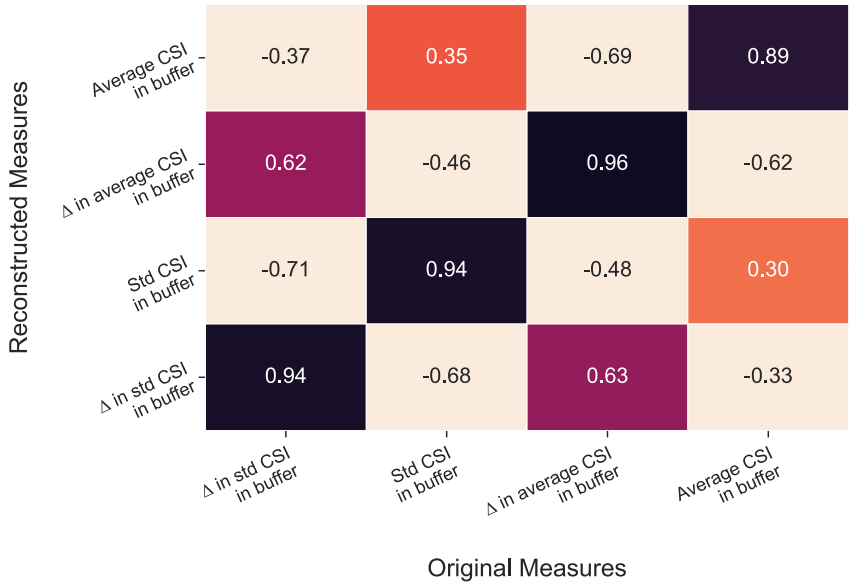

FIGURE B1 Correlation between original and reconstructed independent variables—Ethnologue v17. The figure presents the correlation matrix between original and reconstructed independent variables for different measures of agricultural productivity based on CSI data within buffer zones. Each cell reports the correlation coefficient between the corresponding reconstructed and original variable based on different versions of the Ethnologue (v16 and v17). Variables include the average CSI, change in average CSI, standard deviation of CSI, and change in standard deviation of CSI within the buffer. High correlations (close to 1 or −1) indicate a strong similarity between the original and reconstructed measures, while low correlations suggest distinct patterns in their behavior.

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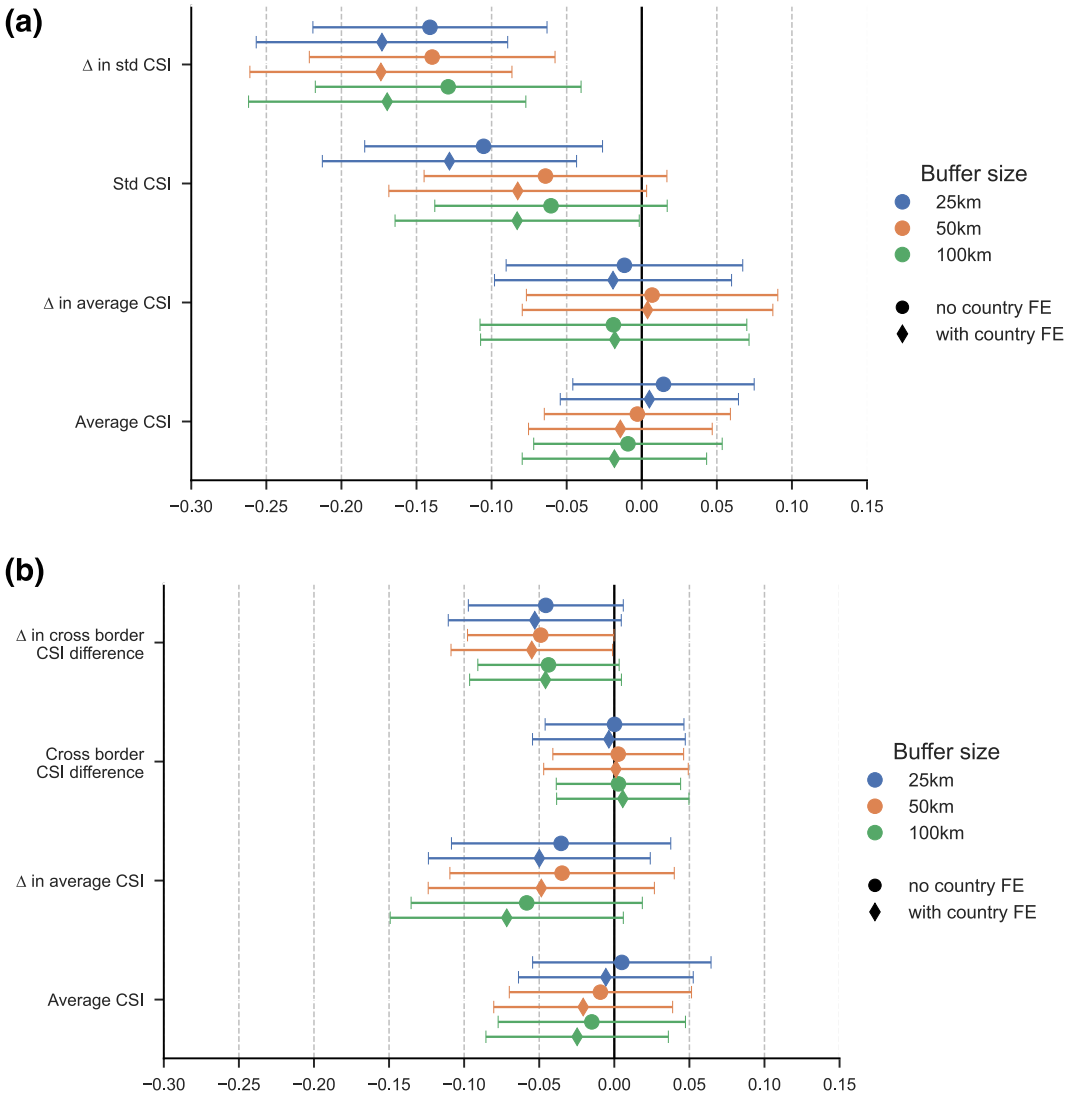

FIGURE B2 Replication and extension of main result—robustness to buffer sizes. (a) Original measure. (b) Alternative measure. The figure examines the robustness of the main results from Table 2 to variations in buffer sizes and the inclusion of country fixed effects. Panel (a) displays the OLS coefficients for the original measure, corresponding to the specification in column (4) of Table 2, while Panel (b) shows the coefficients for the alternative measure, replicating column (6) of Table 2. Different colors represent buffer sizes of 25 km, 50 km, and 100 km. Circles denote estimates without country fixed effects, and diamonds indicate estimates with country fixed effects. Horizontal bars show the 95% confidence intervals.

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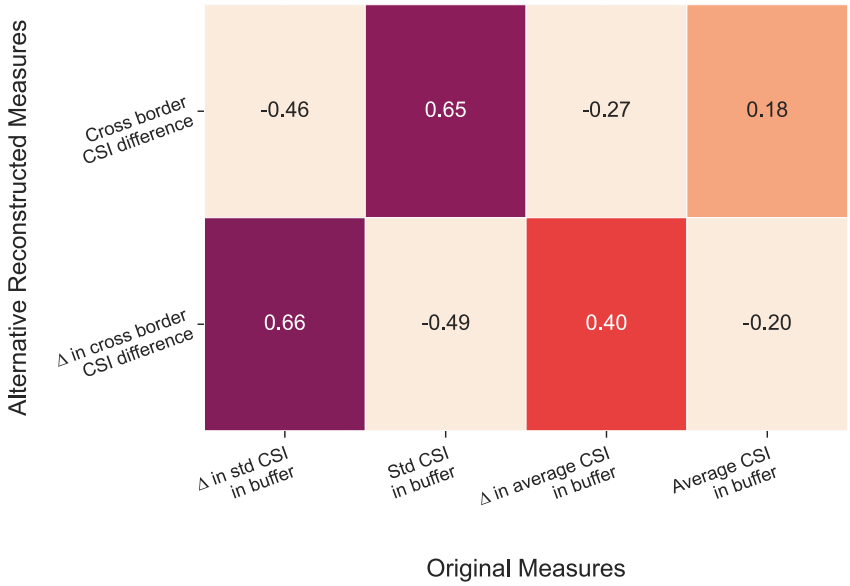

FIGURE B3 Correlation between original and alternative measures of potential gains from inter-ethnic trade—based on Ethnologue v17. The figure presents the correlation matrix between original and alternative measures of potential gains from inter-ethnic trade based on Ethnologue (v16 and v17) data. Each cell shows the correlation coefficient between the corresponding original and alternative measure. Variables include cross-border CSI differences, changes in cross-border CSI differences, as well as within-buffer average CSI, changes in average CSI, standard deviation of CSI, and changes in standard deviation of CSI. High correlations indicate similarity between the original and alternative measures, while low correlations reflect significant differences.

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Advances

Javier Gonzalez and Francisco Parro*

Aggregate Costs of a Gender Gap in the Access to Business Resources

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Abstract: We quantify the aggregate costs of a discriminatory restriction against women in the access to business resources. To do so, we develop a general equilibrium model with an endogenous size distribution of production units, which are run by either female or male entrepreneurs. In this setting, we introduce a distortion that limits the amount of capital that women can use to run their businesses. We calibrate the model to match data from benchmark economies that exhibit relatively egalitarian labor market results between women and men, except in entrepreneurship. Our counterfactual analyses show that a gender-specific capital constraint causes an output loss between 14% and 28% and a fall in aggregate productivity between 12% and 20%. Furthermore, we show that most of the output loss is accounted for by a fall in total factor productivity. Lastly, we show that the aggregate cost of the distortion is mainly triggered by preventing the most skilled women from running bigger businesses, and not the exit of women from entrepreneurship.

Keywords: gender gap, entrepreneurship, aggregate productivity

JEL Classification: E2, J21, J24, O40

1 Introduction

The empowerment of women in many domains of life is one of the most important transformations in the economic and social worlds in recent decades.[1] However,

1 Female labor force participation has notably increased in several regions of the world (World Bank WDI dataset), women are equally or more educated than men worldwide (Parro 2012), and the gender wage gap has also shrunk during past decades (Blau and Kahn 2017).

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a significant gender gap in entrepreneurship still persists. In OECD countries, the proportion of sole-proprietor enterprises owned by women is only between 20% and 40% (OECD 2012). Furthermore, less than 25% of businesses across the world are owned by women according to the Enterprise Surveys elaborated by the World Bank. Discrimination in the access to business resources might be blocking female participation in entrepreneurial activities, as it is implicitly suggested by the literature that studies entrepreneurship in the presence of financial frictions (Buera, Moll, and Shin 2015).

Indeed, descriptive evidence shows that women seem to have access to fewer resources than men to run their businesses (Aidis et al. 2007; OECD 2012, 2016; among others). Some studies also conclude that women entrepreneurs face worse credit conditions than men even when controlling for the firm and entrepreneur characteristics (Agier and Szafarz 2013; Aristei and Gallo 2016; Morazzoni and Sy 2021; Muravyev, Schaefer, and Talavera 2009; among others). Although other supply and demand factors could be blocking the participation of women in business activities, discrimination in access to business resources is definitely a barrier to female entrepreneurship. Furthermore, this type of distortion might be preventing an efficient allocation of talent, with aggregate consequences for the economy as suggested by Hsieh et al. (2019). However, quantifying the aggregate effects of gender barriers in entrepreneurship has received little attention in the macroeconomic literature. Our paper provides a tractable starting point by analyzing the aggregate costs of a gender-specific distortion in the access to business resources.

We model a distortion that limits the amount of capital that female entrepreneurs can employ to run their businesses. The latter distortion is implicitly motivated by the evidence suggesting the existence of a gender gap in loan size and a glass ceiling effect such that the largest female projects face the highest penalty (e.g. Agier and Szafarz 2013). We insert this type of distortion into a model with heterogeneous agents and an endogenous size distribution of production units. In the model, there is a single representative household composed by female and male members who are endowed with heterogeneous managerial skills. Agents must decide whether to run a business or be employed as a production worker in someone else's firm. Production is carried out with a span-of-control technology that uses managerial skills, capital, female labor, and male labor. Hence, more skilled managers run bigger businesses.[2] However, women can employ up to a given amount of capital to start their businesses. As production workers, agents are identical, although women and men are imperfect

2 Hereafter, we will use the terms "entrepreneurs" and "managers" interchangeably.

substitutes in production. Therefore, women and men are otherwise alike, except by their imperfect substitutability in the labor market as production workers, and the capital limit that women face to run their businesses.

We calibrate the model to match data of a *benchmark economy* that exhibits no labor market gender gaps, except regarding entrepreneurship. We choose Denmark, Norway, and Sweden to build the latter benchmark. Hence, we refer hereafter to the average data of Denmark, Norway, and Sweden as the data of the benchmark economy. As observed in Table 1, in the benchmark economy, the gender gap in education is practically null, the ratio of female-to-male labor force participation is close to one, the gender wage gap is tiny; however, a significant gender gap in entrepreneurship is still observed. Specifically, Table 1 shows that only about a third of individuals running a business are women in the benchmark economy. We calibrate the technology parameters of the model to match data on the size distribution of establishments and employment shares by large establishments in the benchmark economy. The gender-specific friction is calibrated to match the observed gender gap in entrepreneurship.

We use the calibrated model to perform counterfactual exercises that quantify the effect of the distortion on aggregate outcomes. Specifically, we build five counterfactual scenarios which gradually and progressively loosen the capital limit for female entrepreneurs. We compare different outcomes between the counterfactual

Table 1: Gender gaps in the benchmark economy.

	Denmark	Norway	Sweden	Average
Gender gap in education	1.5%	0.2%	−1.8%	−0.03%
Female-to-male LFP	0.93	0.94	0.96	0.94
Gender wage gap	0.05	0.05	0.08	0.06
Share of female entrepreneurs (avg.)				0.31
OECD	0.32	0.33	0.29	0.31
ILO	0.30	0.30	0.27	0.29
GEM	0.35*	0.35	0.31	0.34

Sources: Barro-Lee Educational Attainment Dataset, OECD Gender Equality Dataset (OECD), International Labor Organization Statistics (ILO), and Global Entrepreneurship Monitor (GEM). Note: Gender gap in education is the difference between men-women in the percentage of 30–34 years olds who attained tertiary education in 2010; Female-to-male LFP is the ratio between female and male labor force participation in the population 15–64 years old in 2019; Gender wage gap is defined as the difference between male and female median wages divided by the male median wages in 2019; Share of female entrepreneurs is defined as the proportion of women in total self-employment (with or without employees). The Global Entrepreneurship Monitor reports the share of female entrepreneurs in early-stage entrepreneurship. *Latest data available for 2014.

scenarios and the benchmark economy. Then, our quantitative analysis responds to the following question: How costly is a capital limit on women businesses that generates a gender gap in entrepreneurship that is similar to that observed in the benchmark economy?

We find that the gender-specific capital constraint causes an output loss between 13.7% and 28.2%.[3] Furthermore, the estimated losses in average output per efficiency unit of labor ranges from 18.2% to 33.7%, in average output per production worker ranges from 15.1% to 27.3%, and in average output per manager ranges from 24.5% to 49.3%. We also find that the distortion causes a fall in total factor productivity (TFP) between 12.2% and 19.5%. In addition, we quantify the contribution of capital intensity and TFP to the fall in output per worker. We find that the fall in productivity accounts for 65%–80% of the output per worker loss. We also show that the potential aggregate gains of releasing the distortion studied in this paper would be mostly driven by the more intensive capital utilization of existing female entrepreneurs (intensive margin) and not by the entry of more women into entrepreneurship (extensive margin).

Several articles have studied the relationship between gender inequality and economic performance (Doepke and Tertilt 2009; Esteve-Volart 2009; Fernandez 2009; Galor and Weil 1996; Greenwood, Seshadri, and Yorukoglu 2005; Lagerlof 2003; Ngai and Petrongolo 2017; among others).[4] However, few papers quantify the macroeconomic effects of gender gaps in the labor market (Cavalcanti and Tavares 2016; Cuberes and Teigner, 2016; Cuberes and Teigner, 2018; Hsieh et al. 2019). Our goal in this paper is to contribute to this literature by analyzing the macroeconomic effects of a specific distortion that directly impacts the gender gap in entrepreneurship; namely, women's limited access to business resources. As discussed previously, evidence suggests that this distortion is indeed relevant, although their aggregate costs have not been quantified thus far.

The rest of the paper is organized as follows. Section 2 discusses evidence of discrimination against women in the access to business resources. Section 3 develops the theoretical framework. Section 4 describes the calibration strategy. Section 5 presents and discusses the results of this paper. Finally, Section 6 concludes.

3 The range of estimates corresponds to the minimum and maximum effect obtained from the five counterfactual scenarios under consideration.

4 See also Cuberes and Teigner (2014) for a critical review.

2 Literature Review

Evidence shows that gender barriers in entrepreneurship still exist. According to the World Bank Business Surveys, less than 25% of businesses worldwide are owned by women. Meunier et al. (2017) analyze data on female and male entrepreneurship from the World Bank Group Entrepreneurship Database and find that less than a third of the new owners of limited liability companies are women in most of the economies analyzed. Similarly, the OECD (2016) reports that women in the European Union were half as likely as men to be self-employed and women are also less likely than men to participate in business start-ups. Piacentini (2013) shows that, across the majority of OECD countries, the share of women-owned individual enterprises does not exceed 30%. The Global Entrepreneurship Monitor (GEM 2021) also concludes that men are more likely to start new businesses than women. In addition, the 2016 Report of the United Nations Secretary-General's High-Level Panel on Women's Economic Empowerment provides strong evidence that women lag behind men in terms of number of business owners, size of women-owned businesses, and their access to economic resources; specifically, women-owned businesses are smaller and disadvantaged in their access to credit, resources, and assets (UN 2016). Morazzoni and Sy (2021), using data from the U.S. Census for 2018, report that women entrepreneurs make up just over 35% of the entrepreneurial pool in the U.S. Consequently, Morazzoni and Sy conclude that gender gaps in labor market participation are more severe for employers than for salaried workers.

Non-causal evidence suggests that women face worse conditions than men to access business resources, which raises a barrier to female entrepreneurship. The OECD (2012) reports that, in the United States, 60% of female entrepreneurs start with resources less than US$5000, compared to 42% of male entrepreneurs. The same report shows that 39% of U.S. women entrepreneurs started their businesses with credit from a bank compared to 47% of male entrepreneurs; European countries exhibit similar results. OECD data also reveal that women entrepreneurs have less access to business financing than men; for example, only 36% of Danish and 41% of Swedish women business owners report having access to the resources necessary to grow their businesses, compared to 46% of Danish and 50% of Swedish male business owners (OECD 2016). The IFC of the World Bank has estimated that, worldwide, there is a financing gap of US$300 billion for formal small businesses owned by women, and more than 70% of small and medium enterprises owned by women have inadequate or no access to financial services. Along the same lines, Morazzoni and Sy (2021) report that, in 2018, women received only 2.2% of total U.S. venture capital financing. In addition, in developing countries, access to finance is commonly noted as an important barrier to business success. For

example, Aidis et al. (2007), using data from Lithuania and Ukraine, show that access to funds was a more important barrier to advancement for women business owners than for men.

Therefore, non-causal evidence is relatively robust about a fact: on average, women seem to start their businesses with less resources than men. However, the latter is an equilibrium result that entails both supply and demand effects of a diverse nature. In this regard, some studies, mainly outside the U.S., provide causal evidence suggesting the existence of discrimination in access to business resources, a barrier to female participation in business activities. We now briefly discuss some of those studies.

Morazzoni and Sy (2021), using credit data comprising nearly 5000 U.S. entrepreneurs, document evidence of gender differences in business access to credit. Specifically, after controlling for individual and firm characteristics, the authors find that, compared to men, women entrepreneurs raise 35% less business debt[5] and are 10% more likely to be rejected when applying for a bank loan. Furthermore, the authors conclude that the higher rejection rates cannot be empirically explained by higher risk from women-led businesses or lower profitability from these businesses. Therefore, Morazzoni and Sy (2021) provide empirical evidence that credit restrictions appear to penalize women entrepreneurs relatively more; not only do women entrepreneurs report lower levels of business debt than men, but among those who apply for a loan, women are also more likely to be rejected. The authors also develop a model of heterogeneous agents, business choice, and gender-specific financial frictions; relying on a counterfactual analysis, they show that eliminating gender differences in financing leads to a 16% increase in female entrepreneurship rates.

Agier and Szafarz (2013) use a database that includes 34,000 loan applications from a Brazilian microfinance institution over an eleven-year period to study whether male and female borrowers benefit from the same credit conditions.[6] The authors then compare the attribution of loans to male and female applicants with similar creditworthiness characteristics. The authors find a gender gap in loan size; that is, in similar circumstances, the size of the loan is biased in favor of

5 It is the sum of business bank loans, lines of credit, loans from non-financial institutions, business credit card balance, and business loans from various other sources, such as family, employees, federal agencies, etc.

6 The data come from Vivacred, a non-profit microcredit institution operating in Rio de Janeiro's *favelas* that provides credit to tiny and small firms located in Rio's low-income communities and neighborhoods. The typical loan proposed by Vivacred has a fixed monthly interest rate and an additional one-time registration fee depending on the credit duration and the client's repayment history.

male clients and therefore women face harsher loan conditions than men. The authors also confirm the presence of a glass ceiling effect, that is, the largest female projects face the highest penalty.

Aristei and Gallo (2016) provides empirical evidence on gender differences in access to credit for firms in European transition economies. The authors find that the probability of credit rationing against female firms is not explained by observed characteristics of firms. The authors interpret the latter result as evidence of gender discrimination in credit markets.

Using a large sample of small and medium-sized European companies, Moro, Wisniewski, and Mantovani (2017) find that businesses run by women are less likely to apply for a loan, as they anticipate being rejected.[7] As a consequence, companies run by women obtain less bank financing. However, the authors cannot rule out that women exclude themselves due to their perception that banks will discriminate against their gender. In this regard, Bellucci, Borisov, and Zazzaro (2010) find that women entrepreneurs in Italy are more likely to be denied credit and therefore may be more discouraged from applying for a loan.[8]

Muravyev, Schaefer, and Talavera (2009) use the Business Environment and Business Performance Survey to study gender discrimination in small business lending in 26 Central and Eastern European transition economies and several Western European industrialized nations. The authors provide evidence that women-owned businesses are less likely to obtain a bank loan. Cavalluzzo, Cavalluzzo, and Wolken (2002) also find evidence of a credit access gap between firms owned by white males and white females in the U.S., with female denial rates increasing with lender concentration.

Overall, the evidence shows that women do face worse credit conditions than men. Causal evidence supports the conclusion that gender discrimination is a factor that could explain a gender gap in access to credit, although other demand-driven explanations cannot be ruled out. From a quantitative point of view, the aggregate impact of gender barriers in entrepreneurship has received little attention in the macroeconomic literature. Our paper contributes in that direction

7 Data were extracted from the Survey of Access to Finance of Enterprises (SAFE) conducted on behalf of the European Commission and the European Central Bank which collects information about access to finance by enterprises within the European Union. The authors integrate these data with information from the quarterly Bank Loan Survey (BLS) conducted by national central banks on behalf of the European Central Bank. The BLS includes information on the lending activity of banks and the propensity of banks to make loans.

8 The data comprise more than 7800 credit lines made available to individually-owned small businesses (i.e. sole proprietorships) by a major Italian bank belonging to a large Italian banking group quoted on the Milan Stock Exchange.

by providing a tractable model to study the impact on aggregate outcomes of removing a gender-specific capital distortion.

3 Model

We build a one-sector aggregative model in which production is carried out by heterogeneous establishments run by two types of agents: women and men. Each type of agent can also work as a production worker in the establishment run by someone else. This benchmark model is exposed to a supply-driven distortion to female entrepreneurship.

3.1 Preferences

There is a single infinitely lived representative household in the economy. The household comprises at time t a continuum of members of size L_t. The size of the household (population) grows at the constant rate x_L.[9] The members of the household are of two types: women (f) and men (m). We denote by θ_i the fraction of the population of type-i, for $i \in \{f, m\}$. The household preferences are described by a time-additively separable utility function over sequences of per capita consumption, c_t. The representative household gets flow utility from per capita consumption and maximizes:

$$U(\{c_0, \dots, c_\infty\}) = \sum_{t=0}^{\infty} \beta^t L_t \ln c_t, \tag{1}$$

where $\beta \in (0, 1)$ is the discount factor for the future.

3.2 Endowments

The household is endowed with a positive (aggregate) stock of capital at date $t = 0$, that is, $K_0 > 0$. In addition, each household member is endowed with z units of managerial skills. We denote by $g(z)$ the density function of skills, with c.d.f. $G(z)$, and support in $Z = [z^l, z^h]$. The distribution of skills is identical for women and men. Household members are also endowed by one unit of time which they inelastically supply in the labor market as either managers or production workers. We describe later the agents' occupational choice and the associated income in detail.

9 We introduce population growth so the model has standard balanced-growth properties, and thus can be better mapped to data.

3.3 Production Technology

Each establishment – a production unit – produces a homogeneous output. Production is carried out using labor (n), capital (k), and managerial skills (z). Specifically, women's labor, n^f, and men's labor, n^m, are combined in a CES function with an elasticity of substitution given by σ:[10]

$$n_i = \left(\left(n_i^f\right)^{\frac{\sigma-1}{\sigma}} + \left(n_i^m\right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \tag{2}$$

where n_i^j is the quantity of labor type-j hired by manager i.

This nested CES function n that aggregates women's and men's labor is then combined with capital using a Cobb–Douglas aggregator:

$$q_i = k_i^{\alpha} n_i^{1-\alpha}, \tag{3}$$

where $\alpha \in (0,1)$. Output q is combined with managerial skills using a Cobb–Douglas aggregator:

$$y_i(z) = A z^{1-\zeta} q_i^{\zeta}. \tag{4}$$

where the term A is common to all production units, and accounts for exogenous productivity growth at a constant rate x_A. The parameter $\zeta \in (0,1)$ governs returns to scale in variable factors at the establishment level, that is, it is the span-of-control parameter.

3.4 The Problem of a Type-z Manager

A manager with skills z maximizes profits by taking wages and the rental price for capital services as given.[11] The static maximization problem of a type-i agent with skills z is

10 Hereafter, we include a time index t when it is strictly necessary for a clear exposition.

11 The separability that any type of fixed cost would exhibit in the profit function implies that the sum of that cost should not have a significant impact on a baseline-counterfactual comparison. Generally speaking, a fixed cost moves the profit function down both in the frictional scenario (baseline) and in the frictionless scenario (counterfactual). Furthermore, it should be innocuous for the change in the slope of the profit function as a consequence of the capital distortion. Then, the inclusion of an additively separable fixed cost in the profit function only causes equilibrium wages to adjust downward. Therefore, we expect little impact on the main results if this additional parameter were added to the model.

$$\max_{k_i, n_i^f, n_i^m} \left[y_i(z) - w^f n_i^f - w^m n_i^m - R k_i \right], \tag{5}$$

where R is the rental price for capital services and w^j is the wage of labor type-j for $j \in \{f, m\}$.

The first order conditions of the maximization problem are:

$$[k_i]: \zeta \alpha \frac{y_i(z)}{k_i(z)} = R \tag{6}$$

$$\left[n_i^f\right]: \zeta(1-\alpha) \left(\frac{y_i(z)}{n_i(z)}\right) \left(\frac{n_i(z)}{n_i^f(z)}\right)^{\frac{1}{\sigma}} = w^f \tag{7}$$

$$\left[n_i^m\right]: \zeta(1-\alpha) \left(\frac{y_i(z)}{n_i(z)}\right) \left(\frac{n_i(z)}{n_i^m(z)}\right)^{\frac{1}{\sigma}} = w^m \tag{8}$$

We use Eqs. (6)–(8) to compute the relative demand of inputs for a type-i manager with skills z. Appendix A provides a formal derivation of these demands.

3.5 Occupational Choices

Agents must decide in each period whether to operate a single productive unit or work for a wage. We denote by $M = \{w^f, w^m, R\}$ the market prices of labor and capital inputs. Let $\mathcal{I}_i(z; M)$ be an indicator that describes the optimal occupation choice for an agent with abilities z facing market prices M. Specifically, we set $\mathcal{I}_i(z; M) = 1$ if the agent optimally decides to operate a business and $\mathcal{I}_i(z; M) = 0$ if the agent decides to work for a wage. An individual that decides to become a manager must receive in equilibrium a compensation higher than the wage earned by a production worker. Denote by $\pi_i(z;\ M)$ the profits for a type-i manager with skills z. Then,

$$\mathcal{I}_i(z; M) = \begin{cases} 1 & \text{if } \pi_i(z; M) > w^i \\ 0 & \text{if } \pi_i(z; M) \leq w^i, \end{cases} \tag{9}$$

and thus, labor market earnings of an agent i, can be expressed as:

$$e_i(z; M) = \mathcal{I}_i(z; M)\pi_i(z; M) + (1 - \mathcal{I}_i(z; M))w^i \tag{10}$$

Therefore, the income per capita generated by members of type i is

$$v_i(M) = \int_{z \in Z} e_i(z; M) g(z) \mathrm{d}z, \tag{11}$$

3.6 The Household Problem

The representative household must choose the sequence of per capita consumption, the sequence of aggregate capital to carry over to the next period, and the occupation of its members, taking all prices as given:

$$\begin{aligned} &\max_{\{c_t, K_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t L_t \ln c_t \\ &\text{s.t. } C_t + K_{t+1} = \sum_{i \in \{f,m\}} L_{i,t} v_{i,t}(M_t) + R_t K_t + (1-\delta) K_t, \\ &K_0 > 0, \quad L_0 > 0, \end{aligned} \tag{12}$$

where C_t is aggregate consumption at date t, that is, $C_t = L_t c_t$, δ is the depreciation rate of capital, and the occupational choice is described by (9). The first order conditions for per capita consumption and aggregate capital produces the standard Euler equation for capital accumulation:

$$\frac{c_{t+1}}{c_t} = \beta(R_{t+1} + 1 - \delta) \quad \text{for all } t \geq 0. \tag{13}$$

Notice that Eq. (9) satisfies the first order condition for the occupation choice of agents.

3.7 Market Clearing

We derive now the market clearing conditions of the model. The market clearing condition for capital requires that the aggregate capital that the representative household optimally carries over to the next periods equals the aggregate demand for capital by the productive units:

$$K_t(M_t) = L_t \sum_{i \in \{f,m\}} \theta_i \int_{z \in Z} k_{i,t}(z; M_t) \mathcal{I}_{i,t}(z; M_t) g(z) \mathrm{d}z. \tag{14}$$

The market clearing condition for labor market services must equal the supply and demand of production labor. The supply of each type of labor service is determined by the occupational decisions of women and men. Demand is determined by the number of agents of each type that decide to run

an establishment and their demands for each type of production labor (women and men). Then,

$$L_t\theta_j \int_{z\in Z} (1-\mathcal{I}_{j,t}(z;M_t))g(z)\mathrm{d}z = L_t \sum_{i\in\{f,m\}} \theta_i \int_{z\in Z} n^j_{i,t}(z;M_t)\mathcal{I}_{i,t}(z;M_t)g(z)\mathrm{d}z, \tag{15}$$

for $j \in \{f,m\}$. Finally, the market clearing condition for the unique good produced in this economy is given by:

$$L_t \sum_{i\in\{f,m\}} \theta_i \int_{z\in Z} y_{i,t}(z;M_t)\mathcal{I}_{i,t}(z;M_t)g(z)\mathrm{d}z = C_t(M_t) + K_{t+1}(M_t) - (1-\delta)K_t(M_t). \tag{16}$$

The left-hand side of Eq. (16) is aggregate income, whereas the right-hand side is aggregate consumption plus aggregate investment.

3.8 Equilibrium

A competitive equilibrium is a set of allocations $\{c_t, K_{t+1}, \{\mathcal{I}_{i,t}(z)\}_{z\in Z, i\in\{f,m\}}\}_{t=0}^{\infty}$ for the representative household, and a set of prices $\left\{w_t^f, w_t^m, R_t\right\}_{t=0}^{\infty}$ such that (i) the household problem (12) is solved by taking all the prices as given, and subject to the constraint $k_{f,t} \leq \bar{k}$ for women and for all t,[12] (ii) the market for capital services clears for all t (Eq. (14) holds), (iii) the market for labor services clears for all t (Eq. (15) holds), and (iv) the market for goods clears for all t (Eq. (16) holds).

3.9 Capital Constraint for Female Business Owners

In the model economy, women face an exogenous constraint in their access to business resources. The distortion takes the form of a ceiling $\bar{k}$ on the capital amount that women can rent to run their businesses. We now discuss how this distortion impacts female entrepreneurship.

Consider first the partial equilibrium. As a direct consequence of the span-of-control production technology, more skilled women run bigger businesses, and thus, rent larger amounts of capital. Hence, the capital limit $\bar{k}$ is only binding for women with the highest managerial skills. Constrained women face a reduction in their profits as a consequence of the distortion. In contrast, the distortion is innocuous for unconstrained women. Therefore, two cases arise, which are illustrated in panels (a) and (b) of Figure 1.[13] First, suppose low equilibrium wages

12 We discuss this constraint in detail in the next subsection.

13 Figures 1–3 are drawn only to illustrate the effects and are not intended to show what the actual data looks like.

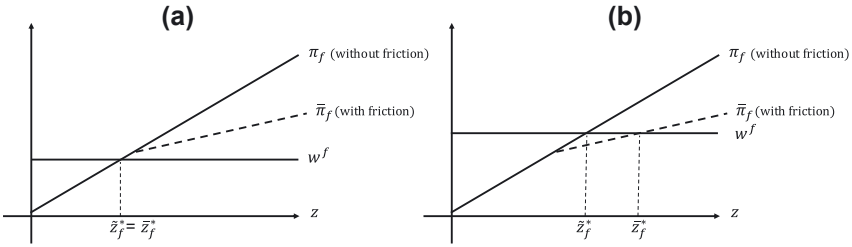

Figure 1: Partial equilibrium analysis.

(panel a). Then, the marginal women entrepreneur is relatively low-skilled, and thus, the capital limit is not binding for her. Hence, the friction is innocuous on the equilibrium number of entrepreneurs, although it certainly impacts the scale of operation of the most productive (constrained) entrepreneurs. Then, the skill threshold above which women start a business in the frictional economy, $\bar{z}_{\mathrm{f}}^*$, is the same as in the frictionless economy, $\tilde{z}_{\mathrm{f}}^*$.[14] Suppose now high equilibrium wages (panel b). In this case, the friction $\bar{k}$ constrains the amount of capital available for the marginal women entrepreneurs, who then experience a fall in their profits (dashed line). Therefore, some female entrepreneurs do not have more incentives to run a business, and thus, they become production workers. That is, $\tilde{z}_{\mathrm{f}}^* < \bar{z}_{\mathrm{f}}^*$, and thus, the number of female entrepreneurs falls. In this partial equilibrium analysis, men's choices are not impacted by the distortion.

We now analyze the general equilibrium. Let us start from the scenario depicted in panel (a) of Figure 1; that is, the case in which the capital limit is not binding for the marginal women entrepreneurs. Thus, the distortion does not exert a direct effect on the number of entrepreneurs. This is the partial equilibrium effect analyzed previously. However, existing female entrepreneurs are constrained in their use of capital and must run smaller businesses as a consequence of the distortion. Those firms demand less production labor which pushes wages down.[15] Hence, profits go up and the choice of running a business becomes more attractive for both women and men. Formally, the threshold managerial skill

14 Throughout this paper, we use the following notation: z^* is the skill size of the marginal entrepreneur; $\bar{z}^*$ is the skill threshold in the baseline (frictional) economy; and, $\tilde{z}^*$ is the skill threshold in the counterfactual (frictionless) economy.

15 The fact that the standard Euler equation for capital accumulation applies in this model implies that the rental rate for capital services is constant across steady states.

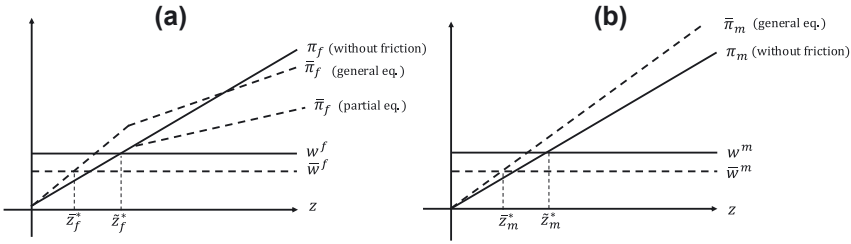

Figure 2: General equilibrium analysis I.

falls; that is $\tilde{z}_{\mathrm{f}}^* > \bar{z}_{\mathrm{f}}^*$ and $\tilde{z}_{\mathrm{m}}^* > \bar{z}_{\mathrm{m}}^*$. Therefore, both female and male entrepreneurship rises. Figure 2 illustrates these first round effects for women (panel a) and men (panel b), where we denote by $\overline{w}$ the frictional wage.

Let us start now from the situation depicted in panel (b) of Figure 1. In this case, the direct effect of the capital limit is a fall in the number of female entrepreneurs, the partial equilibrium effect. Furthermore, existing female entrepreneurs are constrained by the distortion, run smaller businesses, and thus, demand less production labor. Hence, supply of female production labor rises and demand falls which unambiguously pushes wages down and profits up. For women, this is a second round effect, and thus, female entrepreneurship falls, although in a lesser magnitude compared to the partial equilibrium depicted in panel (b) of Figure 1. In the case of men, the fall in wages and rise in profits increases the number of entrepreneurs. Therefore, as first order effects, female entrepreneurship falls, whereas the opposite occurs for men. Figure 3 illustrates this case.

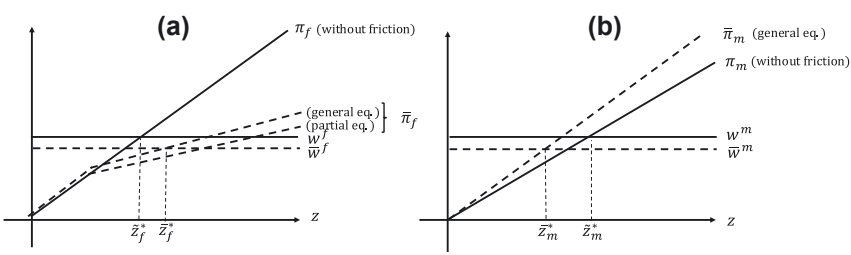

Figure 3: General equilibrium analysis II.

4 Calibration

We calibrate the model to match observations of a benchmark economy, which we build using the average data for Denmark, Norway, and Sweden. As discussed in the introductory section, these economies are close to gender equality in the labor market, except regarding entrepreneurship.[16] We now explain the specific data used to calibrate the model.

We choose a model period of a year. Along the balanced-growth path, consumption per capita and output per capita grow at a common rate x_c. To compute x_c, we consider the average annual growth rate of real GDP per capita in the benchmark economy during the period 1971–2019. Using data from the World Bank WDI dataset, we get $x_c = 0.018$. The U.S. Bureau of Economic Analysis (BEA) provides estimates of depreciation rates for different types of assets. This technological parameter should not be very different between the U.S. economy and our benchmark economy, so we take BEA data to calibrate this parameter. Specifically, we assume that the stock of capital is comprised of private nonresidential equipment and structures. We consider the simple average of the depreciation rates reported by the BEA for these types of assets to calibrate $\delta = 0.0978$. In addition, in the model, $\zeta\alpha$ equals the share of capital. As we will explain later, we follow Guner, Ventura, and Xu (2008) to calibrate ζ. Then, we pick α such that $\zeta\alpha$ equals the capital income share of 30% observed in the U.S. data. The elasticity of substitution between women and men, σ, is calibrated to match the observed gender wage gap in the benchmark economy. Labor force participation is assumed to be egalitarian across gender in order to let the elasticity of substitution σ capture the whole gender wage gap that is observed in the benchmark economy. The discount factor β takes the standard value of 0.95 reported in the literature.

We now explain how we calibrate the span-of-control parameter, ζ, and the distribution of managerial skills. The span-of-control parameter, ζ, and the parameters governing the distribution of managerial abilities $g(z)$ determine the size distribution of establishments, that is, the mean establishment size, the range of employment levels, as well as the share of total labor employed by establishments of different sizes. Following Guner, Ventura, and Xu (2008), we calibrate $g(z)$ and ζ to match data on the fraction of establishments at different employment levels, and the share of total employment accounted for by large establishments. We use data from Eurostat Structural Business Statistics to compute the following targets: (i) mean establishment size, (ii) the fraction of establishments with 0–9

16 Recall that we refer to the average data of Denmark, Norway, and Sweden as the data of the benchmark economy.

employees, 10–19 employees, 20–49 employees, 50 or more employees, and (iii) the share of total employment accounted for by large establishments, defined as those with 50 or more employees. We then select ζ and $g(z)$ to match these statistics. As in Guner, Ventura, and Xu (2008), we assume that log-managerial ability is distributed according to a (truncated) normal distribution with mean m and variance s^2, and we impose that this distribution accounts for the bulk of production units, with a total mass of $1 - \hat{g}$. To account for the remainder of the distribution of establishments, we select a top value for managerial skill, $\hat{z}$ and its corresponding fraction, $\hat{g}$. Hereafter we refer to managers with skills $\hat{z}$ as "superstars." Lastly, the capital limit $\bar{k}$ is calibrated to match the fraction of female entrepreneurs in the benchmark economy.

Hence, there are seven parameters, σ, ζ, m, s^2, $\hat{g}$, $\hat{z}$, and $\bar{k}$ that we choose in order to match eight observations: gender wage gap, mean size, the fraction of establishments corresponding to four different levels of employees, the share of employment accounted for by establishments with 50 or more employees, and the fraction of female entrepreneurs. Table 2 summarizes our choices and Table 3 shows the performance of the model in terms of our targets and two additional untargeted moments.[17]

Table 2: Calibrated parameters.

Parameter	Value	Explanation
Productivity (A)	1	Normalization
Labor force participation ($\theta_f = \theta_m$)	0.5	Normalization based on data
Discount factor (β)	0.95	Extracted from literature
Output per capita growth (x_c)	1.8%	Extracted from data
Depreciation rate (δ)	0.0978	Extracted from data
Share of capital ($\zeta\alpha$)	0.3	Extracted from data
CES elasticity (σ)	1.1357	To match data
Span-of-control (ζ)	0.6219	To match data
Mean log-entrepreneurship skills (μ)	−0.3989	To match data
Variance of log-entrepreneurship skills (s^2)	4.9929	To match data
Highest entrepreneurship ability ($\hat{z}$)	2944	To match data
Mass of highest entrepreneurship ability ($\hat{g}$)	0.0022	To match data
Capital limit ($\bar{k}$)	5.2860	To match data

17 In all simulations we approximate $g(z)$ by a discrete distribution with 5000 grid points. Appendix B explains how we solve the model.

Table 3: Performance of the model.

Statistic	Data	Model
Gender wage gap	0.06	0.06
Fraction of female entrepreneurs	0.31	0.33
Mean firm size	5.90	5.94
***Fraction of establishments at*:**		
0–9 employees	0.92	0.91
10–19 employees	0.043	0.049
20–49 employees	0.025	0.028
50+ employees	0.014	0.016
***Share of employment at*:**		
50+ employees	0.54	0.56
***Untargeted observations*:**		
Total fraction of entrepreneurs	0.09	0.11
Relative proportion of female/male entrepreneurs	0.45	0.49

5 Results

In this section we estimate the aggregate costs of the distortion described in Section 3.9. To do so, we build five counterfactual scenarios which gradually and progressively loosen the capital limit for female entrepreneurs. The first scenario reduces by 10% the gap between $\bar{k}$ and the maximum amount of capital used by non-superstar female entrepreneurs in a frictionless economy (S1).[18] In the second and third scenarios, the latter gap is shortened by 20% (S2) and 50% (S3), respectively. In a fourth counterfactual scenario, all women are unconstrained, except the superstar managers (S4). Lastly, we consider a counterfactual economy where no woman is subject to a capital limit to run her business (S5). Then, we compare different outcomes in the baseline scenario to those in each of the five counterfactual economies. We consider three outcomes: aggregate output, output per worker, and total factor productivity (TFP).

18 The demand of capital by superstars is several times bigger than the amount demanded by the most skilled women in the rest of the distribution. Thus, considering a reduction of 10% in the gap between $\bar{k}$ and the maximum amount of capital used by all women (that is, including superstars) would produce an abrupt rise in the capital limit, which prevents the gradualness that the counterfactual exercise seeks to achieve.

Aggregate output, Y, is the total production carried out by both female and male entrepreneurs:[19]

$$Y = \theta_{\mathrm{f}} \int_{z_{\mathrm{f}}^*} y_{\mathrm{f}}(z) g(z) \mathrm{d}z + \theta_{\mathrm{m}} \int_{z_{\mathrm{m}}^*} y_{\mathrm{m}}(z) g(z) \mathrm{d}z, \tag{17}$$

where z_i^* is the managerial skill of the marginal entrepreneur among type-i agents.

In this first analysis, we also compute the ratio of the average output produced per female manager in the counterfactual scenario to the output produced in the baseline scenario, r_{f}. This outcome is used to discuss the results for aggregate output losses, and we define it as follows:

$$r_{\mathrm{f}} = \frac{\int_{\tilde{z}_{\mathrm{f}}^*} y_{\mathrm{f}}(z) g(z) \mathrm{d}z / \int_{\tilde{z}_{\mathrm{f}}^*} g(z) \mathrm{d}z}{\int_{\bar{z}_{\mathrm{f}}^*} y_{\mathrm{f}}(z) g(z) \mathrm{d}z / \int_{\bar{z}_{\mathrm{f}}^*} g(z) \mathrm{d}z}, \tag{18}$$

where $\bar{z}_{\mathrm{f}}^*$ and $\tilde{z}_{\mathrm{f}}^*$ are the managerial skill of the marginal female entrepreneur in the baseline and counterfactual scenarios, respectively.

Table 4 presents the results. We observe that the distortion causes a fall in aggregate output between 13.7% and 28.2%, depending on the counterfactual scenario considered. Counterfactual S1 implies that the distortion causes an output loss of 13.7%. In the case of the second and third counterfactuals (S2 and S3), we observe a fall in aggregate output by 16.3% and 19.5%, respectively, which is somewhat higher compared to the first scenario. Next, when comparing the baseline economy with one in which all women are unconstrained, except superstar managers, that is, scenario S4, the distortion causes an output loss of 21.7%. Lastly, output loss reaches 28.2% when the baseline economy is compared with a frictionless economy (S5).

Therefore, our counterfactual analysis suggests significant potential output gains derived from a small loosening in the capital limit for women entrepreneurs.

Table 4: Aggregate output losses in benchmark relative to counterfactuals.

	S1	S2	S3	S4	S5
Aggregate output, Y (%)	13.7	16.3	19.5	21.7	28.2
Ratio output per female manager, r_{f} (level)	4.1	4.8	5.9	6.8	9.9

19 Hereafter, we normalize $L = 1$.

However, potential gains are marginally smaller when the capital limit is further loosened. An exception to the latter occurs when the economy moves from a scenario where only the superstars are constrained to one where no women face a capital limit. The intuition behind the latter is as follows. The distortion forces women to run smaller businesses than what would be the optimal size, causing the marginal product of capital to be well above the rental price of it. Indeed, we observe in the second row of Table 4 that the ratio of the average output produced in establishments run by women in the counterfactual scenario to the output produced in the baseline scenario ranges from 4.1 (S1) to 9.9 (S5). Therefore, a small loosening of the distortion allows women to materialize the high marginal product of capital, but these benefits begin to evaporate when capital use becomes more and more intensive. However, superstar managers are highly skilled compared to the mass of managers, and thus, capital-skill complementarity in production counters somewhat the effect of capital intensity on the marginal product of capital. Hence, the marginal productivity of capital is still significant in businesses run by superstar female managers.[20]

Next, we evaluate three measures of output per worker. Concretely, we build the average output per efficiency unit of labor, q_1, as

$$q_1 = \theta_{\mathrm{f}} \frac{\int_{z_{\mathrm{f}}^*} y_{\mathrm{f}}(z) g(z) \mathrm{d}z}{\int_{z_f^*} \left(n_{\mathrm{f}}^{\mathrm{f}}(z) + n_{\mathrm{f}}^{\mathrm{m}}(z)\right) g(z) \mathrm{d}z + \int_{z_{\mathrm{f}}^*} z g(z) \mathrm{d}z} + \theta_{\mathrm{m}} \frac{\int_{z_{\mathrm{m}}^*} y_{\mathrm{m}}(z) g(z) \mathrm{d}z}{\int_{z_{\mathrm{m}}^*} \left(n_{\mathrm{m}}^{\mathrm{f}}(z) + n_{\mathrm{m}}^{\mathrm{m}}(z)\right) g(z) \mathrm{d}z + \int_{z_{\mathrm{m}}^*} z g(z) \mathrm{d}z}, \tag{19}$$

20 In further analysis, we evaluate how the main results change when perfect substitutability is imposed. To do this, we first modify Eq. (2) for the case of perfect substitutability between female and male labor and solve the model following steps analogous to those developed in Appendix A. Then, we recalculate all the results for this alternative model. Specifically, we use the parameter values that were calibrated for the original model, but we modify the capital thresholds that define the five counterfactual scenarios to make them consistent with those of the main analysis. We found that the effect of the distortion on the aggregate outcomes is somewhat less in all counterfactual scenarios, compared to the original model. However, this alternative model, by construction, will not be able to match the observed gender wage gap. Thus, this additional result should only be interpreted as a sensitivity analysis. The complete set of result tables for this alternative model is available upon request.

the average output per production worker, q_2, as

$$q_2 = \theta_{\mathrm{f}} \frac{\int_{z_{\mathrm{f}}^*} y_{\mathrm{f}}(z) g(z) \mathrm{d}z}{\int_{z_{\mathrm{f}}^*} \left(n_{\mathrm{f}}^{\mathrm{f}}(z) + n_{\mathrm{f}}^{\mathrm{m}}(z)\right) g(z) \mathrm{d}z} + \theta_{\mathrm{m}} \frac{\int_{z_{\mathrm{m}}^*} y_{\mathrm{m}}(z) g(z) \mathrm{d}z}{\int_{z_{\mathrm{m}}^*} \left(n_{\mathrm{m}}^{\mathrm{f}}(z) + n_{\mathrm{m}}^{\mathrm{m}}(z)\right) g(z) \mathrm{d}z}, \tag{20}$$

and the average output per manager, q_3, as

$$q_3 = \theta_{\mathrm{f}} \frac{\int_{z_{\mathrm{f}}^*} y_{\mathrm{f}}(z) g(z) \mathrm{d}z}{\int_{z_{\mathrm{f}}^*} g(z) \mathrm{d}z} + \theta_{\mathrm{m}} \frac{\int_{z_{\mathrm{m}}^*} y_{\mathrm{m}}(z) g(z) \mathrm{d}z}{\int_{z_{\mathrm{m}}^*} g(z) \mathrm{d}z}. \tag{21}$$

Table 5 presents the results using the same counterfactual scenarios as in Table 4. We observe that average output per efficiency unit of labor falls by 18.2% when the distortion considered in S1 moves to the level $\bar{k}$. As concluded for aggregate output, this loss gradually increases when considering counterfactuals that progressively loosen the capital limit for women, reaching 33.7% when the baseline economy is compared with a frictionless one. In the case of average output per production worker the losses range from 15.1% to 27.3%, whereas for average output per manager the losses range from 24.5% to 49.3%.

Our next outcome measures aggregate productivity losses. We calculate this variable in two alternative ways. The first measure (TFP_1) considers TFP as the residual from an aggregate technology under a capital share of $\alpha\zeta$ and labor share of $1 - \alpha\zeta$, with no distinctions between workers and managers in the labor force. Concretely, we calculate

$$\mathrm{TFP}_1 = \frac{Y/L}{(K/L)^{\alpha\zeta}}, \tag{22}$$

where

$$L = \theta_{\mathrm{f}} \int_{z_{\mathrm{f}}^*} n_{\mathrm{f}}(z) g(z) \mathrm{d}z + \theta_{\mathrm{m}} \int_{z_{\mathrm{m}}^*} n_{\mathrm{m}}(z) g(z) \mathrm{d}z, \tag{23}$$

$$K = \theta_{\mathrm{f}} \int_{z_{\mathrm{f}}^*} k_{\mathrm{f}}(z) g(z) \mathrm{d}z + \theta_{\mathrm{m}} \int_{z_{\mathrm{m}}^*} k_{\mathrm{m}}(z) g(z) \mathrm{d}z. \tag{24}$$

Table 5: Average output per worker losses in benchmark relative to counterfactuals.

(%)	S1	S2	S3	S4	S5
Output per efficiency unit of labor, q_1	18.2	21.2	24.6	27.0	33.7
Output per production worker, q_2	15.1	17.3	20.0	21.8	27.3
Output per manager, q_3	24.5	29.5	35.0	38.9	49.3

The second measure (TFP_2) considers a separation between workers and managers by their efficiency units and define aggregate labor as $L + \tilde{L}$, where $\tilde{L}$ is

$$\tilde{L} = \theta_{\text{f}} \int_{z_{\text{f}}^*} zg(z)\text{d}z + \theta_{\text{m}} \int_{z_{\text{m}}^*} zg(z)\text{d}z. \tag{25}$$

In this case, we have

$$\text{TFP}_2 = \frac{Y}{K^{\alpha\zeta}\left(L + \tilde{L}\right)^{1-\alpha\zeta}}. \tag{26}$$

We present in Table 6 the results for TFP. We observe that aggregate TFP falls between 12.2% and 19.5% as a consequence of the distortion. In this case, we observe moderate gains, in terms of TFP, from moving across the five counterfactual scenarios considered in Table 6.

Table 6: TFP losses in benchmark relative to counterfactuals.

(%)	S1	S2	S3	S4	S5
TFP_1	12.3	13.9	15.8	17.2	19.4
TFP_2	12.2	13.8	15.8	17.2	19.5

Overall, this subsection shows that the capital limit $\bar{k}$ for women causes a fall in aggregate output between 13.7% and 28.2%, in average output per efficiency unit of labor between 18.2% and 33.7%, and in TFP between 12.2% and 19.5%. How do these effects compare to those found in the literature for other distortions? In this regard, the literature on the aggregate costs of financial frictions is indeed extensive, so we document here only a few studies that will provide an order of magnitude of the effects found in this paper.

Erosa (2001) analyzes financial frictions that arise from costly intermediation. The author estimates gains in output per capita of 40% from eliminating this type of distortion. Buera, Kaboski, and Shin (2011) analyze aggregate cost from financial frictions that take the form of imperfect enforceability contracts. The authors find that the aggregate TFP of the country with the least financial development would be almost 40% below the U.S. level. Greenwood, Sanchez, and Wang (2013) examine what they call the capital deepening and reallocation effects from financial intermediation. The analysis performed by the authors suggests that a country like Uganda could increase its output by 116% percent if it could adopt the world's best practices in the financial sector which, however, amounts to only 29% of the gap between Uganda's potential and actual output. Midrigan and Yi Xu (2014) analyze financial constraints arising from limits on

the amount of debt and equity that agents can issue. They find that financial frictions may reduce the level of aggregate TFP by up to 40%. Hence, this brief discussion suggests that our results are somewhat smaller than those triggered by other frictions in the capital market, although still significant.[21]

5.1 Decompositions

In this section, we analyze the aggregate effects of the distortion triggered by the capital limit $\bar{k}$ along two dimensions. First, we decompose the contribution of capital intensity and TFP to the fall in aggregate output per worker. Second, we assess whether the distortion is mainly channelized through the occupational choice or the production choice of women.

5.1.1 Accounting for Losses

We quantify now the contribution of capital intensity and TFP to the fall in output per worker. To do so, we perform two decompositions that are consistent with the two measures of TFP defined by Eqs. (22) and (26). Implicit to the definition TFP_s, for $s = 1, 2$, we have the following aggregate production function:

$$\frac{Y_s}{L_s} = \left(\frac{K}{L_s}\right)^{\alpha\zeta} \text{TFP}_s, \tag{27}$$

where $L_1 = L$ and $L_2 = L + \tilde{L}$. Hence, we can directly compute:

$$X_{Y_s/L_s} = \alpha\zeta X_{K/L_s} + X_{\text{TFP}_s}, \tag{28}$$

where X_a represents the difference of the logarithm of variable a between the baseline and the counterfactual scenario. Then, we have that the relative contribution of capital intensity and aggregate productivity is $\alpha\zeta X_{K/L_s}/X_{Y_s/L_s} \times 100$ and $X_{\text{TFP}_s}/X_{Y_s/L_s} \times 100$, respectively.

Table 7 shows the decomposition implied by Eq. (28). We observe that aggregate productivity contributes by about 65%–80% to the fall in output per worker, whereas the fall in capital intensity contributes to the rest.[22]

21 The model implicitly attributes all gender differences in entrepreneurship (observed in the data) to discrimination in financial markets. Therefore, our results should be interpreted as an upper bound on the aggregate costs of a gender-specific capital constraint.

22 Appendix C presents the results of a further analysis that considers an economy in which entrepreneurial ability and labor income are correlated.

Table 7: Contribution to output per worker losses.

(%)	S1	S2	S3	S4	S5
		Output Y_1/L_1			
Capital intensity	18.9	21.5	23.7	24.6	34.6
TFP	81.1	78.5	76.3	75.4	65.4
		Output Y_2/L_2			
Capital intensity	18.8	21.4	23.6	24.5	34.6
TFP	81.2	78.6	76.4	75.5	65.4

5.1.2 Intensive versus Extensive Margin

As discussed in Section 3.9, the capital limit of $\bar{k}$ for female entrepreneurs brings two consequences, one at an intensive margin level, and the other at an extensive margin level. First, the distortion may produce an exit of women from entrepreneurship (extensive margin effect). Second, women who stay producing despite the capital limit must run smaller businesses (intensive margin effect). We now compare the output loss generated by the intensive and extensive margin effects. We start by defining the extensive margin effect (EM) in the following way:

$$\mathrm{EM} = \theta_{\mathrm{f}} \int_{\tilde{z}_{\mathrm{f}}^*}^{\bar{z}_{\mathrm{f}}^*} \tilde{y}_{\mathrm{f}}(z) g(z) \mathrm{d}z, \tag{29}$$

where $\bar{z}_{\mathrm{f}}^*$ is the skill level of marginal female entrepreneurs in the baseline economy, $\tilde{z}_{\mathrm{f}}^*$ is the analogous threshold in the counterfactual economy, and $\tilde{y}_{\mathrm{f}}(z)$ denotes the output produced by women in the counterfactual economy.[23] Then, EM shows the output produced by women who are entrepreneurs in the counterfactual economy but who leave entrepreneurship in the baseline economy as a consequence of the distortion. That is, EM is the aggregate output loss triggered by the exit of women with skills $z \in [\tilde{z}_{\mathrm{f}}^*, \bar{z}_{\mathrm{f}}^*]$ from entrepreneurship.

Next, we define the intensive margin effect, IM as

$$\mathrm{IM} = \theta_{\mathrm{f}} \int_{\tilde{z}_{\mathrm{f}}^*} (\tilde{y}_{\mathrm{f}}(z) - \bar{y}_{\mathrm{f}}(z)) g(z) \mathrm{d}z, \tag{30}$$

23 Note that women with skills $z \in [\tilde{z}_{\mathrm{f}}^*, \bar{z}_{\mathrm{f}}^*]$ are not entrepreneurs in the baseline economy, and thus, they do not produce any output.

where $\overline{y}_f(z)$ is the output produced by women in the baseline economy. Therefore, IM measures the change in the output produced by women who run a business even in the presence of a capital limit. Table 8 presents the ratio IM/EM.

Table 8: Extensive versus intensive margin.

	S1	S2	S3	S4	S5
IM/EM	6.5	8.8	12.3	15.4	30.8

We observe that the bulk of output losses come from the fact that female entrepreneurs who stay producing in spite of the distortion run smaller businesses, that is the intensive margin effect. The contribution of women who move to production labor is indeed small, as observed in Table 8. Those are low skilled women who run small businesses, and thus, their exit from entrepreneurship causes small output losses.

Then, this last result implies that the potential aggregate gains from removing the distortion would be driven primarily by an increase in the scale of operation of businesses run by constrained female entrepreneurs. This finding is consistent with the evidence documented by Agier and Szafarz (2013). As discussed in Section 2, the authors find evidence on the existence of a glass ceiling effect that implies that the largest female projects face the highest penalty. The Agier and Szafarz evidence then shows that (highly productive) women run smaller businesses than identical men, which is consistent with the considerable intensive margin effect that we found.

The results presented in Table 8 are also consistent with evidence that shows that women run smaller business than men in terms of different dimensions. For example, Bruhn (2009) finds that women-owned businesses in Latin America tend to be smaller than male-owned companies in terms of sales and number of employees. Sabarwal and Terrell (2009) also study the performance of women-owned businesses in Latin America and find that women-owned businesses are smaller than those owned by men in many dimensions. Furthermore, Bardasi, Sabarwal, and Terrell (2011) find that women-owned businesses in sub-Saharan Africa have sales that are 31% lower than male-owned businesses. Likewise, the research surveyed in Carranza, Dhakal, and Love (2018) shows that women-owned businesses tend to be smaller. OECD data also supports a gender gap in business size. For example, Piacentini (2013), based on the OECD-ORBIS data set, shows that, in all European countries for which data are available, women own less than 15% of the companies with the highest asset value. Specifically, the proportion of companies owned by women in the top 10% of assets ranges from 3% in countries such as Austria and Slovenia to 15% in Italy. In general, the author concludes that

women rarely own large businesses and, when they do, their companies tend to be small and operate with little capital. The OECD (2012) also documents that, when women start businesses, they do so on a smaller scale than men. Then, the evidence reported by these studies is aligned with the significant intensive margin effects that we found.

6 Conclusions

We developed a general equilibrium model to quantify the aggregate consequences of a discriminatory restriction for women in their access to business resources. Our counterfactual analysis shows that a gender-specific capital restriction causes an aggregate output loss between 14% and 28% and a total factor productivity loss between 12% and 20%. About two-thirds of the output loss is explained by the fall in aggregate productivity. Furthermore, we show that the potential aggregate gains of loosening the distortion studied in this paper would be mostly driven by the more intensive use of capital among existing female entrepreneurs and not by the entry of more women into entrepreneurship.

Our findings are policy-relevant. We show significant aggregate gains from more equal access to business resources across genders. Therefore, the evidence provided in this paper supports the need for a more vigorous discussion on mechanisms to reduce sources of discrimination against women entrepreneurs that still persist. For instance, passing and enforcing anti-discrimination laws in the financial system would be a first step. Also, the evaluation of some mechanisms to encourage banks and financial institutions to hire female loan officers or, at least, to reduce discrimination in hiring policies, could create a more favorable environment for women in the capital market. Lastly, some affirmative action policies could also be considered. For example, fiscal subsidies for women-owned businesses could be implemented as a "second best option" to mitigate the aggregate effects of any type of gender-driven misallocation. However, designing support policies for women entrepreneurs is not straightforward, since the social returns of these targeted policies are difficult to quantify and monitor. Hence, potential policies must be supported by better data and stronger institutional frameworks.

Further research should focus on assessing the effectiveness of specific policies targeting women entrepreneurs and analyzing the institutional framework in which these policies are more likely to be successful. Another avenue for future research is to explore an extension of the model developed in this paper to incorporate business risk as a determinant of business activity. One way to do this is by modeling an economy with idiosyncratic uninsurable shocks, decentralized financial markets, and gender-specific financial constraints. In that type of

model, a precautionary motive for saving arises, which would interact with financial restrictions of different magnitude for women and men. It would be interesting to assess whether this latter type of framework reveals new implications regarding the aggregate effects of gender-specific constraints for entrepreneurs.

Appendix A: Inputs Demands

In this appendix, we derive the demand level for each of the inputs of the production technology of our model economy. We first derive the demands from unconstrained agents, and then, the demands from constrained agents.

We use first Eqs. (3), (4), and (6) to get:

$$k(z) = \left(\frac{\zeta\alpha}{R}\right)^{\frac{1}{1-\zeta\alpha}} A^{\frac{1}{1-\zeta\alpha}} z^{\frac{1-\zeta}{1-\zeta\alpha}} n(z)^{\frac{\zeta(1-\alpha)}{1-\zeta\alpha}}. \tag{A1}$$

Then, we plug (A1) in (3) and use (4) to get:

$$\frac{y(z)}{n(z)} = \left(\frac{\zeta\alpha}{R}\right)^{\frac{\zeta\alpha}{1-\zeta\alpha}} A^{\frac{1}{1-\zeta\alpha}} \left(\frac{z}{n(z)}\right)^{\frac{1-\zeta}{1-\zeta\alpha}}. \tag{A2}$$

Let $\Omega_0 = \left(\frac{\zeta\alpha}{R}\right)^{\frac{\zeta\alpha}{1-\zeta\alpha}} A^{\frac{1}{1-\zeta\alpha}}$. Then, we can express Eq. (A2) as:

$$\frac{y(z)}{n(z)} = \Omega_0 \left(\frac{z}{n(z)}\right)^{\frac{1-\zeta}{1-\zeta\alpha}}. \tag{A3}$$

Next we use Eqs. (7) and (A3) to get:

$$n(z) = \Omega_1^{\frac{1-\zeta\alpha}{1-\zeta}} \left(\frac{n(z)}{n^{\mathrm{f}}(z)}\right)^{\frac{1-\zeta\alpha}{(1-\zeta)\sigma}} z, \tag{A4}$$

where $\Omega_1 = \frac{\zeta(1-\alpha)\Omega_0}{w^{\mathrm{f}}}$.

We can substitute back (A4) in (A3) to get:

$$\frac{y(z)}{n(z)} = \frac{\Omega_0}{\Omega_1} \left(\frac{n^{\mathrm{f}}(z)}{n(z)}\right)^{\frac{1}{\sigma}}. \tag{A5}$$

We derive now expressions for $n^{\mathrm{f}}(z)/n(z)$ and $n^{\mathrm{f}}(z)/n^{\mathrm{m}}(z)$. From (2) we get:

$$\frac{n(z)}{n^{\mathrm{f}}(z)} = \left(\left(\frac{n^{\mathrm{m}}(z)}{n^{\mathrm{f}}(z)}\right)^{\frac{\sigma-1}{\sigma}} + 1\right)^{\frac{\sigma}{\sigma-1}}. \tag{A6}$$

From Eqs. (7) and (8) we get:

$$\frac{n^{\mathrm{f}}(z)}{n^{\mathrm{m}}(z)} = \left(\frac{w^{\mathrm{m}}}{w^{\mathrm{f}}}\right)^{\sigma}. \tag{A7}$$

From Eq. (8) we compute:

$$n^{\mathrm{m}}(z) = \left(\frac{\zeta(1-\alpha)(y(z)/n(z))}{w^{\mathrm{m}}}\right)^{\sigma} n(z). \tag{A8}$$

We directly express:

$$n^{\mathrm{f}}(z) = \left(\frac{n^{\mathrm{f}}(z)}{n^{\mathrm{m}}(z)}\right) n^{\mathrm{m}}(z). \tag{A9}$$

From Eq. (6) we can get:

$$k(z) = \left(\frac{\zeta\alpha}{R}\right)\left(\frac{y(z)}{n(z)}\right) n(z). \tag{A10}$$

Let $M = \{w^{\mathrm{f}}, w^{\mathrm{m}}, R\}$. Substituting back the expressions derived for y/n, $n(z)$, $n(z)/n^{\mathrm{f}}(z)$, and (A7) into Eqs. (A8)–(A10) we can express the inputs demand system as:

$$k(z; M) = \Phi_0(M)z, \tag{A11}$$

$$n^{\mathrm{f}}(z; M) = \Phi_1(M)z, \tag{A12}$$

$$n^{\mathrm{m}}(z; M) = \Phi_2(M)z, \tag{A13}$$

We now consider the case in which $k = \overline{k}$. Then, we plug this condition in (3) and use (4) to get:

$$y = Az^{1-\zeta}\overline{k}^{\zeta\alpha} n^{(1-\alpha)\zeta}, \tag{A14}$$

Next we use Eqs. (7) and (A14) to get:

$$n(z) = \overline{\Omega}_0^{\frac{1}{\omega_0}} z^{\frac{1-\zeta}{\omega_0}} \overline{k}^{\frac{\alpha\zeta}{\omega_0}} (n^{\mathrm{f}}(z))^{\frac{-1}{\sigma\omega_0}} \tag{A15}$$

where $\overline{\Omega}_0 = \frac{\zeta(1-\alpha)A}{w^{\mathrm{f}}}$ and $\omega_0 = 1 - \frac{1}{\sigma} - (1-\alpha)\zeta$.

From Eqs. (7) and (8) we get:

$$\frac{n^{\mathrm{m}}(z)}{n^{\mathrm{f}}(z)} = \left(\frac{w^{\mathrm{f}}}{w^{\mathrm{m}}}\right)^{\sigma}. \tag{A16}$$

From (2) we get:

$$n(z) = \left(\left(\frac{n^{\mathrm{m}}(z)}{n^{\mathrm{f}}(z)}\right)^{\frac{\sigma-1}{\sigma}} + 1\right)^{\frac{\sigma}{\sigma-1}} n^{\mathrm{f}}(z). \tag{A17}$$

Let $\overline{\Omega}_1 = \left(\left(\frac{n^{\mathrm{m}}(z)}{n^{\mathrm{f}}(z)}\right)^{\frac{\sigma-1}{\sigma}} + 1\right)^{\frac{\sigma}{\sigma-1}}$. Then, using (A15) and (A16) in (A17), we get:

$$\overline{n}^{\mathrm{f}}(z) = \overline{\Omega}_0^{\frac{\sigma}{1+\sigma\omega_0}} \overline{\Omega}_1^{\frac{-\sigma\omega_0}{1+\sigma\omega_0}} \overline{k}^{\frac{\alpha\zeta\sigma}{1+\sigma\omega_0}} z^{\frac{(1-\zeta)\sigma}{1+\sigma\omega_0}} \tag{A18}$$

Then, substituting back (A18) in (A16), we can get $\overline{n}^{\mathrm{m}}(z)$. Then, the demand of constrained entrepreneurs is given by the following system:

$$k(z; M) = \overline{k}, \tag{A19}$$

$$n^{\mathrm{f}}(z; M) = \overline{n}^{\mathrm{f}}(z; M), \tag{A20}$$

$$n^{\mathrm{m}}(z; M) = \overline{n}^{\mathrm{m}}(z; M), \tag{A21}$$

Appendix B: Solving the Benchmark Model

We first discretize the ability space so it contains Z_n evenly spaced points between z^l and z^h. Then, we use a lognormal density distribution to compute the probability vector for each of the points in the space Z_n. Denote by p the latter vector. Once we have built the ability space and the discrete density function for them, we solve the household problem for fixed wages and rental price of capital. In Appendix A we showed that, for unconstrained entrepreneurs:

$$k(z; M) = \Phi_0(M)z, \tag{B1}$$

$$n^{\mathrm{f}}(z; M) = \Phi_2(M)z, \tag{B2}$$

$$n^{\mathrm{m}}(z; M) = \Phi_1(M)z. \tag{B3}$$

and for constrained entrepreneurs:

$$k(z; M) = \overline{k}, \tag{B4}$$

$$n^{\mathrm{f}}(z; M) = \overline{n}^{\mathrm{f}}(z; M), \tag{B5}$$

$$n^{\mathrm{m}}(z; M) = \overline{n}^{\mathrm{m}}(z; M), \tag{B6}$$

Then, we use the inputs demands to compute the profits, which is the value of being a manager: $\pi_i(z;\ M)$, for $i \in \{f, m\}$. The value of being a production worker is simply w^i. Then, considering the value of being a manager and the

value of being a production worker, we build the indicator function that identifies the members of the household who become a manager:

$$\mathcal{I}_i(z;M) = \begin{cases} 1 & \text{if } \pi_i(z;M) > w^i] \\ 0 & \text{if } \pi_i(z;M) \leq w^i] \end{cases}. \tag{B7}$$

Notice that the vector $\mathcal{I}_i(z;M)$ also identifies the skill size of the marginal manager:

$$z_i^*(M) = \min\{z \in Z\colon \mathcal{I}_i(z;M) = 1\}. \tag{B8}$$

Next, we build the aggregate demands and supply of labor and capital services. Denote by N_i^j the aggregate demand for labor service type j by group i. Analogously, denote by K_i the aggregate demand for capital services by group i. The aggregate demand for labor services type j of agent type i is:

$$N_i^j(M) = \theta_i \sum_{z=z^l}^{z^h} n_i^j(z;M)\mathcal{I}_i(z;M)p(z), \tag{B9}$$

The aggregate demand for capital services by agent type i is:

$$K_i(M) = \theta_i \sum_{z=z^l}^{z^h} k_i(z;M)\mathcal{I}_i(z;M)p(z). \tag{B10}$$

Next, we compute the aggregate demand for each of the four inputs:

$$N^{\mathrm{f}}(M) = \sum_{i\in\{\mathrm{f,m}\}} N_i^{\mathrm{f}}(M), \tag{B11}$$

$$N^{\mathrm{m}}(M) = \sum_{i\in\{\mathrm{f,m}\}} N_i^{\mathrm{m}}(M), \tag{B12}$$

$$K(M) = \sum_{i\in\{\mathrm{f,m}\}} K_i(M), \tag{B13}$$

Then, we compute the supply of labor inputs:

$$S_i(M) = \theta_i \sum_{z=z^l}^{z^h} (1 - \mathcal{I}_i(z;M))p(z) \tag{B14}$$

For the supply of capital we initially set $K = \hat{K}$. Then, given the guess for the capital stock, we compute equilibrium prices. We iterate on prices until all markets clear and find equilibrium prices for a given stock of capital. Then, we evaluate if the resulting rental rate for capital services differs from $\left((1+x_{\mathrm{c}})/\beta\right) + \delta - 1$ and we

iterate on capital until the R is equal to $\left((1+x_c)/\beta\right)+\delta-1$, using an bijective algorithm.

Appendix C: Additional Results

This appendix presents further analysis considering an economy in which entrepreneurial ability and labor income are correlated. This alternative model captures the idea that entrepreneurs with higher ability also have better outside options and are generally more skilled, which would enable them to earn higher wages.[24]

In general, labor income can be modeled as wz^{γ} with $\gamma \geq 0$. In our baseline model, we follow Guner, Ventura, and Xu (2008), Buera and Shin (2013), Buera, Moll, and Shin (2015), among others, and set $\gamma = 0$. The additional analysis presented in this appendix considers $\gamma = 1-\zeta$; that is, we set the same degree of concavity for the production function and the expression for labor income, as in Cagetti and De Nardi (2006). Then, the market clearing condition for production labor was modified accordingly to match the supply of labor in efficiency units with the demand, the model was recalibrated, and the capital thresholds that define the counterfactual scenarios were modified to make them consistent with those of the main analysis. Tables C1–C4 show the results. We observe that the magnitude of the effects is relatively similar to that of the original model, the effect being somewhat lower for some outcomes and higher for other outcomes.

Table C1: Aggregate output losses in benchmark relative to counterfactuals.

	S1	S2	S3	S4	S5
Aggregate output, Y (%)	13.7	17.1	21.0	23.7	34.1
Ratio output per female manager, r_f (level)	2.3	2.8	3.4	4.0	7.0

Table C2: Aggregate output per worker losses in benchmark relative to counterfactuals.

(%)	S1	S2	S3	S4	S5
Output per efficiency unit of labor, q_1	16.1	19.8	24.1	27.0	37.8
Output per production worker, q_2	9.9	11.9	14.0	15.6	21.8
Output per manager, q_3	24.7	30.9	37.9	42.5	58.6

24 We thank an anonymous referee for suggesting this extension of our model.

Table C3: TFP losses in benchmark relative to counterfactuals.

(%)	S1	S2	S3	S4	S5
TFP_1	7.1	8.3	9.8	11.1	12.1
TFP_2	7.1	8.3	9.8	11.1	12.1

Table C4: Contribution to output per worker losses.

(%)	S1	S2	S3	S4	S5
		Output Y_1/L_1			
Capital intensity	29.0	31.8	31.7	30.8	47.5
TFP	71.0	68.2	68.3	69.2	52.5
		Output Y_2/L_2			
Capital intensity	29.0	31.8	31.7	30.8	47.5
TFP	71.0	68.2	68.3	69.2	52.5

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